Linear Algebra I-II
Theorems & Lemmas

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1 Vector Spaces

1.1 Vector Spaces and Linear Subspaces

Definition 1.1: Vector Space

A **vector space** over a field K is a set V, endowed with two operations

$$\begin{aligned} + : V \times V \to V, \qquad & (v_1, v_2) \mapsto v_1 + v_2, \\ \cdot : K \times V \to V, \qquad & (\alpha, v) \mapsto \alpha \cdot v \end{aligned}$$

s.t. the following conditions (axioms) hold:

(V1) $\forall v_1, v_2, v_3 \in V : v_1 + (v_2 + v_3) = (v_1 + v_2) + v_3$,

(V2) $\exists$ an element $0 \in V : 0 + v = v \qquad \forall v \in V$,

(V3) $\forall v \in V\ \exists v' \in V : v + v' = 0$,

(V4) $\forall v_1, v_2 \in V : v_1 + v_2 = v_2 + v_1$,

(V5) $\forall a, b \in K\ \forall v \in V : a \cdot (b \cdot v) = (a \cdot b) \cdot v$,

(V6) $\forall v \in V : 1 \cdot v = v$,

(V7) $\forall a \in K\ \forall v_1, v_2 \in V : a \cdot (v_1 + v_2) = a \cdot v_1 + a \cdot v_2$,

(V8) $\forall a, b \in K\ \forall v \in V : (a + b) \cdot v = a \cdot v + b \cdot v$.

Lemma 1.2: Properties of Vector Spaces

Let V be a vector space over a field K. Then:

(a) The 0 vector from (V2) is unique. To avoid confusion (e.g. with $0 \in K$) we will sometimes denote it by 0_V.

(b) The element v' (corresponding to v) from (V3) is unique. We'll denote it by $-v$. We'll also write $w - v := w + (-v)$ for all $w \in V$,

(c) $\forall v \in V : 0 \cdot v = 0$,

(d) $\forall a \in K : a \cdot 0 = 0$,

(e) $\forall v \in V : -1 \cdot v = -v$,

(f) $\forall v \in V : -(-v) = v$,

(g) If $a \cdot v = 0$ for some $a \in K, v \in V$, then either $a = 0$ or $v = 0$,

(h) Associativity for $+$ holds also for some of n elements. $v_1 + \ldots + v_n$ makes sense no matter how we put the brackets $v_1 + (v_2 + (\ldots + (v_{n-1} + v_n)))$ or $(v_1 + v_2) + (\ldots)$ etc.

Example: The Coordinate Space K^n

Let K be a field and $n \in \mathbb{N}$.

$$K^n = \{(a_1, \dots, a_n) \mid a_i \in K, 1 \leq i \leq n\}.$$

K^n is a vector space over K if we endow it with the following operations:

- Let $v = (a_1, \dots, a_n), w = (b_1, \dots, b_n) \in K^n$.

$$v + w = (a_1, \dots, a_n) + (b_1, \dots, b_n) := (a_1 + b_1, \dots, a_n + b_n),$$

- For $a \in K$

$$a \cdot v = a \cdot (a_1, \dots, a_n) := (a \cdot a_1, \dots, a \cdot a_n)$$

- $0 := (0, \dots, 0) \in K^n$

With these operations K^n becomes a vector space over K.

Definition 1.3: Linear Subspace

Let V be a vector space over K. A subset $W \subseteq V$ is called a **linear subspace** of V (or just subspace of V), if the following holds

(LSS1) $W \neq \emptyset$,

(LSS2) $\forall w_1, w_2 \in W : w_1 + w_2 \in W$,

(LSS3) $\forall a \in K, w \in W : a \cdot w \in W$.

Lemma 1.4: Linear Subspace is a Vector Space

Let V be a vector space over K and $W \subseteq V$ a subspace. Then W is a vector space on its own, when endowed with the operations from V.

Lemma 1.5: Criterion for a Linear Subspace

Let V be a vector space over K, and $W \subseteq V$ a subset. Then W is a linear subspace of V if and only if the following holds:

(1) $0_V \in W$,

(2) $\forall a_1, a_2 \in K, w_1, w_2 \in W : a_1 w_1 + a_2 w_2 \in W$.

Example: Spaces of functions

Fix a field K and a non-empty set S. Define

$$K^S = \{f : S \to K \mid f \text{ is a function}\}.$$

Define $+$ and $\cdot$ on K^S as follows. Let $f, g \in K^S$, $a \in K$. Define

$$(f+g) : S \to K, \qquad S \ni x \xmapsto{f+g} f(x) + g(x),$$
$$(a \cdot f) : S \to K, \qquad S \ni x \xmapsto{a \cdot f} a \cdot f(x).$$

Now Let $S = [0,1]$ and take $K = \mathbb{R}$. Define $C(S) := \{f \in \mathbb{R}^S \mid f \text{ is continuous on } [0,1]\} \subseteq \mathbb{R}^S$. Then $C(S) \subseteq \mathbb{R}^S$ is a linear subspace.

1.2 Span of Sets

Definition 1.6: Span

Let $S \subseteq V$ be a subset. We define

$$\mathrm{Sp}(S) := \bigcap_{W \in \mathcal{N}} W,$$

where

$$\mathcal{N} := \{W \mid W \text{ is a subspace of } V \text{ and } S \subseteq W\},$$

i.e., the collection of all subspaces $W \subseteq V$ that contain S. $\mathrm{Sp}(S)$ is called the span of S (in V).

Lemma 1.7: Properties of the Span

(1) $\mathrm{Sp}(S) \subseteq V$ *is a linear subspace.*

(2) If $W \subseteq V$ *is a linear subspace and* $S \subseteq W$*, then* $\mathrm{Sp}(S) \subseteq W$.

In other words, $\mathrm{Sp}(S)$ *is a linear subspace of* V*, and moreover among those subspaces that contain* S*,* $\mathrm{Sp}(S)$ *is the smallest one.*

Definition 1.8: Linear Combination

Let $n \in \mathbb{N}$, $a_1, \dots, a_n \in K$, $v_1, \dots, v_n \in V$. A **linear combination** of the vectors $v_1, \dots, v_n$ with coefficients $a_1, \dots, a_n$ is the vector

$$a_1 v_1 + \ldots + a_n v_n = \sum_{i=1}^{n} a_i v_i \in V.$$

It is important to note that in this course consider only finitely many elements in the sums of linear combinations.

Lemma 1.9: Alternate Definition of Span

Let $\emptyset \neq S \subseteq V$*. Then*

$$\mathrm{Sp}(S) = \{a_1 v_1 + \ldots + a_n v_n \mid n \in \mathbb{N}, a_i \in K, v_i \in S \quad \forall 1 \leq i \leq n\}.$$

Remark 1.10. *The set* S *can be finite or infinite. Regardless of that, each linear combination uses only finite numbers of elements from* S.

Let $n \in \mathbb{N}$, $v_1, \ldots, v_n \in V$. We'll write

$$\mathrm{Sp}(v_1, \ldots, v_n) := \mathrm{Sp}(\{v_1, \ldots, v_n\}).$$

Remark 1.11. *It obviously holds that*

$$\mathrm{Sp}(v_1, \ldots, v_n) = \left\{ \sum_{i=1}^{n} \alpha_i v_1 \,\middle|\, \alpha_i \in K \quad \forall 1 \le i \le n \right\}.$$

Remark 1.12. *It follows from the definition of span that*

$$\mathrm{Sp}(\emptyset) = \{0_V\}.$$

Definition 1.13: Generating Set

Let V be a vect. sp. over K and $S \subseteq V$. We say that S **generates** (or **spans**) V if $\mathrm{Sp}(S) = V$. S is called a generating set of V if $\mathrm{Sp}(S) = W$, where $W \subseteq V$ is a subspace of V. We say that S **generates** (or **spans**) W.

Definition 1.14: Finite Dimensional

A vector space V is called **finite-dimensional** if $\exists$ a **finite** set $S \subseteq V$ such that $\mathrm{Sp}(S) = V$. Otherwise V is called **infinite-dimensional**.

1.3 Bases of Vector Spaces

1.3.1 Linear Independence

Let V be a vector space over K.

Definition 1.15: Linear Independence

Let $v_1, \ldots, v_n$ be a list of n vectors in V. We say that $v_1, \ldots, v_n$ is **linearly independent** if the only linear combination of $v_1, \ldots, v_n$ that gives $0 \in V$ is the trivial one. In other words,

$$\forall a_1, \ldots, a_n \in K : a_1 v_1 + \ldots + a_n v_n \quad \Longrightarrow \quad a_1 = 0, \ldots a_n = 0.$$

By definition, we say that the empty list of vectors $\emptyset$ is linearly independent. We say that a list of vectors is linearly dependent if, the list is not linearly independent.

Remark 1.16. *(1) 0 can always be written as a linear combination of any finite list of vectors* $0v_1 + \ldots + 0v_n = 0$*, i.e., the trivial linear combination.*

(2) If the list $v_1, \ldots, v_n$ *has two vectors that are the same, say* $v_k = v_l$ *for* $1 \le k < l \le n$*, then* $v_1, \ldots, v_n$ *is* ***not*** *linearly independent. Indeed,* $v_k + (-1)v_l = 0$.

(3) The order of the vectors in the list $v_1, \ldots, v_n$ *has no effect on linear independence. So, if there are no repetitions in* $v_1, \ldots, v_n$ *we can talk about linear independence of the set* $\{v_1, \ldots, v_n\}$.

Definition 1.17: Linear Independence in the Context of Families

Let $\mathcal{F} = \{v_i\}_{i\in I}$ be a familiy of vectors in V, indexed by a set I. We say that $\mathcal{F}$ is linearly independent if $\forall n \in \mathbb{N}$ and any sequence of distinct indices $i_1, \ldots, i_n \in I$, the list of vectors $v_{i_1}, \ldots, v_{i_n}$ is linearly independent. Of course, if in $\mathcal{F}$ there are any repetitions of vectors, i.e., if $\exists i, j, i \neq j$, s.t. $v_i = v_j$, then $\mathcal{F}$ cannot be linearly independent.

Lemma 1.18: Subsets of Linearly Independent Sets are Linearly Independent

If $\emptyset \neq S \subseteq V$ is linearly independent. Then every non-empty subset $S' \subseteq S$ is also linearly independent.

Definition 1.19: Linear Dependence

A family $\mathcal{F} = \{v_i\}_{i\in I}$ of vectors in V is called **linearly dependent** if it is not linearly independent.

1.3.2 Basis of a Vector Space

Definition 1.20: Basis of a Vector Space

A subset $S \subseteq V$ of a vector space is called a basis of V if the following holds:

(1) S is linearly independent.

(2) S generates (or spans) V, i.e., $\mathrm{Sp}(S) = V$.

Proposition 1.21: Equivalent Formulation of basis

A subset $S \subseteq V$ is a basis of V if and only if every $v \in V$ can be written in a unique way as a linear combination of vectors in S.

Proof. For simplicity we will assume that $|S| < \infty$. The case $|S| = \infty$ is similar.

<u>$\Longleftarrow$:</u> We assume that every $v \in V$ can be written in a unique way as a linear combination of elements from S. Now, "can be written" means that $\mathrm{Sp}(S) = V$. Next write $S = \{v_1, \ldots, v_n\}$. Consider the vector $0 \in V$. Clearly

$$0 = 0 \cdot v_1 + \ldots + 0 \cdot v_n.$$

Since 0 can be written in a unique way as a linear combination of $v_1, \ldots, v_n$ it follows that

$$0 = a_1 v_1 + \ldots + a_n v_n \qquad \Longrightarrow \qquad a_1 = \ldots = a_n = 0.$$

Therefore, S is linearly independent, which proves that S is a basis of V.

<u>$\Longrightarrow$:</u> We assume that S is a basis for V. By assumption $\mathrm{Sp}(S) = V$, so this means every $v \in V$ can be expressed as a linear combination of vectors from S. It remains to show that this unique.

Assume by contradiction that $\exists v \in V$ which has two different ways to be written as a linear combination of vectors from S, i.e.,

$$v = a_1 v_1 + \ldots + a_n v_n = b_1 v_1 + \ldots + b_n v_n$$

and $a_k \neq b_k$ for some $1 \leq k \leq n$. Then

$$0 = (a_1 - b_1)v_1 + \ldots + \underbrace{(a_k - b_k)}_{\neq 0} + \ldots + (a_n - b_n)v_n$$

and we deduce that 0 can be written as a non-trivial linear combination of elements from S. Therefore, S is linearly dependent, which is a contradiction to the fact that S is a basis of V. □

1.3.3 Example: Bases

$V = M_{m\times n}(K)$, $\{E_{ij} \mid 1 \leq i \leq m, 1 \leq j \leq n\}$ is a basis, where

$$E_{ij} = \begin{pmatrix} 0 & \ldots & 0 & \ldots & 0 \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & \ldots & 1 & \ldots & 0 \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & \ldots & 0 & \ldots & 0 \end{pmatrix},$$

where the 1 is at the position (i, j) in the matrix.

Lemma 1.22: A

Let $v_1, \ldots, v_m \in V$ be a list of vectors which is linearly dependent. Then $\exists 1 \leq j \leq m$ s.t.:

(1) $v_j \in \mathrm{Sp}(v_1, \ldots, v_{j-1})$.

(2) $\mathrm{Sp}(v_1, \ldots, v_{j-1}, v_{j+1}, \ldots, v_m) = \mathrm{Sp}(v_1, \ldots, v_m)$. *In other words, we can drop one of the vectors from the list without changing the span.*

Lemma 1.23: B

Same assumptions an in Lemma 1.22, but we assume in addition that $v_1, \ldots, v_k$ are linearly independent, where $1 \leq k \leq m$ is some index. Then $k < j$.

Lemma 1.24: C

Let $w_1, \ldots, w_n \in V$ s.t. $\mathrm{Sp}(w_1, \ldots, w_n) = V$, and let $v \in V$. Then $v, w_1, \ldots, w_n$ is linearly dependent.

Lemma 1.25: D

Let $v_1, \ldots, v_n \in V$ be s.t. $\mathrm{Sp}(v_1, \ldots, v_n) = V$, and let $u_1, \ldots, u_m \in V$ be linearly independent. Then $m \leq n$. Moreover, one can delete m vectors from the list $v_1, \ldots, v_n$ s.t. the remaining list, say $v'_1, \ldots, v'_{n-m}$ together with $u_1, \ldots, u_m$ still span V, i.e., $\mathrm{Sp}(u_1, \ldots, u_m, v'_1, \ldots, v'_n) = V$.

Lemma 1.26: E

Let $w_1, \ldots, w_l \in V$ and assume that $w_j \notin \mathrm{Sp}(w_1, \ldots, w_{j-1})$ $\forall 1 \leq j \leq l$. Then the list $w_1, \ldots, w_l$ is linearly independent.

Lemma 1.27: F

Suppose that $\mathrm{Sp}(v_1, \ldots, v_n) = V$*. Then* $\exists$ *a subset of* $\{v_1, \ldots, v_n\}$ *which is a basis for* V*.*

Theorem 1.28: Bases Have the Same Number of Elements

Let V *be a finite dimensional vector space over* K*. Then* V *has a basis with finitely many elements. Moreover, every basis of* V *is finite and has the same number of elements.*

Definition 1.29: Dimension of a Vector Space

Let V be a finite dimensional vector space over K. We define the dimension of V to be the number $n \in \mathbb{N}$ of elements in a basis (any basis !) of V. We write $\dim V := n$. In case V is not finite dimensional we write $\dim V = \infty$.

Remark 1.30.

- $\dim V$ *is well defined as it does not depend on a specific choice of basis for* V*.*
- $\dim\{0\} = 0$*, b.c.* $\emptyset$ *is a basis for the space* $\{0\}$*, and* $|\emptyset| = 0$.

Summary

Let V be a finite dimensional vector space over K. Then:

(1) Every finite list of vectors that spans V contains a sublist which is a basis for V.

(2) Every linearly independent list can be extended to a basis of V.

Theorem 1.31: Equivalent Statements for Bases

Let V *be a vector space of* $\dim V = n \in \mathbb{N}$*. Let* $v_1, \ldots, v_n \in V$*. Then the following statements are equivalent:*

(1) $v_1, \ldots, v_n$ *are linearly independent.*

(2) $v_1, \ldots, v_n$ *generate* V*, i.e.,* $\mathrm{Sp}(v_1, \ldots, v_n) = V$*.*

(3) $v_1, \ldots, v_n$ *is a basis for* V*.*

Theorem 1.32: Consequences of Bases

Suppose that $n = \dim V < \infty$*. Let* $v_1, \ldots, v_k \in V$*. Then:*

(1) If $k < n$*, the list* $v_1, \ldots, v_k$ ***cannot*** *span* V*.*

(2) If $k > n$*, the list* $v_1, \ldots, v_k$ ***cannot*** *be linearly independent.*

Proposition 1.33: Dimension of Subspaces

Let V *be a finite dimensional vector space over* K*. Then every subspace* $U \subseteq V$ *is also finite dimensional. Moreover,* $\dim U \leq \dim V$ *with equality if and only if* $U = V$*.*

1.4 Row and Column Spaces

Row and Column Spaces

Definition 1.34: Row and Column Space

Let $A \in M_{m\times n}(K)$ and $u_1,\dots,u_m \in K^n$ the rows of A and $v_1,\dots,v_n \in K^m$ the columns of A, i.e.,

$$A = \begin{pmatrix} — & u_1 & — \\ & \vdots & \\ — & u_m & — \end{pmatrix} = \begin{pmatrix} | & & | \\ v_1 & \dots & v_n \\ | & & | \end{pmatrix}.$$

We define two spaces

$$\mathrm{RowS}(A) := \mathrm{Sp}(u_1,\dots,u_m) \subseteq K^n$$
$$\mathrm{ColS}(A) := \mathrm{Sp}(v_1,\dots,v_n) \subseteq K^m.$$

Lemma 1.35: Row Equivalent Matrices

If $A, B \in M_{m\times n}(K)$ are row-equivalent matrices then $\mathrm{RowS}(A) = \mathrm{RowS}(B)$.

Remark 1.36. *The statement of the Lemma does **not** hold for the column space.*

1.4.1 Transpose of a Matrix

Definition 1.37: Transpose of a Matrix

Let $A = (a_{i,j}) \in M_{m\times n}(K)$. We define the transposed matrix of A as $A^T := (b_{i,j}) \in M_{n\times m}(K)$, where $b_{i,j} = a_{j,i}$.

Lemma 1.38: Properties of the Transpose

or every $A, B \in M_{m\times n}(K)$, $\lambda \in K$ we have

1. $(A+B)^T = A^T + B^T$.
2. $(\lambda A)^T = \lambda A^T$.
3. $(A^T)^T = A$.
4. $\mathrm{RowS}(A^T) = \mathrm{ColS}(A)$, $\mathrm{ColS}(A^T) = \mathrm{RowS}(A)$.

1.5 Sums of Vector Spaces

Definition 1.39: Sum of Vector Spaces

Let V be a vector space over K and $U, W \subseteq V$ linear subspaces. We define the sum $U+W$ of U, W as

$$U + W := \{u + w \mid u \in U, w \in W\}.$$

In a similar way we can define sums of more than two subspaces. If $U_1,\dots,U_r \subseteq V$ are subspaces, define

$$U_1 + \ldots + U_r := \{u_1 + \ldots + u_r \mid u_1 \in U_1, \dots, u_r \in U_r\}.$$

Proposition 1.40: Properties of the Sum of Vector Spaces

Let $U, W \subseteq V$ be linear subspaces. Then:

1. $U + W = \mathrm{Sp}(U \cup W)$. *In particular,* $U + W \subseteq V$ *is a subspace and* $U, W \subseteq U + W$ *are subspaces of* $U + W$.

2. *If* U, W *are finite dimensional the so is* $U+W$. *Moreover, one can find a basis for* $U+W$ *as follows:*

   (a) *Choose a basis* $p_1, \ldots, p_k$ *for* $U \cap W$, *where* $k = \dim(U \cap W)$.

   (b) *Extend* $p_1, \ldots, p_k$ *to a basis* $p_1, \ldots, p_k, u_1, \ldots u_1, \ldots, u_{l-k}$ *of* U, *where* $l = \dim U$.

   (c) *Extend* $p_1, \ldots, p_k$ *to a basis* $p_1, \ldots, p_k, w_1, \ldots, w_{m-k}$ *of* W, *where* $m = \dim W$.

   Then $p_1, \ldots, p_k, u_1, \ldots, u_{l-k}, w_1, \ldots, w_{m-k}$ *is a basis for* $U + W$.

3. *In particular, we have* $\dim(U + W) = \dim U + \dim W - \dim(U \cap W)$.

Corollary 1.41: Consequences of the Properties

Let $U, W \subseteq V$ *be two finite dimensional linear subspaces of a vector space* V. *The following are equivalent:*

1. $\dim(U + W) = \dim U + \dim W$.

2. $\dim(U \cap W) = 0$.

3. $U \cap W = \{0\}$.

4. *Every* $v \in U + W$ *can be written in a* ***unique*** *way as* $v = u + w$ *with* $u \in U$, $w \in W$

5. *If* $u + w = 0$, *where* $u \in U$ *and* $w \in W$, *then* $u = w = 0$.

Definition 1.42: Linear Complement

Let V be a vector space over K and $U \subseteq V$ a linear subspace. A linear subspace $W \subseteq V$ is called a **linear complement** of U if $U + W = V$ and $U \cap W = \{0\}$. Note that U, W satisfy all the five equivalent statements from the corollary 1.41.

Proposition 1.43: Existence of a Complement

Let V *be a finite dimensional vector space and* $U \subseteq V$ *a subspace. Then there exists a subspace* $W \subseteq V$ *which is a complement to* U.

2 Linear Maps

Definition 2.1: Linear Map

Let V, W be vector spaces over K. A map $T : V \to W$ is called linear map if:

1. $\forall v_1, v_2 \in V : \; T(v_1 + v_2) = T(v_1) + T(v_2)$
2. $\forall \alpha \in K, \forall v \in V : \; T(\alpha \cdot v) = \alpha \cdot T(v)$

This means that $T : V \to W$ is a map that respects the structures of vector spaces on $(V, +, \cdot), (W, +, \cdot)$. We denote the set of all linear maps $T : V \to W$ by $\mathrm{Hom}(V, W)$. Sometimes $\mathrm{hom}_K(V, W)$.

Proposition 2.2: Properties of Linear Maps

Let V, W be vector spaces over a field K, and let $T : V \to W$ be a linear map. Then:

1. *$\forall n \in \mathbb{N}, a_1, \dots, a_n \in K, v_1, \dots, v_n \in V$ we have*

$$T\Big(\sum_{i=1}^{n} a_i v_i\Big) = \sum_{i=1}^{n} a_i T(v_i).$$

2. *$T(0_V) = 0_W$.*

Theorem 2.3: A unique Linear Map

Let V be a finite dimensional vector space and $v_1, \dots, v_n \in V$ a basis for V. Let $w_1, \dots, w_n \in W$ be arbitrary vectors. Then there exists a unique linear map $T : V \to W$ s.t.

$$T(v_i) = w_i \qquad \forall 1 \leq i \leq n.$$

Lemma 2.4: Matrix Representation of the Unique Linear Map

For every linear transformation $T : K^n \to K^m$ there exists a unique matrix $A \in M_{m \times n}(K)$ s.t. $T = T_A$.

2.1 Kernel & Image

Definition 2.5: Kernal and Image

Let V, W we vector spaces over K and $T : V \to W$ a linear map.

1. The set
$$\ker(T) := \{v \in V \mid Tv = 0\} \subseteq V$$
is a linear subspace of V. It is called the **kernel of T**.

2. The set
$$\mathrm{Im}(T) := \{Tv \mid v \in V\} \subseteq W$$
is a linear subspace of W. It is called the **image of T**.

With the notation from set theory, we have

$$\ker(T) = T^{-1}(\{0\}), \qquad \mathrm{Im}(T) = T(V).$$

Proposition 2.6: Equivalences for Injective and Surjective

Let $T : V \to W$ be a linear map. Then:

1. *T is injective* $\iff$ $\ker(T) = \{0\}$.
2. *T is surjective* $\iff$ $\mathrm{Im}(T) = W$.

2.1.1 About Linear Maps

Definition 2.7: Isomorphism

A linear map $T : V \to W$ is called an **isomorphism** if there exists a linear map $S : W \to V$ s.t. $S \circ T = \mathrm{id}_V$, $T \circ S = \mathrm{id}_W$. We say that V is isomorphic to W if there exists an isomorphism $T : V \to W$. We write $V \cong W$.

Lemma 2.8: Linearity of the Inverse Map

Let $T : V \to W$ be a linear map which is bijective. Then the inverse map $T^1 : W \to V$ is linear. In other words, a linear bijective map is an isomorphism.

Lemma 2.9: Composition of Linear Maps

Let $T : V \to W$ and $S : W \to U$ be linear maps. Then the composition $S \circ T : V \to U$ is also linear.

Definition 2.10: Endomorphism / Automorphism

A linear map $T : V \to V$ is sometimes called an **endomorphism** of V. We denote the set of all endomorphisms of V by $\mathrm{End}(V)$. (So, $\mathrm{End}(V) = \mathrm{Hom}(V, V)$.)
A linear map $T : V \to V$ which is an isomorphism is also called an **automorphism** of V.

Lemma 2.11: Span of a Basis

Let $T : V \to W$ be a linear map. Let $v_1, \dots, v_n$ be a basis of V. Then $\mathrm{Im}(T) = \mathrm{Sp}(Tv_1, \dots, Tv_n)$

Theorem 2.12: Rank Theorem

Let V, W be vector spaces, where V is finite dimensional and let $T : V \to W$ be a linear map. Then

$$\dim \ker(T) + \dim \mathrm{Im}(T) = \dim V.$$

Proof. Define $n := \dim V$. Let $u_1, \dots, u_k$ be a basis for $\ker(T)$. Note that $\ker(T)$ is finite dimensional, b.c. it is a subspace of V, which is finite dimensional. Extend $u_1, \dots, u_k$ to a basis $u_1, \dots, u_k, v_1, \dots, v_{n-k}$ of V, which we can do by a previous result. By Lemma 2.11, we have that

$$\mathrm{Im}(T) = \mathrm{Sp}(\underbrace{Tu_1, \dots Tu_2}_{=0}, Tv_1, \dots, Tv_{n-k}) = \mathrm{Sp}(Tv_1, \dots, Tv_{n-k}).$$

We'll show now that $Tv_1, \dots, Tv_{n-k}$ is linearly independent. To see this, assume that

$$\sum_{i=1}^{n-k} a_i T v_i = 0,$$

for some $a_1, \dots a_{n-k} \in K$. Since T is linear we can write this linear combination as

$$T\Big(\sum_{i=1}^{n-k} a_i v_i\Big) = 0 \qquad \Longrightarrow \qquad \sum_{i=1}^{n-k} a_i v_i \in \ker(T).$$

Now, $u_1, \dots, u_k$ form a basis for $\ker(T)$. So, $\exists\, b_1, \dots, b_k \in K$ s.t.

$$\begin{aligned} \sum_{i=1}^{n-k} a_i v_i &= b_1 u_1 + \ldots + b_k u_k \\ \Longrightarrow \quad b_1 u_1 + \ldots + b_k u_k &+ (-a_1) v_1 + \ldots + (-a_{n-k}) v_{n-k} = 0. \end{aligned}$$

Since, $u_1, \dots, u_k, v_1, \dots, v_{n-k}$ form a basis for V, we have that $b_1 = \ldots = b_k = a_1 = \ldots = a_{n-k} = 0$. This shows that $Tv_1, \dots, Tv_{n-k}$ are linearly independent. Therefore, $Tv_1, \dots, Tv_{n-k}$ form a basis for $\mathrm{Im}(T)$. Thus,

$$\begin{aligned} \dim \mathrm{Im}(T) &= n - k = \dim V - \dim \ker(T) \\ \Longrightarrow \quad \dim \ker(T) &+ \dim \mathrm{Im}(T) = \dim V. \end{aligned}$$

$\square$

Corollary 2.13: Consequences of the Rank Theorem

Let $T : V \to W$ be a linear map, where V, W are finite dimensional vector spaces. Then the following holds:

1. *If* $\dim W < \dim V$ *then T is **not** injective.*
2. *If* $\dim W > \dim V$ *then T is **not** surjective.*
3. *If* $\dim W = \dim V$ *then T is bijective $\iff$ T is surjective $\iff$ T is injective.*

Corollary 2.14: Equivalence for an Isomorphism

Two finite dimensional vector spaces V, W are isomorphic if and only if $\dim V = \dim W$.

Theorem 2.15: Image of a Set

Let $T : V \to W$ be an isomorphism between two vector spaces V, W. Let S be a family of vector in V. Write $T(S) := \{Tv\}_{v\in S}$. Then:

1. *S is linearly independent if and only if $T(S)$ is linearly independent.*
2. *S spans V if and only if $T(S)$ spans W.*
3. *S is a basis of V if and only if $T(S)$ is a basis of W.*

What is the significance of isomorphisms? If two vector spaces V, W are isomorphic, $V \cong W$ (,i.e., $\exists T : V \to W$ bijection), then V and W have the same linear algebraic properties (like dimension). We can think of V and W as the same space represented in two different ways.

However, identifying V with W usually requires a choice of an isomorphism $T : V \to W$. Usually there is no preferred (canonical) choice. It is therefore better, in general, not to identify different vector spaces that are isomorphic.

Definition 2.16: Rank

Let $T : V \to W$ be a linear map. We define the **rank** of T by

$$\operatorname{rank}(T) := \dim \operatorname{Im}(T)$$

Lemma 2.17: Rank of the Composition

Let $T : V \to W$, $S : W \to U$ be linear maps, where U, V, W are finite dimensional vector spaces. Then

1. $\operatorname{rank}(S \circ T) \leq \min\{\operatorname{rank}(S), \operatorname{rank}(T)\}$.
2. *If S is injective, then* $\operatorname{rank}(S \circ T) = \operatorname{rank}(T)$
3. *If T is surjective, then* $\operatorname{rank}(S \circ T) = \operatorname{rank}(S)$.

2.2 Writing Linear Maps using Bases/Coordinates

In this Section we'll assume all vector spaces to be finite dimensional unless otherwise stated. We'll write bases of a vector space V as an ordered list $\mathcal{B} = (v_1, \ldots, v_n)$.

Definition 2.18: Coordinate Map

Let V be a vector space over K and $\mathcal{B} = (v_1, \dots, v_n)$ a basis of V. Let $v \in V$. We define the coordinate-vector of v w.r.t. the basis $\mathcal{B}$ as

$$[v]_{\mathcal{B}} = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} \in K^n,$$

where $a_1, \dots, a_n \in K$ are the unique scalars in K s.t. $v = a_1 v_1 + \ldots + a_n v_n$. Note that $[v]_{\mathcal{B}}$ is uniquely defined since every $v \in V$ has a unique representation as a linear combination of $v_1, \dots, v_n$.
Define a map $\Phi_{\mathcal{B}} : V \to K^n$ as

$$\Phi_{\mathcal{B}}(v) := [v]_{\mathcal{B}} \in K^n.$$

Proposition 2.19: $\Phi_{\mathcal{B}}$ is an Isomorphism

$\Phi_{\mathcal{B}}$ is an isomorphism

Remark 2.20. *The inverse map to $\Phi_{\mathcal{B}}$ is given by*

$$\Phi_{\mathcal{B}}^{-1} : K^n \to V, \quad \Phi_{\mathcal{B}}^{-1} \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} := a_1 v_1 + \ldots + a_n v_n.$$

Corollary 2.21: Isomorphism Coordinate Space

Let V be a vector space over K with $\dim V = n \in \mathbb{Z}_{\geq 0}$. Then $V \cong K^n$.

Proof. Choose a basis $\mathcal{B}$ for V. Then $\Phi_{\mathcal{B}} : V \to K^n$ is an isomorphism. □

Remark 2.22.
1. *The corollary does not hold for $\dim V = \infty$, at least not as stated.*
2. *We've seen that $V \cong K^n$, if V is finite dimensional, but in general there does* ***not*** *exist a canonical isomorphism $V \xrightarrow{\cong} K^n$. For example $\Phi_{\mathcal{B}}$ depends on the choice of basis $\mathcal{B}$.*

Now we arrive at the main question. Let V, W be vector spaces over K, $n := \dim V, m := \dim W$, and let $T : V \to W$ be a linear map. Fix a basis $\mathcal{B}$ for V and a basis $\mathcal{C}$ for W. Consider the diagram

$$\begin{tikzcd} V \arrow[r, "T"] \arrow[dr, phantom, "C"] & W \arrow[d, "\Phi_{\mathcal{C}}"] \\ K^n \arrow[u, "\Phi_{\mathcal{B}}^{-1}"] \arrow[r, "\Phi_{\mathcal{C}} \circ T \circ \Phi_{\mathcal{B}}^{-1}"'] & K^m \end{tikzcd}$$

The 'C' means that the diagram is commutative.

Since $\Phi_{\mathcal{B}}^{-1}, T, \Phi_{\mathcal{C}}$ are all linear maps, the map $\Phi_{\mathcal{C}} \circ T \circ \Phi_{\mathcal{B}}^{-1} : K^n \to K^m$ is also linear. By Lemma 2.4, any linear map $K^n \to K^m$ can be written using a matrix $A \in M_{m \times n}(K)$. In other words, there exists a unique matrix $A \in M_{m \times n}(K)$ s.t. $\Phi_{\mathcal{C}} \circ T \circ \Phi_{\mathcal{B}}^{-1} = T_A$. The matrix A is called the **representation matrix of T w.r.t. the bases $\mathcal{B}$ and $\mathcal{C}$**. We denote this matrix by $[T]_{\mathcal{C}}^{\mathcal{B}} \in M_{m \times n}(K)$.

2.2.1 Entries of the Representation Matrix

Let $T : V \to W$ be a linear map and $\mathcal{B} = (v_1, \ldots, v_n)$ a basis for V, and $\mathcal{C} = (w_1, \ldots, w_m)$ a basis for W.

$$
\begin{tikzcd}
v_i \in \arrow[rrr, bend left, mapsto] & V \arrow[r, "T"] \arrow[dr, phantom, "\textcircled{c}"] & W \arrow[d, "\Phi_{\mathcal{C}}"'] & \ni Tv_i \arrow[d, mapsto] \\
e_i \in \arrow[u, mapsto] & K^n \arrow[u, "\Phi_{\mathcal{B}}^{-1}"'] \arrow[r, "\Phi_{\mathcal{C}} \circ T \circ \Phi_{\mathcal{B}}^{-1}"'] & K^m & \ni [Tv_i]_{\mathcal{C}}
\end{tikzcd}
$$

$T_A(e_i) = \Phi_{\mathcal{C}} \circ T \circ \Phi_{\mathcal{B}}^{-1}(e_i) = \Phi_{\mathcal{C}} \circ Tv_i = [Tv_i]_{\mathcal{C}}$. But these are precisely the columns of A. In other words

$$A = [T]_{\mathcal{C}}^{\mathcal{B}} = \begin{pmatrix} | & & | \\ [Tv_1]_{\mathcal{C}} & \ldots & [Tv_n]_{\mathcal{C}} \\ | & & | \end{pmatrix} \in M_{m \times n}.$$

Another way to describe this matrix is: Let $A = (a_{i,j})$, where $a_{i,j}$ are defined uniquely by

$$Tv_j = \sum_{i=1}^{m} a_{ij} w_j.$$

Proposition 2.23: Represenation matrix

Let $T : V \to W$ be a linear map between two finite dimensional vector spaces V, W. Let $\mathcal{B}, \mathcal{C}$ be bases for V, W respectively. Then

$$\forall v \in V : [T]_{\mathcal{C}}^{\mathcal{B}} \cdot [v]_{\mathcal{B}} = [Tv]_{\mathcal{C}}.$$

2.2.2 Representing Composition of Linear Maps by Matrices

Let $T : V \to W, S : W \to U$ be linear maps. Let $\mathcal{A} = (v_1, \ldots, v_n)$ be a basis for V, $\mathcal{B} = (w_1, \ldots, w_m)$ a basis for W and $\mathcal{C} = (u_1, \ldots, u_p)$ a basis for U.

Write $[T]_{\mathcal{B}}^{\mathcal{A}} = (a_{ij}), [S]_{\mathcal{C}}^{\mathcal{B}} = (b_{ij}), [S \circ T]_{\mathcal{C}}^{\mathcal{A}} = (c_{ij})$. By definition we have that

$$T(v_j) = \sum_{k=1}^{m} a_{kj} w_k \qquad \forall 1 \leq j \leq n \tag{2.1}$$

$$S(w_j) = \sum_{i=1}^{p} b_{ij} u_i \qquad \forall 1 \leq j \leq m \tag{2.2}$$

$$S \circ T(v_j) = \sum_{i=1}^{p} c_{ij} u_i \qquad \forall 1 \leq j \leq n \tag{2.3}$$

But we can calculate $S \circ T(v_j)$ by using Equations (2.1), (2.2). Then we have

$$\begin{aligned} S \circ T(v_j) &= S\big(\sum_{k=1}^{m} a_{kj} w_k\big) = \sum_{k=1}^{m} a_{kj} S(w_k) = \sum_{k=1}^{m} a_{kj} \big(\sum_{i=1}^{p} b_{ik} u_i\big) \\ &= \sum_{k=1}^{m} \sum_{i=1}^{p} a_{kj} b_{ik} u_i = \sum_{i=1}^{p} \sum_{k=1}^{m} a_{kj} b_{ik} u_i = \sum_{i=1}^{p} \big(\sum_{k=1}^{m} b_{ik} a_{kj}\big) u_i \\ \implies \quad c_{ij} &= \sum_{k=1}^{m} b_{ik} a_{kj}. \end{aligned}$$

3 Matrices

Definition 3.1: Matrix Multiplication

Let $A \in M_{m \times n}(K), B \in M_{n \times p}(K)$. We define a new matrix $A \cdot B \in M_{m \times p}(K)$ which is called the product of A and B. Write $A = (a_{ij}), B = (b_{ij})$. The matrix $A \cdot B$ has entries (c_{ij}), where

$$c_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}.$$

Important: $A \cdot B$ is defined only when the number of columns of A equals the number of rows of B.

Conclusion: Let $T : V \to W, S : W \to U$ be lineear maps between finite dimensional vector spaces. Let $\mathcal{A}, \mathcal{B}, \mathcal{C}$ be bases for V, W, U respectively. Then

$$[S \circ T]_{\mathcal{C}}^{\mathcal{A}} = [S]_{\mathcal{C}}^{\mathcal{B}} \cdot [T]_{\mathcal{B}}^{\mathcal{A}}.$$

Definition 3.2: Kronecker-Delta/Identity Matrix

Let $\mathcal{I}$ be an index set. Define for all $i, j \in \mathcal{I}$

$$\delta_{ij} := \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}$$

This is called the **kronecker-delta**.
We define the **identity matrix** $I_n \in M_{n\times n}(K)$ as

$$I_n := \delta_{ij}, \qquad I_n = \begin{pmatrix} 1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & 1 \end{pmatrix}.$$

Remark 3.3. *Let V be a finite dimensional vector space and $n := \dim V$. Let $id : V \to V$ be the identity map. Let $\mathcal{B}$ be any basis for V. Then $[id]^{\mathcal{B}}_{\mathcal{B}} = I_n$. In other words, the matrix representation of the identity map w.r.t $\mathcal{B}\&\mathcal{B}$ is the identity matrix I_n.*

Proposition 3.4: Associativity and Distributivity of Matrix Multiplication

1. $\forall A \in M_{m\times n}(K), B \in M_{n\times p}(K), C \in M_{p\times q}(K):\ (AB)C = A(BC) \in M_{m\times q}$.
2. $\forall A, B \in M_{m\times n}(K), C \in M_{n\times p}(K), C' \in M_{q\times m}(K)$ *we have*

$$(A+B)C = AC + BC \in M_{m\times p}(K),$$
$$C'(A+B) = C'A + C'B \in M_{q\times n}(K).$$

3. $\forall A \in M_{m\times n}(K):\ I_m A = A I_n = A$.
4. $\forall \alpha \in K, A \in M_{m\times n}(K), B \in M_{n\times p}(K):\ \alpha(AB) = (\alpha A)B = A(\alpha B)$.

Remark 3.5. *Matrix multiplication is in general **not** commutative, i.e., $\exists A, B$ s.t. $AB \neq BA$.*

*In $M_{n\times n}(K)$ we can add/subtract matrices and also multiply matrices. There is also a unit I_n. These operations satisfy properties like associativity, distributivity, etc. From thsi we get that $M_{n\times n}(K)$ is a (non-commutative) **ring** with a unity.*

Definition 3.6: Invertibility

An $n \times n$ matrix $A \in M_{n\times n}(K)$ is **invertible** if $\exists B \in M_{n\times n}(K)$ s.t. $AB = BA = I_n$.

Remark 3.7. *If A is invertible, then there exists only one matrix B s.t. $AB = BA = I_n$.*

Proof. Assume $AB = BA = I_n$ and $AC = CA = I_n$. Then

$$B = BI_n = B(AC) = (BA)C = I_n C = C.$$

□

Definition 3.8: The Inverse Matrix

If A is invertible, the unique matrix B s.t. $AB = BA = I_n$ is called **the inverse of** A and is denoted by A^{-1}. So, we have $AA^{-1} = A^{-1}A = I_n$.
We denote the set of all invertible $n \times n$ matrices by

$$GL_n(K) = GL(n, K) := \{A \in M_{n\times n}(K) \mid A \text{ is invertible}\}.$$

This set is called the general linear group.

Proposition 3.9: Structure of the General Linear Group

$GL_n(K)$, endowed with matrix multiplication $(A, B) \mapsto A \cdot B$ is a group. The unity is I_n. The inverse of $A \in GL_n(K)$ is A^{-1}. Moreover,

$$\forall A, B \in GL_n(K):\ (A^{-1})^{-1} = A, \quad (AB)^{-1} = B^{-1}A^{-1}.$$

Corollary 3.10: Unique Solution for Linear System of Equations

Let $A \in GL_n(K)$. Then $\forall b \in K^n$ the linear system of equations $Ax = b$ has a unique solution, namely $x = A^{-1}b$.

Proposition 3.11: Invertibility and Isomorphisms

Let $A \in M_{n\times n}(K)$. Then $A \in GL_n(K)$ if and only if $T_A : K^n \to K^n$ is an isomorphism. Moreover, $(T_A)^{-1} = T_{A^{-1}}$.

Definition 3.12: Upper/Lower Triangular and Diagonal Matrices

A matrix $A \in M_{n\times n}(K)$, $A = (a_{ij})$, is called

- **Upper triangluar** if $a_{ij} = 0 \quad \forall i > j$, i.e.,

$$\begin{pmatrix} * & \dots & \dots & * \\ 0 & \ddots & & \vdots \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \dots & 0 & * \end{pmatrix}.$$

- **Lower Triangular** if $a_{ij} = 0 \quad \forall i < j$, i.e.,

$$\begin{pmatrix} * & 0 & \dots & 0 \\ \vdots & \ddots & \ddots & \vdots \\ \vdots & & \ddots & 0 \\ * & \dots & \dots & * \end{pmatrix}.$$

- **Diagonal** if $a_{ij} = 0 \quad \forall i \neq j$, i.e.,

$$\begin{pmatrix} 1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & 1 \end{pmatrix}.$$

Lemma 3.13: Multiplication of Triangular/Diagonal Matrices

Let A, B be two diagonal (respectively both upper triangular, respectively both lower triangular). Then $A \cdot B$ is of the same type.

Corollary 3.14: Change of Bases for a Linear Map

Let V, W be finite dimensional vector spaces over K. Let $n := \dim V, m := \dim W$. Let $\mathcal{B}, \mathcal{B}'$ be two bases for V, and $\mathcal{C}, \mathcal{C}'$ be two bases for W. Then

$$[id_V]_{\mathcal{B}}^{\mathcal{B}'} \in GL_n(K),\ [id_W]_{\mathcal{C}'}^{\mathcal{C}} \in GL_m(K), \textit{ and}$$
$$[id_V]_{\mathcal{B}}^{\mathcal{B}'} = ([id_V]_{\mathcal{B}'}^{\mathcal{B}})^{-1}, [id_V]_{\mathcal{C}}^{\mathcal{C}'} = ([id_V]_{\mathcal{C}'}^{\mathcal{C}})^{-1}.$$

Let $T : V \to W$ be a linear map. Then

$$[T]_{\mathcal{C}'}^{\mathcal{B}'} = [id_W]_{\mathcal{C}'}^{\mathcal{C}} \cdot [T]_{\mathcal{C}}^{\mathcal{B}} \cdot [id_V]_{\mathcal{B}}^{\mathcal{B}'}.$$

Definition 3.15: Transition Matrix/Change of Bases Matrix

Let V be a finite dimensional vector space over K, $n := \dim V$. Let $\mathcal{B}, \mathcal{B}'$ be two bases of V. The matrix $[id_V]^{\mathcal{B}}_{\mathcal{B}'} \in GL_n(K)$ is called the **change of bases matrix** between $\mathcal{B}$ and $\mathcal{B}'$, or the **transition matrix** from basis $\mathcal{B}$ to basis $\mathcal{B}'$.
If $\mathcal{B} = (v_1, \dots, v_n), \mathcal{B}' = (v'_1, \dots, v'_n)$ then

$$[id_V]^{\mathcal{B}}_{\mathcal{B}'} = \begin{pmatrix} | & & | \\ [v_1]_{\mathcal{B}'} & \dots & [v_n]_{\mathcal{B}'} \\ | & & | \end{pmatrix}.$$

Proposition 3.16: Inverse of a Matrix with Bases

Let $T : V \to W$ bw an isomorphism between two finite dimensional vector spaces. Let $\mathcal{B}, \mathcal{C}$ be bases for V, W respectively. Then $[T]^{\mathcal{B}}_{\mathcal{C}}$ is an invertible matrix and

$$[T^{-1}]^{\mathcal{C}}_{\mathcal{B}} = ([T]^{\mathcal{B}}_{\mathcal{C}})^{-1}.$$

Corollary 3.17: Change of Basis for an Endomorphism

Let V be a finite dimensional vector space and $T : V \to V$ a linear map (an endomorphism). Let $\mathcal{B}, \mathcal{B}'$ be two bases for V. Then

$$[T]^{\mathcal{B}'}_{\mathcal{B}'} = [id_V]^{\mathcal{B}}_{\mathcal{B}'} \cdot [T]^{\mathcal{B}}_{\mathcal{B}} \cdot [id_V]^{\mathcal{B}'}_{\mathcal{B}} = ([id_V]^{\mathcal{B}'}_{\mathcal{B}})^{-1} \cdot [T]^{\mathcal{B}}_{\mathcal{B}} \cdot [id_V]^{\mathcal{B}'}_{\mathcal{B}}.$$

Definition 3.18: Equivalence and Similarity of Matrices

1. Let $A, B \in M_{m\times n}(K)$. We say that A and B are **equivalent** if $\exists\, P \in GL_m(K), Q \in GL_n(K)$ s.t. $B = PAQ$.
2. Let $A, B \in M_{n\times n}(K)$. We say that A and B are **similar** if $\exists\, P \in GL_n(K)$ s.t. $B = P^{-1}AP$.

Remark 3.19. *1. Equivalence between $m \times n$ matrices gives an equivalence relation on $M_{m\times n}(K)$.*

2. $A, B \in M_{m\times n}(K)$ are equivalent if and only if there exists two vector spaces V, W with $n := \dim V$, $m := \dim W$ and two bases $\mathcal{B}, \mathcal{B}'$ of V and two bases $\mathcal{C}, \mathcal{C}'$ for W s.t. $A = [T]^{\mathcal{B}}_{\mathcal{C}}$, $B = [T]^{\mathcal{B}'}_{\mathcal{C}'}$. In other words, A is equivalent to B if and only if A and B are matrix representations of the same linear map w.r.t. different choices of bases for V and W.

3. Similarity between $n \times n$ matrices gives an equivalence relation on $M_{n\times n}(K)$.

4. $M, N \in M_{n\times n}(K)$ are similar if and only if there exists and n-dimensional vector space V and $T : V \to V$ and two bases $\mathcal{B}, \mathcal{B}'$ of V s.t. $M = [T]^{\mathcal{B}}_{\mathcal{B}}$, $N = [T]^{\mathcal{B}'}_{\mathcal{B}'}$.

Proposition 3.20: Existence of 'Nice' Bases

Let V, W be finite dimensional vector spaces over K, of dimension $n := \dim V$, $m := \dim W$. Let $T : V \to W$ be a linear map with $\operatorname{rank}(T) = r$. (Recall that $\operatorname{rank}(T) := \dim \operatorname{Im}(T)$.) Then there exists a basis $\mathcal{B} = (v_1, \dots, v_n)$ of V and a basis $\mathcal{C} = (w_1, \dots, w_m)$ of W s.t.

$$[T]^{\mathcal{B}}_{\mathcal{C}} = \begin{pmatrix} I_r & 0_{r,n-r} \\ 0_{m-r,r} & 0_{m-r,n-r} \end{pmatrix}.$$

Corollary 3.21

Let $A \in M_{m\times n}(K)$. Then $\exists P \in GL_m(K), Q \in GL_n(K)$ s.t.

$$PAQ = \begin{pmatrix} I_r & 0_{r,n-r} \\ 0_{m-r,r} & 0_{m-r,n-r} \end{pmatrix},$$

where $r = \text{col-rank}(A)$. In particular, $\text{col-rank}(A) \leq \min\{m, n\}$.

Corollary 3.22: Condition for Equivalence if Matrices

Let $A, B \in M_{m\times n}(K)$. Then $A \sim B$ if and only if $\text{col-rank}(A) = \text{col-rank}(B)$. Moreover, $\forall\, 0 < r \leq \min\{m, n\}$ there is precisely one equivalence class of matrices A with $\text{col-rank} = r$.

3.1 col-rank = row-rank

Theorem 3.23: col-rank = row-rank

Let $A \in M_{m\times n}(K)$. Then $\text{col-rank}(A) = \text{row-rank}(A)$.

Remark 3.24.

1. *Because of this Theorem we'll denote from now on $\operatorname{rank}(A) := \text{col-rank}(A) = \text{row-rank}(A)$.*
2. *Suppose A' is a row-reduced echelon matrix which is row-equivalent to A. Then we have that $\operatorname{rank}(A) = \#$ pivots in A'. Indeed, $\operatorname{rank}(A) = \text{row-rank}(A) = \text{row-rank}(A') = \#$ pivots in A'.*
3. *Recall that for $T : V \to W$ we defined $\operatorname{rank}(T) := \dim \operatorname{Im}(T)$. Also, if $A \in M_{m\times n}(K)$, then $\operatorname{rank}(T_A) = \dim \operatorname{Im}(T) = \text{col-rank}(A)$.*

We now present some Lemmas for the proof of the Theorem.

Lemma 3.25: Isomorphisms preserve the Rank

Let $T : V \to W$ be a linear map, and $S : W \xrightarrow{\cong} U$, $L : P \xrightarrow{\cong} V$ be isomorphisms. Then

1. *$\operatorname{rank}(S \circ T) = \operatorname{rank}(T)$.*
2. *$\operatorname{rank}(T \circ S) = \operatorname{rank}(T)$.*

Lemma 3.26

Let $A \in M_{m\times n}(K)$. Then A is invertible if and only if $T_A K^n \to K^n$ is in isomorphism.

Lemma 3.27: Equivalent Matrices Have the Same Rank

Let $A, B \in M_{m\times n}(K)$ s.t. $B = PAQ$, where $P \in GL_m(K)$, $Q \in GL_m(K)$. Then

1. $\text{col-rank}(A) = \text{col-rank}(B)$
2. $\text{row-rank}(A) = \text{row-rank}(B)$.

Proof. **Proof of Theorem 3.23:** By Corollary 3.21, $\exists P \in GL_m(K), Q \in GL_n(K)$ s.t.

$$PAQ = \begin{pmatrix} I_r & 0_{r,n-r} \\ 0_{m-r,r} & 0_{m-r,n-r} \end{pmatrix}.$$

Denote this matrix by B. For B we have $\text{col-rank}(B) = \text{row-rank}(B) = r$. By Lemma 3.27, we have that $\text{col-rank}(A) = \text{col-rank}(B) = r$, and $\text{row-rank}(A) = \text{row-rank}(B) = r$. Therefore, $\text{col-rank}(A) = \text{row-rank}(A)$. □

Remark 3.28. *Although* $\text{row-rank}(A) = \text{col-rank}(A)$, *the spaces* $\text{ColS}(A)$ *and* $\text{RowS}(A)$ *are* ***different***. *First of all if $A \in M_{m\times n}$, then $\text{ColS}(A) \subseteq K^m$ and $\text{RowS}(A) \subseteq K^n$. But even when $n = m$, these two spaces are* ***different***. *They have the same dimension, but are different.*

3.2 More on Matrices and Linear Equations

Let $A \in M_{m\times n}(K)$, and consider the system of linear equations $Ax = 0$. Denote by

$$\text{Sol}(A, 0) := \{x \in K^n \mid Ax = 0\} = \ker(T_A),$$

the **space** of solutions $Ax = 0$. Note that $\dim \text{Sol}(A, 0) = n - \text{rank}(A)$ (b.c. $\dim \ker(T_A)) = n - \dim \text{Im}(T_A)$). Let A' be a row-reduced echelon matrix which is row-equivalent to A (e.g. A' is obtained from A by Gauss-elimination). # pivots of $A' = \text{rank}(A)$, $n - \text{rank}(A) = $ # free variables. Also, we have that $\text{Sol}(A, 0) = \text{Sol}(A', 0)$.

$$\begin{pmatrix} 0 & \dots & 0 & 1 & * & \dots & * & & & & \\ 0 & \dots\dots\dots & & & 0 & 1 & * & \dots & * & & \\ 0 & \dots\dots\dots\dots & & & & 0 & 1 & * & \dots & * \\ 0 & \dots\dots\dots\dots\dots\dots\dots\dots & & & & & & & & 0 \\ 0 & \dots\dots\dots\dots\dots\dots\dots\dots & & & & & & & & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}.$$

Denote by $x_{i_1}, \dots, x_{i_l}$ the free variables, where $1 \le i_1 < \dots < i_l \le n$, $l = n - \text{rank}(A)$ and by $x_{j_1}, \dots, x_{j_k}$, where $1 \le j_1 < \dots < j_k \le n$, $r = \text{rank}(A)$ the pivots. For each choice of values for the free variables $x_{i_1}, \dots, x_{i_l} \in K$, the pivot variables $x_{j_1}, \dots, x_{j_k}$ are uniquely determined. So, we obtain a map

$$\Phi : K^l \to \text{Sol}(A, 0)$$
$$\Phi(\lambda_1, \dots, \lambda_l) = \text{ the unique solution of } Ax = 0,$$

with $x_{i_1} = \lambda_1, \dots, x_{i_l} = \lambda_l$.

Proposition 3.29: Φ is an Isomorphism

The map Φ is linear. Moreover, it is an isomorphism.

Corollary 3.30

Let $b \in K^m$, $A \in M_{m\times n}(K)$. Then

1. $\mathrm{Sol}(A,b) := \{x \in K^n \mid Ax = b\} \neq \emptyset$ *iff* $b \in \mathrm{ColS}(A)$.
2. *If* $\mathrm{Sol}(A,b) \neq \emptyset$ *and* $y \in \mathrm{Sol}(A,b)$ *then*

$$\mathrm{Sol}(A,b) = y + \mathrm{Sol}(A,0) := \{y + x \mid x \in \mathrm{Sol}(A,0)\}.$$

Proposition 3.31

Let $A \in M_{m\times n}(K)$, $b \in K^m$. The following statements are equivalent:

1. $\mathrm{rank}(A|b) = \mathrm{rank}(A)$.
2. *The system of equations $Ax = b$ has a solution.*

Proposition 3.32

1. $\mathrm{rank}(A|b) = \mathrm{rank}(A) = n$.
2. *The system of equations $Ax = b$ has a unique solution.*

3.3 Elementary Row Operations Revisited

Let $A \in M_{n\times p}(K)$. We learned about elementary row operations which are used as part of the Gauss elimination. All of these operations can be expressed by matrix multiplications.

Denote by E_{ij} the matrix in Example 1.3.3.

Type I: Let $1 \leq i \leq n$, $1 \leq j \leq n$, $i \neq j$. Let $\alpha \in K$. Define $Q_{ij}(\alpha)$ to be the matrix obtained from I_n after applying $R_i + \alpha R_j \longrightarrow R_i$. We have $Q_{ij}(\alpha) = I_n + \alpha E_{ij}$.

Type II: Let $1 \leq i \leq n, 1 \leq j \leq n$. Define P_{ij} to be the matrix obtained from I_n by permuting (exchanging) row # i and row # j, i.e. applying $R_i \longleftrightarrow R_j$. We have $P_{ij} = I_n - E_{ii} - E_{jj} + E_{ij} + E_{ji}$.

Type III: Let $1 \leq i \leq n$ and $0 \neq \alpha \in K$. Denote by $S_i(\alpha)$ the matrix obtained from I_n by applying $\alpha R_i \longrightarrow R_i$.

Lemma 3.33: Elementary Row Operations as Matrix Multiplication

Let $A \in M_{n\times p}(K)$.

1. *Multiplying A from the left with $Q_{ij}(\alpha)$ results in the row operation $R_i + \alpha R_j \longrightarrow R_i$: $A \overset{R_i+\alpha R_j \to R_i}{\to} Q_{ij}(\alpha)A$.*
2. *Multiplying A from the left with P_{ij} results in the row operation $R_i \longleftrightarrow R_j$: $A \overset{R_i \leftrightarrow R_j}{\longrightarrow} P_{ij}A$.*
3. *Multiplying A from the left with $S_i(\alpha)$ results in the row operation $\alpha R_i \longrightarrow R_i$: $A \overset{\alpha R_i \to R_i}{\longrightarrow} S_i(\alpha)A$.*

The same holds for elementary column operations, which is done by right hand side multiplications of the matrices.

Lemma 3.34: Elementary Matrices are Invertible

Every elementary matrix is invertible, and the inverse of each of them is an elementary matrix of the same type, i.e.

$$Q_{ij}(\alpha)^{-1} = Q_{ij}(-\alpha), \qquad P_{ij}^{-1} = P_{ji}, \qquad S_{ij}(\alpha)^{-1} = S_{ij}(\alpha^{-1}).$$

Theorem 3.35

$\forall\, A \in GL_n(K)$, $\exists\, k \geq 1$ *and elementary matrices* $T_1, \ldots, T_k$ *s.t.* $T_k \cdot \ldots \cdot T_1 \cdot A = I_n$. *Furthermore, we have that*

$$A = T_1^{-1} \cdot \ldots \cdot T_k^{-1}, \qquad A^{-1} = T_k \cdot \ldots \cdot T_1.$$

In particular, A can be written as a finite product of elementary matrices.

Theorem 3.36: Equivalent condition for Invertibility

Let $A \in M_{n\times n}(K)$. Place A together with I_n in the following way $(A|I_n)$. Then A is invertible if and only if $(A|I_n)$ can be brought via a sequence of elementary row-operations to the form $(I_n|B)$ for some matrix $B \in M_{n\times n}(K)$. In this case we have $A^{-1} = B$.

4 Constructing New Vector Spaces Out of Old Ones

Proposition 4.1: The Vector Space of Linear Functions

Let V, W be two vector spaces over K. Then $\operatorname{Hom}(V,W)$ has the structure of a vector space over K, if we endow it with the following operations:

- $\forall\, T_1, T_2 \in \operatorname{Hom}(V,W)$, $(T_1 + T_2)(v) := T_1(v) + T_2(v) \quad \forall\, v \in V$.
- $\forall\, T \in \operatorname{Hom}(V,W), \alpha \in K$, $(\alpha T)(v) := \alpha T(v) \quad \forall\, v \in V$.

The neutral element of $\operatorname{Hom}(V,W)$ is the map $0 : V \to W, \quad v \mapsto 0$.

Theorem 4.2: The map $\Psi_{\mathcal{C}}^{\mathcal{B}}$

Let V, W be finite dimensional vector spaces over K, $\mathcal{B} = (v_1, \ldots, v_n)$ a basis for V, and $\mathcal{C} = (w_1, \ldots, w_n)$ a basis for W. Then the map

$$\Psi_{\mathcal{C}}^{\mathcal{B}} : \operatorname{Hom}(V,W) \to M_{m\times n}(K)$$

defined by

$$T \mapsto [T]_{\mathcal{C}}^{\mathcal{B}} := \begin{pmatrix} | & & | \\ [Tv_1]_{\mathcal{C}} & \ldots & [Tv_n]_{\mathcal{C}} \\ | & & | \end{pmatrix}$$

is a linear map and moreover an isomorphism.

Corollary 4.3: Dimesnion of $\mathrm{Hom}(V, W)$

If V, W are finite dimensional vector spaces then so is $\mathrm{Hom}(V, W)$ and $\dim \mathrm{Hom}(V, W) = \dim V \cdot \dim W$.

4.0.1 Rings

Recall that Rings have the same axioms as Fields except that we do **not** require that every non-zero element has an inverse.

Definition 4.4: Homomorphism of Rings

Let R, S be rings. A **homomorphism of rings** is a map $\varphi : R \to S$ s.t.:

- $\forall r_1, r_2 \in R : \varphi(r_1 + r_2) = \varphi(r_1) + \varphi(r_2)$.
- $\forall r_1, r_2 \in R : \varphi(r_1 \cdot r_2) = \varphi(r_1) \cdot \varphi(r_2)$
- $\varphi(1) = 1$.
- $\varphi(0) = 0$.

Corollary 4.5: Ψ for an Endomorphism is an isomorphism of Rings

Let V be a vector space over K. Then $\mathrm{End}(V) := \mathrm{Hom}(V, V)$ is a ring with a unity. This ring is, in general, not commutative. The multiplication (product) in $\mathrm{End}(V)$ is given by composition, i.e.,

$$T_1 \cdot T_2 := T_1 \circ T_2 \qquad \forall T_1, T_2 \in \mathrm{Hom}(V, V).$$

The unity is id_V. Moreover, if V is finite dimensional and $\dim V = n$, and $\mathcal{B} = (v_1, \ldots, v_n)$ is a basis for V, then $\Psi^{\mathcal{B}}_{\mathcal{B}} : \mathrm{Hom}(V, V) \to M_{n\times n}(K)$ is an isomorphism of rings

4.1 Dual Space

An important special case of $\mathrm{Hom}(V, W)$ is when $W = K$.

Definition 4.6: Dual Space

Let V be a vector space over K. The **dual space** of V, is defined to be the vector space $V^* := \mathrm{Hom}(V, K)$. The elements of V^* are linear maps $\ell : V \to K$, and are sometimes called **functionals** (on V).

Corollary 4.7: Dimension of the Dual Space

Let V be a finite dimensional vector space over K. Then $\dim V^* = \dim V$.

Let $V = K^n_{\text{col}}$. Then we can identify V^* with K^n_{col}, as follows: If $\ell \in K^n_{\text{row}}$, then ℓ defines a functional $f_\ell : K^n_{\text{col}} \to K$ by $f_\ell(v) := \ell \cdot v \in K$. The map $D : K^n_{\text{row}} \to (K^n_{\text{col}})^*$ defined by $D(\ell) := f_\ell$ is a linear map and is an isomorphism.

Let V be a finite dimensional vector space over K and $\mathcal{B} = (v_1, \ldots, v_n)$ a basis for V. $\forall\, 1 \le i \le n$, define a functional $v_i^* \in V^*$, i.e., $v_i^* : V \to K$ linear map, as follows:

$$v_i^*(v_j) := \delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}.$$

Since $v_1, \ldots, v_n$ form a basis for V, our definition determines uniquely a well-defined functional $v_i^* : V \to K$.

Lemma 4.8: The Dual Basis

*The elements $v_1^*, \ldots, v_n^* \in V^*$ form a basis for V^*. This basis is called the **dual basis** to $\mathcal{B}$. We denote this basis by $\mathcal{B}^* = (v_1^*, \ldots, v_n^*)$.*
Moreover,

$$\forall\, \ell \in V^* : \ell = \sum_{i=1}^{n} \ell(v_i) \cdot v_i^*.$$

Proof. Let $\ell \in V^*$. We claim that

$$\ell = \sum_{i=1}^{n} \ell(v_i) \cdot v_i^*. \tag{4.1}$$

Or in other words, we claim that

$$\forall\, v \in V : \ell(v) = \sum_{i=1}^{n} \ell(v_i) \cdot v_i^*(v). \tag{4.2}$$

Indeed, both sides of Equation (4.1) are linear maps $V \to K$, so it's enough to check that Equation (4.2) holds for $v = v_1, \ldots, v = v_n$, b.c. $v_1, \ldots, v_n$ form a basis for V. And indeed, for $v = v_k$ we have

$$\sum_{i=1}^{n} \ell(v_i) \cdot v_i^*(v_k) = \sum_{i=1}^{n} \ell(v_i) \cdot \delta_{ik} = \ell(v_k).$$

The claim just proven shows that $v_1^*, \ldots, v_n^*$ span V^*. Since $\dim V^* = \dim V = n$, it follows that $v_1^*, \ldots, v_n^*$ form a basis for V^*. $\square$

Question: Let V be a finite dimensional vector space over K. Let $\mathcal{B}$ and $\mathcal{C}$ be two bases for V. Therefore, $\mathcal{B}^*$ and $\mathcal{C}^*$ are two bases for V^*. What is the relation between $[id_{V^*}]_{\mathcal{B}^*}^{\mathcal{C}^*}$ and $[id_V]_{\mathcal{B}}^{\mathcal{C}}$.

Proposition 4.9: Relation between $[id_{V^*}]_{\mathcal{B}^*}^{\mathcal{C}^*}$ and $[id_V]_{\mathcal{B}}^{\mathcal{C}}$

$$[id_{V^*}]_{\mathcal{B}^*}^{\mathcal{C}^*} = \left([id_V]_{\mathcal{C}}^{\mathcal{B}}\right)^T = \left(\left([id_V]_{\mathcal{B}}^{\mathcal{C}}\right)^T\right)^{-1}.$$

4.2 The Dual Map

Definition 4.10: The Dual Map

Let V, W be vector spaces over K. Let $T : V \to W$ be a linear map. Define a new map

$$T^* : W^* \to V^*, \quad T^*(\ell) = \ell \circ T \quad \forall\, \ell \in W^*.$$

Then T^* is a linear map. It is called **the dual map** to T.

Proof. Let $\ell \in W^*$, i.e., $\ell : W \to K$ a linear map. Clearly

$$\ell \circ T : V \to K$$

is a linear map, b.c. both ℓ and T are linear. So, $T^*(\ell) := \ell \circ T \in V^*$. This shows that $T^* : W^* \to V^*$ indeed has as it's target V^* as claimed, see also the Figure 4.2. We'll show now that T^* is a linear

$$\begin{tikzcd}
W \arrow[r, "\ell"] & K \\
V \arrow[u, "T"] \arrow[ur, "\ell \circ T = T^*(\ell)"'] &
\end{tikzcd}$$

Figure 1: Map diagram of the dual map

map. Let $\ell \in W^*$, $\alpha \in K$. We have that $T^*(\alpha\ell) \in V^*$ is the map

$$\begin{aligned}
& T^*(\alpha\ell)(v) = \big((\alpha\ell) \circ T\big)(v) = (\alpha\ell)(Tv) = \alpha\ell(Tv) = \alpha(\ell \circ T)(v) = \alpha\big(T^*(\ell)\big)(v) \\
\implies \quad & T^*(\alpha\ell) = \alpha T^*(\ell).
\end{aligned}$$

Now, let $\ell_1, \ell_2 \in W^*$. We have that $T^*(\ell_1 + \ell_2) \in V^*$ is the map

$$\begin{aligned}
& T^*(\ell_1 + \ell_2)(v) = \big((\ell_1 + \ell_2) \circ T\big)(v) = (\ell_1 \circ T + \ell_2 \circ T)(v) = (\ell_1 \circ T)(v) + (\ell_2 \circ T)(v) \\
& \qquad = \big(T^*(\ell_1)\big)(v) + \big(T^*(\ell_2)\big)(v) \\
\implies \quad & T^*(\ell_1 + \ell_2) = T^*(\ell_1) + T^*(\ell_2).
\end{aligned}$$

This shows that T^* is linear as claimed. □

Proposition 4.11: Composition of the Dual Map

Let U, V, W be vector spaces over K. Let $T : V \to W$, $S : W \to U$ be linear maps. Consider $T^* : W^* \to V^*$, $S^* : U^* \to W^*$, the dual maps. Then

$$(S \circ T)^* = T^* \circ S^*.$$

Lemma 4.12: Interpretation of the Matrix Transpose

Let $S : V \to W$ be a linear map between two finite dimensional vector spaces V and W over K. Let $\mathcal{B}$ be a basis for V and $\mathcal{C}$ be a basis for W. Consider $S^* : W^* \to V^*$ and the bases $\mathcal{C}^*, \mathcal{B}^*$ for W^*, V^* respectively. Then

$$[S^*]_{\mathcal{B}^*}^{\mathcal{C}^*} = \big([S]_{\mathcal{C}}^{\mathcal{B}}\big)^T.$$

Proof. Write $\mathcal{B} = (v_1, \dots, v_n), \mathcal{C} = (w_1, \dots, w_m)$. Define $A := [S]_{\mathcal{C}}^{\mathcal{B}}, A = (a_{ij})$. A satisfies

$$Sv_j = \sum_{i=1}^{m} a_{ij} w_i.$$

Now, for $w_j^* \in \mathcal{C}^*$ we have by Lemma 4.8

$$\begin{aligned} S^* w_j^* = w_j^* \circ S &= \sum_{i=1}^n (w_j^* \circ S)(v_i) \cdot v_i^* = \sum_{i=1}^n w_j^*\big(S(v_i)\big) \cdot v_i^* \\ &= \sum_{i=1}^n w_j^* \left(\sum_{k=1}^m a_{ki} w_k \right) \cdot v_i^* = \sum_{i=1}^n \sum_{k=1}^m a_{ki} w_j^*(w_k) \cdot v_i^* = \sum_{i=1}^n a_{ji} v_i^*. \end{aligned}$$

Therefore, the coordinates of $S^* w_j^*$ in the basis $\mathcal{B}^* = (v_1^*, \dots, v_n^*)$ are

$$\begin{pmatrix} a_{j1} \\ \vdots \\ a_{jn} \end{pmatrix},$$

which is the row $\#j$ in A, represented as a column, or equivalently the column $\#j$ in the matrix A^T. □

4.3 Reflexitvity

Let V be a vector space over K. We have V^*, but also $V^{**} := (V^*)^*$ by applying 'dual' one more time, to V^*. $(V^*)^*$ is sometimes called the bidual space of V. Is there any relation between $(V^*)^*$ and V.

Theorem 4.13: The Canonical Isomorphism between $(V^*)^*$ and V

Let v be a finite dimensional vector space over K. Then there exists a canonical isomorphism $\tau : V \to (V^*)^*$, given by

$$\forall\, v \in V,\ \tau(v) : V^* \to K$$

is the map

$$\tau(v)(\ell) := \ell(v) \quad \forall\, \ell \in V^*.$$

In other words,

$$V \ni v \overset{\tau}{\longmapsto} \big(V^* \ni \ell \longmapsto \ell(v) \in K\big).$$

Remark 4.14. *The map $\tau : V \to (V^*)^*$ exists and is defined in the same way as above regardless of V being finite or infinite dimensional. Moreover, τ is always injective. However, when V is infinite dimensional, τ fails to be surjective.*

4.4 Naturality

The map $\tau : V \to (V^*)^*$ exists for every space V, and is defined in the same way. To distinguish τ's for different spaces, denote it now by $\tau_V : V \to (V^*)^*$. Let V, W be two vector spaces over K. Then for every linear map $T : V \to W$ we have

$$\begin{tikzcd} V \arrow[r, "\tau_V"] \arrow[d, "T"'] \arrow[dr, phantom, "\circledcirc"] & (V^*)^* \arrow[d, "(T^*)^*"] \\ W \arrow[r, "\tau_W"'] & (W^*)^* \end{tikzcd}$$

which we call naturality.

Proposition 4.15: Image of τ

Let V be a finite dimensional vector space over K, let $\mathcal{B} = (V_1, \dots, v_n)$ be a basis for V. Let $\mathcal{B}^* = (v_1^*, \dots, v_n^*)$ be the dual basis and $(\mathcal{B}^*)^* = ((V_1^*)^*, \dots, (v_n^*)^*)$ the basis for $(V^*)^*$, which is dual to $\mathcal{B}^*$. Then $\tau(\mathcal{B}) = (\mathcal{B}^*)^*$, i.e.,

$$\tau(v_i) = (v_i^*)^* \qquad \forall\, 1 \leq i \leq n.$$

4.5 Direct Sums

Proposition 4.16: The Vector Space of the Cartesian Product

et V, W be a vector space over K. Then the cartesian product $V \times W$ becomes a vector space over K if we endow it with the following operations:

- $\forall\, (v_1, w_1), (v_2, w_2) \in V \times W :\ (v_1, w_1) + (v_2, w_2) := (v_1 + v_2, w_1 + w_2)$.
- $\forall\, \alpha \in K, (v, w) \in V \times W :\ \alpha \cdot (v, w) := (\alpha v, \alpha w) \in V \times W$.
- $0_{V \times W} := (0_V, 0_W)$.

This new vector space $V \times W$ is usually denoted $V \oplus W$ and is called the direct sum of V and W. We write the elements of $V \oplus W$ as (v, w) or $v \oplus w$.

Remark 4.17. *$V \oplus W$ comes together with the following* ***canonical maps****:*

1) $i_V : V \to V \oplus W, \quad i_V(v) := (v, 0)$.

2) $i_W : W \to V \oplus W, \quad i_W(w) := (0, w)$.

3) $p_V : V \oplus W \to V, \quad p_V(v, w) := v$.

4) $p_W : V \oplus W \to W, \quad p_W(v, w) := w$.

The blue marked maps are canonical embeddings, while the orange marked maps are canonical projections. These maps are all linear. The maps p_V, p_W are surjective, and i_V, i_W are injective. Becuase of the maps i_V, i_W we can sometimes view V and W as subspaces of $V \oplus W$.

Proposition 4.18: The map φ

Let V be a vector space over K, $U \subseteq V$ a subspace and $U' \subseteq V$ another subspace which is a complement of U in V (recall Definition 1.42). Then the map

$$\varphi : U \oplus U' \to V, \quad \varphi(u, u') := u + u'$$

is a linear isomorphism.

Remark 4.19. *When U' is a complement of U in V, then $U + U' = V$ and we also have a canonical isomorphism $U \oplus U' \xrightarrow{\cong} V$. But formally speaking $U \oplus U'$ is a different vector space than V.*

4.5.1 Generalization to more Summands

Let $U_1, \dots, U_m$ be **finitely** many vector spaces over K. We can define $U_1 \oplus \ldots \oplus U_m := U_1 \times \ldots \times U_m$, with the coordinatewise/componentwise operations.

For **infinite** families U_i, $i \in \mathcal{I}$ one can define

$$\bigoplus_{i \in \mathcal{I}} U_i := \left\{ t : \mathcal{I} \to \bigcup_{i \in \mathcal{I}} U_i \mid t(i) \in U_i \quad \forall\, i \in \mathcal{I},\ t(i) \neq 0 \text{ for at most finitely many } i\text{'s} \right\}.$$

4.6 Quotient Spaces

Let V be a vector space over K, and $U \subseteq V$ a subspace. We will define a new vector space over K, V/U, together with a linear map $\pi : V \to V/U$ s.t. π is surjective and $\ker(\pi) = U$. V/U will have some other good properties. V/U is called the **quotient of V by U** or V **modulo** U. π is called the **quotient map** or the projection onto the quotient space.

We begin with an equivalence relation on v. Define for $v_1, v_2 \in V$

$$v_1 \sim v_2 \quad :\Longleftrightarrow \quad v_1 - v_2 \in U.$$

We claim that $\sim$ defines an equivalence relation on V. Indeed

- **Refelxivity:** $v \sim v$, b.c. $v - v = 0 \in U$.
- **Symmetry:** If $v_1 \sim v_2 \implies v_2 \sim v_1$, b.c. $v_2 - v_1 = -(v_1 - v_2) \in U$.
- **Transitivity:** If $v_1 \sim v_2$ and $v_2 \sim v_3$, then $v_1 \sim v_3$, b.c. $v_1 - v_3 = \underbrace{(v_1 - v_2)}_{\in U} + \underbrace{(v_2 - v_3)}_{\in U} \in U$.

We denote the equivalence class of $v \in V$ by $[v]$. So

$$[v] = v + U = \{v + u \mid u \in U\},$$

or equivalently

$$[v] = \{w \in V \mid v \sim w\} = \{w \in W \mid v - w \in U\}.$$

Definition 4.20: Quotient Space

Let V be a vector space over K and $U \subseteq V$ a subspace. Define $V/U := V/\sim =$ the set of equivalence classes of the equivalence relation $\sim$. So, the elements of V/U are $[v]$, where $v \in V$. Alternatively, each element in V/U is a set of the form $v + U$, where $v \in V$. V/U is called the **quotient of V by U** (or modulo U).

We want to turn V/U into a vector space over K. To achieve this, we define:

- **Addition:** $\forall [v_1], [v_2] \in V/U, \quad [v_1] + [v_2] := [v_1 + v_2] \in V/U.$
- **Mult. by a scalar:** $\forall \alpha \in K, [v] \in V/U, \quad \alpha \cdot [v] := [\alpha v] \in V/U.$ (Alternatively written: $\alpha(v + U) := \alpha v + U$.)
- **The zero:** $0_{V/U} := [0_V]$. (Alternatively written: $0_{V/U} := 0_V + U$.)

Proposition 4.21: Well Definedness of The Operations

The operations above are well defined. Moreover, they give V/U the structure of a vector space over K.

We now define the map

$$\pi : V \to V/U, \quad v \mapsto [v] \in V/U.$$

Proposition 4.22: Properties of π

π is a linear map. Moreover, $\operatorname{Im}(\pi) = V/U$ (,i.e., π is surjective), and $\ker(\pi) = U$.

Proposition 4.23: Dimension of V/U

Suppose that V is finite dimensional. Then V/U is finite dimensional too, and $\dim V/U = dim V - \dim U$.

Theorem 4.24: The Isomorphism Theorem

Let V, W be vector spaces over K and $T : V \to W$ a linear map. Define a new map $\overline{T} : V/\ker(T) \to \operatorname{Im}(T)$, as follows. Let $x \in V/\ker(T)$. Choose a representative $v \in V$ for x, i.e., $x = [v]$. Define $\overline{T}(x) = \overline{T}([v]) := T(v)$. Then:

1. *$\overline{T}$ is well defined, and is a linear map.*
2. *$\overline{T} : V/\ker(T) \to \operatorname{Im}(T)$ is an isomorphism.*
3. *The Diagramm in Figure 4.6 commutes, where $i : \operatorname{Im}(T) \to W$ is the inclusion.*

$$\begin{tikzcd}
V \arrow[r, "T"] \arrow[d, "\pi"'] \arrow[dr, phantom, "©"] & W \\
V/\ker(T) \arrow[r, "\overline{T}"'] & \operatorname{Im}(T) \arrow[u, "i"']
\end{tikzcd}$$

Figure 2: Map diagram of the map $\overline{T}$.

Proof. **1) $\overline{T}$ is well defined:**. Suppose $x = [v] = [v'] \in V/\ker(T)$, we need to show that $T(v) = T(v')$. Indeed, $[v] = [v']$ implies that $v - v' \in \ker(T)$. Therefore, $T(v - v') = 0$. Since T is linear we conclude that $T(v) = T(v')$.

$\overline{T}$ **is linear:** Let $\alpha \in K, [v] \in V/\ker(T)$. By the linearity of T, the definition of the operations on the quotient space $V/\ker(T)$ and the definition of $\overline{T}$, we have that

$$\overline{T}(\alpha[v]) = \overline{T}([\alpha v]) = T(\alpha v) = \alpha T(v) = \alpha\overline{T}([v]).$$

Now, let $[v_1], [v_2] \in V/\ker(T)$. Then

$$\overline{T}([v_1] + [v_2]) = \overline{T}([v_1 + v_2]) = T(v_1 + v_2) = T(v_1) + T(v_2) = \overline{T}([v_1]) + \overline{T}([v_2]).$$

2) $\overline{T}$ **is injective:** Let $x \in \ker(\overline{T})$. Write x as $x = [v]$ for some $v \in V$. Then

$$\underbrace{\overline{T}(x)}_{=0} = T(v).$$

Therefore, $v \in \ker(T)$. Thus, $x = [v] = [0_V] = 0_{V/\ker(T)}$.

$\overline{T}$ **is surjective:** Let $w \in \mathrm{Im}(T)$. Then there exists a $v \in V$ s.t. $T(v) = w$. Therefore, $\overline{T}([v]) = T(v) = w$. This shows that $\mathrm{Im}(T) \subseteq \mathrm{Im}(\overline{T})$.

But by the definition of $\overline{T}$ we also have that $\mathrm{Im}(\overline{T}) \subseteq \mathrm{Im}(T)$.

3) The diagram commutes: Let $v \in V$ and $w \in \mathrm{Im}(T) \subseteq W$ s.t. $T(v) = w$. Then by the definition of $\pi, \overline{T}$ and i, we have that

$$(i \circ \overline{T} \circ \pi)(v) = (i \circ \overline{T})([v]) = i\big(T(v)\big) = i(w) = w.$$

□

4.6.1 Universal Propertiy of Quotient Spaces

Theorem 4.25: Universal Property of Quotient Spaces

Let V be a vector space over K and $U \subseteq V$ a subspace. The quotient space V/U has the following universal property:
For every space W and every linear map $T : V \to W$, for which $T(U) = 0$(, i.e., for which $U \subseteq \ker(T)$), there exists a ***unique*** *linear map $T' : V/U \to W$ s.t. the diagram in Figure 4.6.1 commutes. Moreover, $\ker(T') = \ker(T)/U$.*

$$\begin{tikzcd}
V \arrow[r, "T"] \arrow[d, "\pi"'] \arrow[dr, phantom, "©"] & W \\
V/U \arrow[ur, "T'\ (\exists!)"'] &
\end{tikzcd}$$

Figure 3: Map diagram of the map T'.

Jargon: Every linear map $T : V \to W$ that maps U to 0 factors through V/U(, i.e., $T = T' \circ \pi$), and moreover in a unique way. We say that T **induces** a map $T' : V/U \to W$. We also say that T **descends** to a map $T' : V/U \to W$.

4.6.2 Quotient Spaces & Complements

Proposition 4.26: A Canonical Isomorphism to a Complement

Let V be a vector space over K and $U \subseteq V$ a subspace. Let $W \subseteq V$ be a complement of U. Then there exists a canonical isomorphism $Q : W \xrightarrow{\cong} V/U$, defined by

$$Q(w) := [w] \in V/U \qquad \forall w \in W.$$

Remark 4.27. *The isomorphism Q is canonical, only once W is specified. In general there is no canonical choice of a complement W to U. (In general there exist many choices of complements W to U, and there is no preferred choice.)*

4.7 Relations to Dual Spaces

Proposition 4.28: Perpendicular Subspace

Let V be a vector space over K and $U \subseteq V$ a subspace. Define

$$U^{\perp} := \big\{\ell \in V^* \mid \ell|_U = 0\big\},$$

i.e., all the functionals on V that vanish on U. Then:

1. *$U^{\perp} \subseteq V^*$ is a linear subspace.*
2. *There exists a **canonical** isomorphism $\big(V/U\big)^* \cong U^{\perp}$. If we assume that V is finite dimensional then there exists a **canonical** isomorphism $U^* \cong V^*/U^{\perp}$.*

Proposition 4.29

Let $T : V \to W$ be a linear map. Then

$$(\mathrm{Im}(T))^{\perp} = \ker(T^*).$$

5 Determinants

Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_{2\times 2}(K).$$

There is a very simple criterion to check wether A is invertible or not.

Proposition 5.1: Criterion for Invertibility of a 2×2 Matrix

A is invertible if and only if $ad - bc \neq 0$. Moreover, if A is invertible, then

$$A^{-1} = \frac{1}{ad - bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

The number $ad - bc$ is $\det(A)$.

Definition 5.2: N-Linear Function

Let $D : M_{n\times n}(K) \to K$ be a function. We say that D is n-**linear** if for every $1 \leq i \leq n$, D is a linear function of the i-th row when the other $(n-1)$ rows are held fixed.

We can think of $M_{n\times n}(K)$ as $K^n_{\text{row}} \times \ldots \times K^n_{\text{row}}$ and if

$$A = \begin{pmatrix} - & \alpha_1 & - \\ & \vdots & \\ - & \alpha_n & - \end{pmatrix}$$

we write $D(A) = D(\alpha_1, \ldots, \alpha_n)$. Saying that D is n-linear means that for all $1 \leq i \leq n$, we have that

1. $$D(\alpha_1, \ldots, \alpha_i + \alpha_i', \ldots, \alpha_n) = D(\alpha_1, \ldots, \alpha_i, \ldots, \alpha_n) + D(\alpha_1, \ldots, \alpha_i', \ldots, \alpha_n)$$
$$\forall\, \alpha_1, \ldots, \alpha_i, \alpha_i', \ldots, \alpha_n \in K^n_{\text{row}}$$

2. $$D(\alpha_1, \ldots, c \cdot \alpha_i, \ldots, \alpha_n) = c \cdot D(\alpha_1, \ldots, \alpha_i, \ldots, \alpha_n)$$
$$\forall\, c \in K,\ \alpha_1, \ldots \alpha_i, \ldots, \alpha_n \in K^n_{\text{row}}$$

Lemma 5.3: Linear Combinations of n-linear Functions

Any linear combination of n-linear functions is an n-linear function again.

Definition 5.4: Alternating Functions

Let $D : M_{n\times n}(K) \to K$ be an n-linear function. We say that D is **alternating** if for every matrix $A \in M_{n\times n}(K)$ in which some two rows are equal we have that $D(A) = 0$.

Proposition 5.5

Let D be an n-linear function. Assume that D has the property that whenever A has two equal adjacent (aufeinanderfolgende) rows then $D(A) = 0$. Then

1. *D is alternating.*
2. *For every matrix B, if B' is obtained from B by interchanging any two of its rows, then $D(B') = -D(B)$.*

Definition 5.6: Determinant Function

A function $D : M_{n\times n}(K) \to K$ is called **determinant function** if D is n-linear, is alternating and $D(I) = 1$.

Definition 5.7: Row&Column-Deletion Matrix

Let $A \in M_{n\times n}(K)$, $n \geq 2$, $1 \leq i \leq n$, $1 \leq j \leq n$. We denote by

$$A(i|j) \in M_{(n-1)\times(n-1)}(K)$$

the matrix obtained from A by deleting row $\#i$ and column $\#j$.
If D is an $(n-1)$-linear function and $A \in M_{n\times n}(K)$ we denote

$$D_{ij}(A) := D\big(A(i|j)\big).$$

Notation: Let $A \in M_{m\times n}(K)$ and $1 \leq i \leq m$, $1 \leq j \leq n$. Denote the element at position (i,j) in the matrix A by $A(i,j)$ or A_{ij}.

Theorem 5.8: Entwicklungssatz

Let $n \geq 2$ and D an $(n-1)$-linear function, which is alternating. Let $1 \leq j \leq n$. Define a function $E_j : M_{n\times n}(K) \to K$ by

$$E_j(A) := \sum_{i=1}^{n} (-1)^{i+j} A_{ij}\, D_{ij}(A).$$

Then E_j is an n-linear function and is alternating. If D is a determinant function then so is E_j.

Corollary 5.9: Existence of a Determinant Function

$\forall\, n \geq 1$, *there exists at least one determinant function $M_{n\times n}(K) \to K$.*

5.1 Uniqueness of the Determinant

Let $D : M_{n\times n}(K) \to K$ be an n-linear alternating function. Let

$$A = \begin{pmatrix} — & \alpha_1 & — \\ & \vdots & \\ — & \alpha_n & — \end{pmatrix} \in M_{n\times n}(K).$$

Denote

$$I = \begin{pmatrix} 1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & 1 \end{pmatrix} = \begin{pmatrix} — & \varepsilon_1 & — \\ & \vdots & \\ — & \varepsilon_n & — \end{pmatrix}.$$

(So, $\varepsilon_1, \dots, \varepsilon_n$ is the standard basis of K^n_{row}.) Clearly,

$$\alpha_i = \sum_{j=1}^{n} A(i,j)\, \varepsilon_j \qquad \forall\, 1 \leq i \leq n.$$

Therefore,

$$D(A) = D(\alpha_1, \dots, \alpha_n) = D\left(\sum_{j=1}^{n} A(1,j)\,\varepsilon_j, \alpha_2, \dots, \alpha_n\right) = \sum_{j=1}^{n} A(1,j)\cdot D(\varepsilon_j, \alpha_2, \dots, \alpha_n)$$
$$= \sum_{j=1}^{n} A(1,j)\cdot D\left(\varepsilon_j, \sum_{k=1}^{n} A(2,k)\,\varepsilon_k, \dots, \alpha_n\right) = \sum_{j,k=1}^{n} A(1,j)\cdot A(2,k)\cdot D(\varepsilon_j, \varepsilon_k, \dots, \alpha_n).$$

Applying these steps to every row we get that

$$D(A) = \sum_{k_1,\dots,k_n=1}^{n} A(1,k_1)\cdot A(2,k_2)\cdot \ldots \cdot A(n,k_n)\cdot D(\varepsilon_{k_1}, \varepsilon_{k_2} \dots, \varepsilon_{k_n}), \tag{5.1}$$

where each of the indices $k_1, \dots, k_n$ runs between 1 and n. So far we haven't used the fact that D is alternating. Now, since D is alternating $D(\varepsilon_{k_1}, \dots, \varepsilon_{k_n}) = 0$, whenever two of the indices $k_1, \dots, k_n$ are equal. So, we are interested only in ordered sequences of indices $(k_1, \dots, k_n)$ in which $k_i \in \{1, \dots, n\} \quad \forall i$ and $k_i \neq k_j \quad \forall i \neq j$. Such a sequence is called a permutation of $\{1, \dots, n\}$, or a permutation of degree n.

Definition 5.10: Permutation

A **permutation** is a bijective map $\sigma : \{1, \dots, n\} \to \{1, \dots, n\}$.

So Alternatively, we can think of a permutation of degree n as a bijective function $\sigma : \{1, \dots, n\} \to \{1, \dots, n\}$. Then the sequence $(k_1, \dots, k_n) = (\sigma 1, \dots, \sigma 2)$ is a permutation as defined earlier. (Notation: $\sigma(i) := \sigma i$.). In other words, a permutation σ of degree n is a re-ordering of a list with n elements $(1, \dots, n) \xrightarrow{\sigma} (\sigma 1, \dots, \sigma n)$.

We can rewrite Equation (5.1) as

$$D(A) = \sum_{\sigma \in S_n} A(1, \sigma 1)\cdot \ldots \cdot A(n, \sigma n)\cdot D(\varepsilon_{\sigma 1}, \dots, \varepsilon_{\sigma n}),$$

where S_n is the set of all permutations of degree n.

5.1.1 Important Thing About Permutations

Let σ be a permutation of degree n. Then one can pass from $(1, \dots, n)$ to $(\sigma 1, \dots, \sigma n)$ by performing a finite sequence of interchanges of pairs, called transpositions.

Definition 5.11: Transposition

A **transposition** is a permutation $\theta : \{1, \dots, n\} \to \{1, \dots, n\}$, where $\exists\, i, j \in \{1, \dots, n\}$, $i \neq j$ s.t. $\theta(i) = j$, $\theta(j) = i$ and $\theta(k) = k \quad \forall\, k \neq i, j$.

Now, every permutation $\sigma : \{1, \dots, n\} \to \{1, \dots, n\}$ can be written as a composition of transpositions

$$\sigma = \theta_1 \circ \ldots \circ \theta_m, \tag{5.2}$$

where $\theta_1, \dots, \theta_m$ are transpositions. But these θ_i's and m are not unique. In other words, the same σ can be written as in Equation (5.2) in many different ways

$$\sigma = \theta_1 \circ \ldots \circ \theta_m = \theta'_1 \circ \ldots \circ \theta'_{m'}.$$

Why Can σ be written as in (5.2). Start with $(1, \dots, n)$ and if $\sigma 1 \neq 1$ then interchange 1 and

$\sigma 1$. We now have $(\sigma 1, \ldots, 1, \ldots, n)$. If $\sigma 2 \neq 2$, then interchange 1 and $\sigma 2$; otherwise proceed to 3, etc. But of course there are other ways to do it, e.g., $(3,2,1)$ can be obtained in 3 steps: $(1,2,3) \to (2,1,3) \to (2,3,1) \to (3,2,1)$. But also in one step $(1,2,3) \to (3,2,1)$.

Definition 5.12: Sign of a Permutation

Let σ be a permutation and let $\theta_1, \ldots, \theta_m$ be transpositions s.t. $\sigma = \theta_1 \circ \ldots \circ \theta_m$. The permutation σ is called **even** if m is even, and is called **odd** if m is odd. The number $(-1)^m \in \{-1, 1\}$ is called the **sign** of σ, i.e., $\operatorname{sgn}(\sigma) := (-1)^m$.

Lemma 5.13: Well Definedness of the Sign of a Permutation

If $(\sigma 1, \ldots, \sigma n)$ is obtained from $(1, \ldots, n)$ in two different ways, one time by a sequence of m transpositions, and another time by a sequence of m' transpositions, then m' and m have the same parity, i.e., $(-1)^{m'} = (-1)^m$.
(So the parity of m depends only on σ and not on the particular way we performed a sequence of transpositions $(1, \ldots, n) \to \ldots \to (\sigma 1, \ldots, \sigma n)$.)

Proof. We know that determinant functions $M_{n\times n}(K) \to K$ exists for every $n \geq 1$. Fix a determinant function $E : M_{n\times n}(K) \to K$. Consider $E(\varepsilon_{\sigma 1}, \ldots, \varepsilon_{\sigma n})$. If $(\sigma 1, \ldots, \sigma n)$ can be obtained from $(1, \ldots, n)$ by a sequence of m transpositions, then the matrix

$$\begin{pmatrix} — & \varepsilon_{\sigma 1} & — \\ & \vdots & \\ — & \varepsilon_{\sigma n} & — \end{pmatrix}$$

can be obtained from

$$I = \begin{pmatrix} — & \varepsilon_{1} & — \\ & \vdots & \\ — & \varepsilon_{n} & — \end{pmatrix}$$

by a sequence of m interchanges of pairs of rows. Therefore, since E is a determinant function, we have that

$$E(\varepsilon_{\sigma 1}, \ldots, \varepsilon_{\sigma n}) = (-1)^m E(\varepsilon_1, \ldots, \varepsilon_n) = (-1)^m E(I) = (-1)^m.$$

But this applies also to m'. So,

$$E(\varepsilon_{\sigma 1}, \ldots, \varepsilon_{\sigma n}) = (-1)^{m'} E(\varepsilon_1, \ldots, \varepsilon_n) = (-1)^{m'} E(I) = (-1)^{m'}.$$

So we have that

$$(-1)^m = (-1)^{m'}.$$

□

5.1.2 Back to our $D(A)$

We have seen that if D is n-linear and alternating then

$$D(A) = \sum_{\sigma \in S_n} A(1, \sigma 1) \cdot \ldots \cdot A(n, \sigma n) \cdot D(\varepsilon_{\sigma 1}, \ldots, \varepsilon_{\sigma n}).$$

But

$$D(\varepsilon_{\sigma 1}, \varepsilon_{\sigma n}) = (\operatorname{sgn}\sigma) \cdot D(\varepsilon_1, \dots, \varepsilon_n) = (\operatorname{sgn}\sigma) \cdot D(I) \tag{5.3}$$

Theorem 5.14: Uniqueness of the Determinant Function

$\forall\, n \geq 1$, there exists a unique determinant function $M_{n\times n}(K) \to K$. We will denote it by det*. It is given by the formula*

$$\det(A) = \sum_{\sigma \in S_n} A(1, \sigma 1) \cdot \ldots \cdot A(n, \sigma n).$$

Corollary 5.15

For every n-linear and alternating function $D : M_{n\times n}(K) \to K$ we have that

$$D(A) = \det(A) \cdot D(I) \qquad \forall\, A \in M_{n\times n}(K).$$

5.1.3 A Bit more on Permutations

Denote by S_n the set of permutations $\sigma : \{1, \dots, n\} \to \{1, \dots, n\}$. S_n has the structure of a group:

- We can compose permutations, i.e., if $\sigma, \tau : \{1, \dots, n\} \to \{1, \dots, n\}$, then $\sigma \cdot \tau := \sigma \circ \tau$, where $\sigma \circ \tau : \{1, \dots, n\} \to \{1, \dots, n\}$.
- The identity $id : \{1, \dots, n\} \to \{1, \dots, n\}$ is the neutral element $1 \in S_n$.
- The inverse σ^{-1} of σ, is given by the inverse of the function σ (recall that σ is bijective, so σ^{-1} does exist).

Also, note that $|S_n| = n!$

Lemma 5.16: Sign of the Composition of Permutations

$\forall\, \sigma, \tau \in S_n$ we have

$$\operatorname{sgn}(\sigma \cdot \tau) = \operatorname{sgn}(\sigma) \cdot \operatorname{sgn}(\tau).$$

Theorem 5.17: The Determinant of the Product of Two Matrices

$\forall\, A, B \in M_{n\times n}(K)$ we have that

$$\det(A \cdot B) = \det(A) \cdot \det(B).$$

Proof. Fix a matrix $B \in M_{n\times n}(K)$, and consider a function $D : M_{n\times n}(K) \to K$ defined by

$$D(A) := \det(A \cdot B).$$

Claim: D is n-linear and alternating.
Proof of Claim: Write

$$A = \begin{pmatrix} — & \alpha_1 & — \\ & \vdots & \\ — & \alpha_n & — \end{pmatrix}.$$

If we view α_i as a $1 \times n$ matrix, then $\alpha_i B$ is also $1 \times n$, and in fact

$$A \cdot B = \begin{pmatrix} — & \alpha_1 B & — \\ & \vdots & \\ — & \alpha_n B & — \end{pmatrix}.$$

So,

$$D(A) = \det(\alpha_1 B, \ldots, \alpha_n B).$$

Clearly, by the rules of matrix multiplication, for every two row vectors α_i, α_i' we have that $(\alpha_i + \alpha_i')B = \alpha_i B + \alpha_i' B$ and

$$\forall\, c \in K: \ (c \cdot \alpha_i)B = c \cdot (\alpha_i B).$$

Since det is n-linear it follows that D is also n-linear.

If $\alpha_i = \alpha_j$, then $\alpha_i B = \alpha_j B$. Therefore,

$$D(A) = \det(\alpha_1 B, \ldots, \alpha_n B) = 0.$$

This proves the claim.

We continue with the proof of the Theorem. Since we found that D is n-inear and alternating, by Corollary 5.15, we have that

$$D(A) = \det(A) \cdot D(I).$$

But $D(I) = \det(I \cdot B) = \det(B)$. Therefore,

$$D(A) = \det(A \cdot B) = \det(A) \cdot \det(B). \qquad \square$$

Corollary 5.18: Equivalent Condition for The Invertibility of a Matrix

If A is invertible then $\det(A) \neq 0$. Moreover,

$$\det(A^{-1}) = \frac{1}{\det(A)}.$$

5.2 More on Determinants

Theorem 5.19: Determinant of a Transposed Matrix

$\forall\, A \in M_{n \times n}(K): \ \det(A^T) = \det(A).$

Corollary 5.20

If $D : M_{n \times n}(K) \to K$ is an alternating n-linear function (of the rows of a matrix) then D is also an alternating and n-linear function of the columns of a matrix. In fact, $D(A^T) = D(A) \quad \forall\, A \in M_{n \times n}(K)$. In particular, this holds also for $D = \det$.

Theorem 5.21: Determinant after Row/Column Operations

1. *If B is obtained from A by adding a multiple of one row of A to another row (or by doing the analogous operation on columns), then* $\det(B) = \det(A)$.
2. *If* $A \xrightarrow{R_i \leftrightarrow R_j} B$, $i \neq j$, *(or the analogous operation on columns) then* $\det(B) = -\det(A)$.
3. *If* $A \xrightarrow{c \cdot R_i \to R_i} B$, *then* $\det(B) = c \cdot \det(A)$.

5.2.1 Determinants of Block Matrices

Consider a block matrix M of the type

$$M = \begin{pmatrix} A & B \\ 0 & C \end{pmatrix},$$

where $A \in M_{r \times r}(K)$, $C \in M_{s \times s}(K)$, $B \in M_{r \times s}(K)$.

Proposition 5.22: Determinant of a Block Matrix

Let M be as above. Then

$$\det(M) = \det(A) \cdot \det(C).$$

Corollary 5.23: Entwicklungssatz with Cofactors

$\forall\, A \in M_{n \times n}(K)$ *we have the following n different formulas for* $\det(A)$, *i.e.,*

$$\det(A) = \sum_{i=1}^{n} (-1)^{i+j} A_{ij} \det\big(A(i|j)\big), \quad j = 1, \ldots, n.$$

The scalars $c_{ij} := (-1)^{i+j} \det\big(A(i|j)\big)$ *are called the i, j* ***cofactor*** *of A. We have*

$$\det(A) = \sum_{i=1}^{n} A_{ij} \cdot c_{ij}. \tag{5.4}$$

Claim: If we replace A_{ij} in Equation (5.4) by A_{ik}, where k is fixed, we always get 0. In other words, $\forall\, A \in M_{n \times n}(K)$, $\forall\, j$ we have

$$\sum_{i=1}^{n} A_{ik} \cdot c_{ij} = \begin{cases} \det(A) & j = k \\ 0 & j \neq k \end{cases}.$$

Proof of Claim: Let $B \in M_{n \times n}(K)$ be the matrix obtained from A by replacing column j of A by column k of A. So, in B column j equals column k. Clearly, $\det(B) = 0$ since B has two equal columns ($\Longleftrightarrow$ B^T has two equal rows).

$$\Longrightarrow \qquad 0 = \det(B) = \sum_{i=1}^{n} (-1)^{i+j} B_{ij} \det\big(B(i|j)\big) = \sum_{i=1}^{n} = \sum_{i=1}^{n} A_{ik} \det\big(A(i|j)\big) = \sum_{i=1}^{n} A_{ik} \cdot c_{ij}.$$

This proves the claim.

So, $\forall A \in M_{n\times n}(K)$, $\forall j, k$ we have

$$\sum_{i=1}^{n} A_{ik} \cdot c_{ij} = \delta_{jk} \cdot \det(A). \tag{5.5}$$

Definition 5.24: Classical Adjoint of A

The matrix $(c_{ij})^T$ is called the **classical adjoint of** A, i.e.,

$$\operatorname{adj}(A) \in M_{n\times n}(K), \qquad \big(\operatorname{adj}(A)\big)_{ij} = c_{ji} = (-1)^{i+j} \cdot \det\big(A(j|i)\big).$$

The Equation (5.5) says that

$$(\operatorname{adj} A) \cdot A = \det(A) \cdot I.$$

Indeed,

$$\begin{aligned} & \det(A) \cdot \delta_{jk} = \sum_{i=1}^{n} c_{ij} \cdot A_{ik} = \sum_{i=1}^{n} \big(\operatorname{adj}(A)\big)_{ji} \cdot A_{ik} \\ \Longrightarrow \quad & \det(A) \cdot I = \operatorname{adj}(A) \cdot A. \end{aligned}$$

Theorem 5.25: Formula for the inverse of A

Let $A \in M_{n\times n}(K)$. Then A is invertible if and only if $\det(A) \neq 0$. When A is invertible we have

$$A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A).$$

5.3 Cramer's Rule

Let $A \in M_{n\times n}(K)$. We would like a formula to solve the system of equations $Ax = b$, where

$$x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$$

are the unknowns and

$$b = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} \in K^n$$

is given. Assume $Ax = b$. Then

$$\begin{aligned} & \operatorname{adj} A \cdot Ax = \operatorname{adj} A \cdot b \\ \Longrightarrow \quad & \det(A) \cdot x = \operatorname{adj} \cdot b \\ \Longrightarrow \quad & \forall\, 1 \leq j \leq n: \ \det(A) \cdot x_j = \sum_{i=1}^{n} (\operatorname{adj} A)_{ji} \cdot b_i = \sum_{i=1}^{n} (-1)^{i+j} b_i \det\big(A(i|j)\big), \end{aligned}$$

where the last expression is the determinant of the matrix obtained from A by replecing column j by b.

Theorem 5.26: Cramer's Rule

Let $A \in M_{n\times n}(K)$ with $\det(A) \neq 0$. Let $b \in K^n$. Then the system $Ax = b$ has a unique solution, and it is given by the formula

$$x_j = \frac{\det\big(A(j,b)\big)}{\det(A)} \qquad \forall\, 1 \leq j \leq n,$$

where $A(j,b)$ is the matrix obtained from A by replacing column j by b.

5.4 Determinants of Endomorphisms

Let $A \in M_{n\times n}(K)$ and $P \in GL_n(K)$, then

$$\det(P^{-1}AP) = \det(P^{-1}) \cdot \det(A) \cdot \det(P) = \frac{1}{\det(P)} \cdot \det(A) \cdot \det(P) = \det(A).$$

In other words, similar matrices have the same determinant.

Let V be a finite dimensional vector space over K. Put $n := \dim V$. Let $T : V \to V$ be a linear map. Pick a basis $\mathcal{B}$ for V. We have a matrix $[T]_{\mathcal{B}}^{\mathcal{B}} \in M_{n\times n}(K)$. Consider $\det[T]_{\mathcal{B}}^{\mathcal{B}} \in K$.

Question: Does $\det[T]_{\mathcal{B}}^{\mathcal{B}}$ depend on the basis $\mathcal{B}$?

Answer: Let $\mathcal{B}'$ be another basis for V. Denote by $P := [id_V]_{\mathcal{B}}^{\mathcal{B}'}$ the transition matrix between the basis $\mathcal{B}'$ and $\mathcal{B}$. Recall that

$$[T]_{\mathcal{B}'}^{\mathcal{B}'} = [id_V]_{\mathcal{B}'}^{\mathcal{B}} \cdot [T]_{\mathcal{B}}^{\mathcal{B}} \cdot [id_V]_{\mathcal{B}}^{\mathcal{B}'},$$

and we also have $[id_V]_{\mathcal{B}'}^{\mathcal{B}} = \big([id_V]_{\mathcal{B}}^{\mathcal{B}'}\big)^{-1}$. So,

$$[T]_{\mathcal{B}'}^{\mathcal{B}'} = P^{-1} \cdot [T]_{\mathcal{B}}^{\mathcal{B}} \cdot P. \qquad \Longrightarrow \qquad \det[T]_{\mathcal{B}'}^{\mathcal{B}'} = \det[T]_{\mathcal{B}}^{\mathcal{B}}.$$

Recall also that if $T, S : V \to V$ are two linear maps then $[S \circ T]_{\mathcal{B}}^{\mathcal{B}} = [S]_{\mathcal{B}}^{\mathcal{B}} \cdot [T]_{\mathcal{B}}^{\mathcal{B}}$. Recall also that if $T : V \to V$ is an isomorphism then $[T^{-1}]_{\mathcal{B}}^{\mathcal{B}} = \big([T]_{\mathcal{B}}^{\mathcal{B}}\big)^{-1}$. Finally, $[id_V]_{\mathcal{B}}^{\mathcal{B}} = I$.

Corollary 5.27: Well Definedness of the Determinant of an Endomorphism

*Let V be a finite dimensional vector space over K. Let $T : V \to V$ be a linear map. Then $\det[T]_{\mathcal{B}}^{\mathcal{B}}$ does **not** depend on the choice of basis $\mathcal{B}$ of V. We denote this scalar by $\det(T) \in K$. The function*

$$\det : \mathrm{End}(V) \to K, \quad T \mapsto \det(T)$$

has the following properties:

1. $\det(S \circ T) = \det(S) \cdot \det(T) \qquad \forall\, S, T \in \mathrm{End}(V)$.
2. *T is an isomorphism if and only if $\det(T) \neq 0$. If T is an isomorphism then*
$$\det(T^{-1}) = \frac{1}{\det(T)}.$$
3. $\det(id_V) = 1$.
4. $\det(0) = 0 \in K$.

Remark 5.28. *In contrast to $T : V \to V$, if we take $T : V \to W$, and $\mathcal{B}, \mathcal{C}$ bases of V, W, then* $\det[T]_{\mathcal{C}}^{\mathcal{B}}$ ***depends*** *on $\mathcal{B}, \mathcal{C}$ (unless $W = V$ and $\mathcal{C} = \mathcal{B}$).*

6 Eigenvalues & Eigenvectors

Definition 6.1: Eigenvalues & Eigenvectors

Let V be a vector space over K and $T : V \to V$ a linear map (an endomorphism of V). We say that $\lambda \in K$ is an **eigenvalue** of T if $\exists\, v \in V \setminus \{0\}$ s.t.

$$Tv = \lambda v.$$

Any vector $v \in V \setminus \{0\}$ with the property that $Tv = \lambda v$ is called an **eigenvector** of T for the eigenvalue λ.

Definition 6.2: Diagonalizability

Let V be a finite dimensional vector space over K and $T : V \to V$ a linear map. We say that T is diagonalizable if there exists a basis $\mathcal{B}$ of V s.t.

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} \lambda_1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \lambda_n \end{pmatrix}. \tag{6.1}$$

Lemma 6.3: Equivalent Condition for Diagonalizability

T is diagonalizable if and only if there exists a basis $\mathcal{B} = (v_1, \dots, v_n)$ consisting of eigenvectors of T (i.e., $\forall\, i$, $Tv_i = \lambda_i v_i$ for some $\lambda_i \in K$). Moreover, if for some basis $\mathcal{B} = (v_1, \dots, v_n)$ the matrix $[T]_{\mathcal{B}}^{\mathcal{B}}$ has the shape as in Equation (6.1), then $\forall\, i$, λ_i is an eigenvalue of T and v_i is an eigenvector of λ_i.

Remark 6.4. *We saw that $\forall\, A \in M_{n\times n}(K)$ we have a linear map $T_A : K^n \to K^n$, $T_A(v) := Av$. We say that $A \in M_{n\times n}(K)$ is diagonalizable if the linear map T_A is diagonalizable.*

Proposition 6.5: Linear Independence of Eigenvectors of Distinct Eigenvalues

Let V be a vector space over K and $T : V \to V$ a linear map. Suppose that $\lambda_1, \dots, \lambda_n \in K$ are eigenvalues of T which are pairwise distinct (i.e., $\lambda_i \neq \lambda_j \quad \forall\, i \neq j$). Let $v_1, \dots, v_n$ be eigenvectors for $\lambda_1, \dots, \lambda_n$ respectively. Then $v_1, \dots, v_n$ are linearly independent.

Proof. Induction on n.

$n = 1$**:** v_1 is an eigenvector for λ_1. By definition of $v_1 \neq 0$. $\implies$ v_1 is linearly independent.

Induction step: Let $n \geq 1$ and suppose the proposition holds for any list of n eigenvalues and eigenvectors. We'll prove now that the propostion holds also fro a list of $n+1$ eigenvalues and eigenvectors. Let $\lambda_1, \dots, \lambda_n, \lambda_{n+1}$ be pairwise distinct eigenvalues of T, and $v_1, \dots, v_n, v_{n+1}$ a given list of eigenvectors of $\lambda_1, \dots, \lambda_n, \lambda_{n+1}$, respectively. We need to prove that $v_1, \dots, v_n, v_{n+1}$ are linearly independent.

Assume that

$$a_1 v_1 + \ldots + a_n v_n + a_{n+1} v_{n+1} = 0. \tag{6.2}$$

We need to show that $a_1 = \ldots = a_n = a_{n+1} = 0$. Apply T to Equation (6.2). Since T is linear, we have that

$$a_1 Tv_1 + \ldots + a_n Tv_n + a_{n+1} Tv_{n+1} = 0.$$

But $\forall\, i: \; Tv_i = \lambda_i v_i$. Therefore,

$$a_1\lambda_1 v_1 + \ldots, a_n\lambda_n v_n + a_{n+1}\lambda_{n+1} v_{n+1} = 0. \tag{6.3}$$

Consider now Equation (6.3) - $\lambda_{n+1}\cdot$(6.2), i.e.,

$$a_1(\lambda_1 - \lambda_{n+1})v_1 + \ldots + a_n(\lambda_n - \lambda_{n+1})v_n + a_{n+1}\underbrace{(\lambda_{n+1} - \lambda_{n+1})}_{=0} v_{n+1} = 0.$$

By the induction hypothesis $v_1, \ldots, v_n$ are linearly independent. Therefore,

$$a_1(\lambda_1 - \lambda_{n+1}) = \ldots = a_n(\lambda_n - \lambda_{n+1}) = 0.$$

By assumption $\lambda_i \neq \lambda_{n+1} \quad \forall\, 1 \leq i \leq n$. So we have that $a_1 = \ldots = a_n = 0$. Going back to Equation (6.2) we get that $a_{n+1} \cdot v_{n+1} = 0$. But by assumption we also have that $v_{n+1} \neq 0$. Therefore, $a_{n+1} = 0$. □

6.1 The Characteristic Polynomial

Proposition 6.6: Equivalent Condition for Eigenvalue

*Let V be a vector space over K and $T: V \to V$ a linear map. Then $\lambda \in K$ is an eigenvalue of T if and only if the linear map $T - \lambda \cdot id$ is **not** injective.*

Definition 6.7: The Characterstic Polynomial

Let V be a finite dimensional vector space over K and $T: V \to V$ a linear map. We define the **characteristic polynomial of** T as

$$p_T(x) := \det(T - x \cdot id_V).$$

We define the characteristic polynomial of a matrix $A \in M_{n\times n}(K)$ in the same way, i.e.,

$$p_A(x) := \det(A - x \cdot I_n).$$

Definition 6.8: Eigenspace

Let V be a vector space over K and $T \in \mathrm{End}(V)$. Let $\lambda \in K$ be an eigenvalue of T. The **eigenspace** corresponding to λ is defined as

$$\mathrm{Eig}_T(\lambda) := \{v \in V \mid Tv = \lambda v\} = \ker(T - \lambda id_V) = \{v \in V \mid v \text{ is an eigenvector of } \lambda\} \cup \{0\}.$$

Clearly, since $\mathrm{Eig}_T(\lambda)$ is the kernel of a linear map, $\mathrm{Eig}_T(\lambda) \subseteq V$ is a linear subspace of V.

Lemma 6.9: Properties of the Characteristic Polynomial

Let $A \in M_{n\times n}(K)$. Then:

1. *$p_A(x)$ is a polynomial of degree n, with leading coefficient $(-1)^n$, i.e., $p_A(x) = c_n x^n + c_{n-1}x^{n-1} + \ldots + c_1 x + c_0$, with $c_i \in K \quad \forall i$, and $c_n = (-1)^n$.*

2. *$p_{A^T}(x) = p_A(x)$.*

3. *If A has upper triangular form as in Definition 3.12, then*

$$p_A(x) = (\lambda_1 - x) \cdot \ldots \cdot (\lambda_n - x).$$

4. *If*

$$A = \begin{pmatrix} B_1 & * \\ 0 & B_2 \end{pmatrix}$$

is a block matrix, then

$$p_A(x) = p_{B_1}(x) \cdot p_{B_2}(x).$$

5. *If A and B are similar matrices, i.e., $B = P^{-1}AP$ for some $P \in GL_n(K)$, then*

$$p_A(x) = p_B(x).$$

Remark 6.10. *If V is an n-dimensional vector space over K and $T \in \mathrm{End}(V)$, then $p_T(x)$ is a polynomial of degree n with leading coefficient $(-1)^n$.*

Definition 6.11: The Trace

Let $A \in M_{n\times n}(K)$. Define the **trace** (Spur) of A as

$$\mathrm{Tr}(A) := \sum_{i=1}^{n} A_{ii} \in K,$$

i.e., the sum of the entries along the diagonal of A. If V is a finite dimensional vector space over K and $T \in \mathrm{End}(V)$, we define the trace of T as

$$\mathrm{Tr}(T) := \mathrm{Tr}\left([T]_{\mathcal{B}}^{\mathcal{B}}\right),$$

where $\mathcal{B}$ is some basis for V. We'll see soon that $\mathrm{Tr}(T)$ is well defined, i.e., it does not depend on the choice of $\mathcal{B}$.

Lemma 6.12: Properties of the Trace

1.
$$\forall\, A, B \in M_{n\times n}(K): \ \mathrm{Tr}(A \cdot B) = \mathrm{Tr}(B \cdot A).$$

2. *If A and B are similar matrices, then*
$$\mathrm{Tr}(A) = \mathrm{Tr}(B).$$
In particular, if $\mathcal{B}$ and $\mathcal{B}'$ are two bases for V, then
$$\mathrm{Tr}\left([T]_{\mathcal{B}}^{\mathcal{B}}\right) = \mathrm{Tr}\left([T]_{\mathcal{B}'}^{\mathcal{B}'}\right).$$
Thus, $\mathrm{Tr}(T)$ is well defined.

Proof. **1):**
$$\begin{aligned}\mathrm{Tr}(A \cdot B) &= \sum_{i=1}^{n} (A \cdot B)_{ii} = \sum_{i=1}^{n} \sum_{j=1}^{n} A_{ij} \cdot B_{ji} \\ &= \sum_{j=1}^{n} \sum_{i=1}^{n} A_{ij} \cdot B_{ji} = \sum_{j=1}^{n} \sum_{i=1}^{n} B_{ji} A_{ij} = \sum_{j=1}^{n} (B \cdot A)_{jj} \\ &= \mathrm{Tr}(B \cdot A).\end{aligned}$$

2): If $B = P^{-1}AP$ for some $P \in GL_n(K)$, then by 1) we have that
$$\mathrm{Tr}(B) = \mathrm{Tr}(P^{-1}AP) = \mathrm{Tr}\left(P^{-1} \cdot (AP)\right) = \mathrm{Tr}\left((AP) \cdot P^{-1}\right) = \mathrm{Tr}(A).$$
Regarding the claim about $\mathrm{Tr}(T)$, recall that
$$[T]_{\mathcal{B}'}^{\mathcal{B}'} = [id_V]_{\mathcal{B}'}^{\mathcal{B}} \cdot [T]_{\mathcal{B}}^{\mathcal{B}} \cdot [id_V]_{\mathcal{B}}^{\mathcal{B}'} \tag{6.4}$$
and $[id_V]_{\mathcal{B}'}^{\mathcal{B}} = \left([id_V]_{\mathcal{B}}^{\mathcal{B}'}\right)^{-1}$. So, $[T]_{\mathcal{B}'}^{\mathcal{B}'}$ and $[T]_{\mathcal{B}}^{\mathcal{B}}$ are similar matrices. Therefore,
$$\mathrm{Tr}\left([T]_{\mathcal{B}'}^{\mathcal{B}'}\right) = \mathrm{Tr}\left([T]_{\mathcal{B}}^{\mathcal{B}}\right).$$
□

Lemma 6.13: Coefficients of the Characteristic Polynomial

Let $A \in M_{n\times n}(K)$ and $p_A(x) = c_n x^n + c_{n-1}x^{n-1} + \ldots + c_1 x + c_0$ its characteristic polynomial. We've already seen that $c_n = (-1)^n$. We also have that
$$c_{n-1} = (-1)^{n-1} \cdot \mathrm{Tr}(A), \qquad c_0 = \det(A).$$
In other words,
$$p_A(x) = (-1)^n x^n + (-1)^{n-1} x^{n-1} + c_{n-2} x^{n-2} + \ldots + c_1 x + \det(A).$$

Definition 6.14: Direct Sum

Let V be a vector space over K and $U_1, \ldots, U_r \subseteq V$ subspaces. Denote $W := U_1 + \ldots + U_r \subseteq V$. We say that W is a **direct sum** of $U_1, \ldots, U_r$ (or that $U_1, \ldots, U_r$ is a direct sum) if every $w \in W$ has a **unique** representation as $w = u_1 + \ldots + u_r$ with $u_i \in U_i \quad \forall i$.

Proposition 6.15: Eigenspaces are a Direct Sum

Let V be a vector space over K and $T \in \mathrm{End}(V)$. Let $\lambda_1, \ldots, \lambda_r \in K$ be pairwise distinct eigenvalues of T. Then

$$\mathrm{Eig}_T(\lambda_1) + \ldots + \mathrm{Eig}_T(\lambda_r)$$

is a direct sum.

Corollary 6.16: Equivalent Condition for Diagonalizability of an Endomorphsim

Assume V is finite dimensional, $T \in \mathrm{End}(V)$ and let $\lambda_1, \ldots, \lambda_r \in K$ be all the eigenvalues of T. Assume additionally that $\lambda_1, \ldots, \lambda_r$ are pairwise distinct. Then T is diagonalizable if and only if

$$\sum_{i=1}^{r} \dim \mathrm{Eig}_T(\lambda_i) = \dim V.$$

Lemma 6.17: Decomposition of the Characteristic Polynomial

Let $A \in M_{n \times n}(K)$ (or $T \in \mathrm{End}(V)$) be diagonalizable. Then $p_A(x)$ decomposes as a product of linear factors, i.e., degree 1

$$p_A(x) = (\lambda_1 - x) \cdot \ldots \cdot (\lambda_n - x),$$

where $\lambda_1, \ldots, \lambda_n$ are the eigenvalues of A (but there might be repetitions among the λ_i's).

6.1.1 A Quick Digression about Polynomials

Let $0 \neq p(x) \in K[x]$, $\lambda \in K$. By dividing $p(x)$ by $(x - \lambda)$ we can write

$$p(x) = (x - \lambda) \cdot q(x) + R,$$

where $\deg q(x) = \deg p(x) - 1$ and $R \in K$. But if we substitute $x = \lambda$, we obtain

$$p(\lambda) = 0 \cdot q(\lambda) + R = R \quad \Longrightarrow \quad p(x) = (x - \lambda) \cdot q(x) + p(\lambda).$$

Conclusion: $p(\lambda) = 0 \iff p(x) = (x - \lambda) \cdot q(x)$ for some $q(x) \in K[x]$. We say that λ is a zero of $p(x)$ (Nullstelle) or a root of $p(x)$.

Sometimes λ is also a zero of $q(x)$, so $q(x) = (x - \lambda) \cdot h(x)$. Therefore, $p(x) = (x - \lambda)^2 \cdot h(x)$, etc.

Definition 6.18: Multiplicity of Zeroes

Let $0 \neq p(x) \in K[x]$, $\lambda \in K$. We say that λ is a **zero of multiplicity** $m \in \mathbb{N}_{>0}$ if

$$p(x) = (x - \lambda)^m \cdot g(x),$$

where $g(x) \in K[x]$ and $g(\lambda) \neq 0$.
If $m \geq 2$ we say λ is a **multiple zero** of $p(x)$.
If $m = 1$ we say λ is a **simple zero** of $p(x)$.

6.2 Geometric & Algebraic Multiplicities of Eigenvalues

Definition 6.19: Geometric & Algebraic Multiplicities

Let V be a vector space over K and $T \in \mathrm{End}(V)$. Let $\lambda \in K$ be an eigenvalue of T. We define the **geometric multiplicity** of λ to be

$$m_g(T;\lambda) := \dim \mathrm{Eig}_T(\lambda).$$

Assume now, in addition, that V is finite dimensional. We define the **algebraic multiplicity** of λ to be the multiplicity of the zero λ in the characteristic polynomial $p_T(x)$ of T. We denote it by $m_A(T;\lambda)$.

Proposition 6.20: Inequality Regarding Geometric & Algebraic Multiplicity

$$m_g(T;\lambda) \leq m_a(T;\lambda).$$

Proof. Let $v_1, \dots, v_k$ be a basis for $\mathrm{Eig}_T(\lambda)$. We extend this basis to a basis $\mathcal{B} = (v_1, \dots, v_k, w_{k+1}, \dots, w_n)$ of V. Then we have

$$[T]^{\mathcal{B}}_{\mathcal{B}} = \begin{pmatrix} | & & | & | & & | \\ [Tv_1]_{\mathcal{B}} & \dots & [Tv_n]_{\mathcal{B}} & [Tw_{k+1}]_{\mathcal{B}} & \dots & [Tw_n]_{\mathcal{B}} \\ | & & | & | & & | \end{pmatrix} = \begin{pmatrix} \lambda & \dots & 0 & * \\ \vdots & \ddots & \vdots & \vdots \\ 0 & \dots & \lambda & * \\ 0 & \dots & 0 & C \end{pmatrix},$$

where $C \in M_{(n-k)\times(n-k)}(K)$. Therefore,

$$p_T(x) = \det\left([T]^{\mathcal{B}}_{\mathcal{B}} - xI_n\right) = (\lambda - x)^k \cdot \det(C - xI_{n-k}) = (\lambda - x)^k \cdot p_C(x) = (-1)^k \cdot (x - \lambda)^k \cdot p_C(x).$$

This shows that the multiplicity of λ as a zero of $p_T(x)$ is at least k. In other words,

$$m_a(T;\lambda) \geq k = m_g(T;\lambda).$$

□

6.3 Diagonalizability

Theorem 6.21: Equivalent Statements for Diagonalizablility

et V be an n-dimensional vector space over K and $T \in \mathrm{End}(V)$. The following statements are equivalent:

1. *T is diagonalizable.*
2. *There exists a basis for V whose elements are eigenvectors of T.*
3. *The characteristic polynomial $p_T(x)$ splits into a product of linear factors, and for every eigenvalue of T the geometric and algebraic multiplicities are equal.*
4. $$\sum_{i=1}^{k} \dim \mathrm{Eig}_T(\lambda_i) = \dim V,$$
   where $\lambda_1, \ldots, \lambda_k \in K$ are the eigenvalues of T.
5. $$V \cong \bigoplus_{i=1}^{k} \mathrm{Eig}_T(\lambda_i).$$

Corollary 6.22

If $p_T(x)$ splits as a product of linear factors and each eigenvalue λ of T has algebraic multiplicity, then T is digaonolizable.

Corollary 6.23

If $p_T(x)$ splits as a product of linear factors and each eigenavalue λ of T has algebraic multiplicity of 1, then T is diagonalizable.

6.4 Trigonalization

Definition 6.24: Trigonalizability

We say that $T \in \mathrm{End}(V)$ is **trigonalizable** if there exists a basis $\mathcal{B}$ for V s.t. $[T]_{\mathcal{B}}^{\mathcal{B}}$ is upper triangular, i.e.,

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} \lambda_1 & \ldots & * \\ \vdots & \ddots & \vdots \\ 0 & \ldots & \lambda_n \end{pmatrix}.$$

Theorem 6.25: Equivalent Condition for Trigonalizability

et V be a finite dimensional vector space over K and $T \in \mathrm{End}(V)$. Then T is trigonalizable if and only if $p_T(x)$ splits as a product of linear factors, i.e.,

$$p_T(x) = (\lambda_1 - x)^{l_1} \cdot \ldots \cdot (\lambda_k - x)^{l_k}.$$

Theorem 6.26: Fundamental Theorem of Algebra

Every polynomial $p(x) \in \mathbb{C}[x]$ with complex coefficients, splits as a product of linear factors and a constant, i.e.,

$$p(x) = c \cdot (x - a_1)^{n_1} \cdot \ldots \cdot (x - a_r)^{n_r},$$

with $c \in \mathbb{C}$, $r \geq 0$, $a_i \in \mathbb{C}$, $n_j \in \mathbb{N}_{>0}$.

Corollary 6.27: Trigonalizability over $\mathbb{C}$

Let V be a finite dimensional vector space over $\mathbb{C}$. Then every $T \in \mathrm{End}(V)$ trigonalizable.

6.5 Minimal Polynomial & Cayley-Hamilton Theorem

Definition 6.28: Matrix Polymomial

Let $A \in M_{n\times n}(K)$. Let $g(x) = a_d x^d + \ldots + a_1 x + a_0 \in K[x]$ be a polynomial. We can define $g(A) \in M_{n\times n}(K)$, as

$$g(A) := a_d A^d + \ldots + a_1 A + a_0 I_n.$$

Remark 6.29. *$\forall\, A \in M_{n\times n}(K)\, \exists\, g \in K[x]$ s.t. $g(A) = 0$. Indeed, consider*

$$I_n, A, A^2, \ldots, A_r \in M_{n\times n}(K). \tag{6.5}$$

Since $M_{n\times n}(K)$ is a vector space of dimension n^2, then if we take $r > n^2$ then the elements in (6.5) cannot be linearly independent. Therefore, $\exists\, a_0, a_1, \ldots, a_r \in K$ s.t.

$$a_0 I_n + a_1 A + \ldots + a_r A^r = 0.$$

A similar thing applies to $T \in \mathrm{End}(V)$, where V is a finite dimensional vector space over K. ($T^k = \underbrace{T \circ \ldots \circ T}_{\times k}$, $g(T) := a_d T^d + \ldots + a_1 T + a_0 id_V$.)

Definition 6.30: Minimal Polynomial

Let V be a vector space over K and $T \in \mathrm{End}(V)$. A **minimal polynomial** for T is a polynomial $g(x) \in K[x]$ of the minimal possible degree s.t. $g(T) = 0$.

Theorem 6.31: Cayley-Hamilton

1. *Let $A \in M_{n\times n}(K)$. Then*
$$p_A(A) = 0.$$
2. *Let V be a finite dimensional vector space over K, and $T \in \mathrm{End}(V)$. Then*
$$p_T(T) = 0.$$

The proof of this Theorem may be important for the exam but will be skipped in this Document. The proof can be found in the lecture notes.

7 Euclidean & Hermitian (unitary) Spaces

At last we endow our spaces with a geometric structure. We work here with vector spaces V over $K = \mathbb{R}$ or $K = \mathbb{C}$. Over $\mathbb{R}$ we'll have Euclidean spaces. Over $\mathbb{C}$ we'll have Hermitian (a.k.a unitary) spaces.

Definition 7.1: Scalar Product

Let V be a vector space over $\mathbb{R}$. A **scalar product** (a.k.a inner product) is a function $V \times V \to \mathbb{R}, \quad (v, w) \mapsto \langle v, w \rangle$ that has the following properties:

1. Linearity in the first variable:

$$\langle v_1 + v_2, w \rangle = \langle v_1, w \rangle + \langle v_2, w \rangle \qquad \forall\, v_1, v_2, w \in V$$
$$\langle \alpha v, w \rangle = \alpha \langle v, w \rangle \qquad \forall\, \alpha \in \mathbb{R}, v, w \in V.$$

2. Linearity in the second variable:

$$\langle v, w_1, w_2 \rangle = \langle v, w_1 \rangle + \langle v, w_2 \rangle \qquad \forall\, v, w_1, w_2 \in V$$
$$\langle v, \alpha w \rangle = \alpha \langle v, w \rangle \qquad \forall\, \alpha \in \mathbb{R}, v, w \in V.$$

3. Symmetry:
$$\langle v, w \rangle = \langle w, v \rangle \qquad \forall\, v, w \in V.$$

4. Positivity:
$$\langle v, v \rangle > 0 \qquad \forall\, v \in V \setminus \{0\}.$$

We call $\big(V, \langle \cdot, \cdot \rangle\big)$ an **Euclidean space**, or a space endowed with a scalar product.

Definition 7.2: Hermitian Product

Let V be a vector space over $\mathbb{C}$. A **Hermitian product** (or inner product) on V is a function $V \times V \to \mathbb{C}, \quad (v,w) \mapsto \langle v,w\rangle$ with the following properties:

1. Linearity in the first variable:
$$\langle v_1+v_2,w\rangle = \langle v_1,w\rangle + \langle v_2,w\rangle \qquad \forall\, v_1,v_2,w \in V$$
$$\langle \alpha v,w\rangle = \alpha\,\langle v,w\rangle \qquad \forall\, \alpha \in \mathbb{C}, v,w \in V.$$

2. Sesquilinearity (complex anti-linearity) in the second variable:
$$\langle v,w_1,w_2\rangle = \langle v,w_1\rangle + \langle v,w_2\rangle \qquad \forall\, v,w_1,w_2 \in V$$
$$\langle v,\alpha w\rangle = \overline{\alpha}\,\langle v,w\rangle \qquad \forall\, \alpha \in \mathbb{C}, v,w \in V.$$

3. Hermitian property:
$$\langle v,w\rangle = \overline{\langle w,v\rangle} \qquad \forall\, v,w \in V.$$

4. Positivity:
$$\langle v,v\rangle > 0 \qquad \forall\, v \in V.$$

We call $\big(V,\langle\cdot,\cdot\rangle\big)$ a **Hermitian (or unitary) space**.

Lemma 7.3: Properties of Euclidean/Hermitian Spaces

Let V be an Euclidean/Hermitian vector space. Then:

1. $\forall\, v \in V: \ \langle 0,v\rangle = \langle v,0\rangle = 0.$
2. *Let $W \in V$. If $\langle v,w\rangle = 0 \quad \forall\, v \in V$, then $w = 0$.*
3. *Let $w_1,w_2 \in V$. If $\langle v,w_1\rangle = \langle v,w_2\rangle \quad \forall\, v \in V$, then $w_1 = w_2$.*

Definition 7.4: Some Definitions to Euclidean/Hermitian Spaces

1. Let V be an Euclidean/Hermitian vector space. $\forall v \in V$, define $\|v\| := \sqrt{\langle v, v\rangle} \in \mathbb{R}_{\geq 0}$. $\|\cdot\|$ is called the **norm** on V induced by $\langle \cdot, \cdot \rangle$.

2. A vector $u \in V$ is called a **unit vector** if $\|u\| = 1$. Note that if $0 \neq v \in V$ is any vector then
$$\frac{v}{\|v\|}$$
is a unit vector (which is positively proportional to v, so it points in the same direction as v). We call $\frac{v}{\|v\|}$ the normalization of v.

3. For $v, w \in V$ we define the **distance** between v and w by
$$d(v, w) := \|v - w\|.$$

4. Two vectors $u, v \in V$ are called **orthogonal** (or perpendicular one to the other) if $\langle u, v\rangle = 0$. In this case we sometimes write $u \perp v$.

5. A subset $S \subseteq V$ is called an **orthogonal system** if $0 \notin S$ and for every two distinct elements $u, v \in S$ we have $u \perp v$.

6. An orthogonal system $S \subseteq V$ is called **orthonormal** if $\forall v \in S$ we have $\|v\| = 1$.

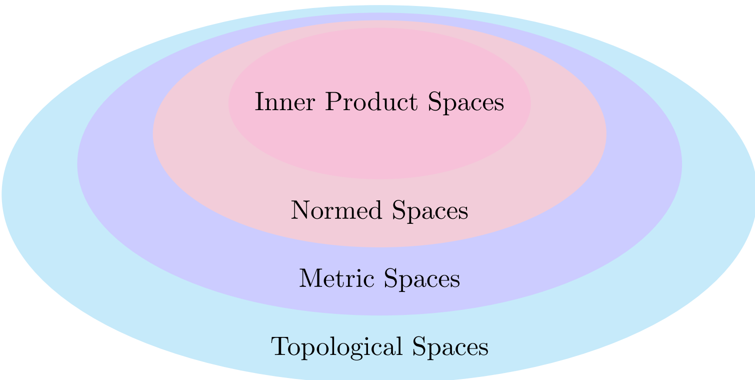

Figure 4: Overview of the generality of different spaces

Theorem 7.5: Pythagoras Theorem

Let V be an Euclidean/Hermitian vector space and $u, v \in V$ with $u \perp v$. Then
$$\|u + v\|^2 = \|u\|^2 + \|v\|^2.$$

Lemma 7.6: Induced Norm

Let V be an Euclidean space. Then

$$\forall\, u, v \in V:\ \langle u, v\rangle = \frac{1}{2}\big(\|u+v\|^2 - \|u\|^2 - \|v\|^2\big).$$

This means that $\|\cdot\|$ determines $\langle\cdot,\cdot\rangle$.
If V is an Hermitian space, we have that

$$\forall u, v \in V:\ \frac{1}{2}\big(\langle u, v\rangle + \overline{\langle u, v\rangle}\big) = \frac{1}{2}\big(\|u+v\|^2 - \|u\|^2 - \|v\|^2\big).$$

*Warning: There exist norms that do **not** come from scalar products.*

Definition 7.7: Complex Conjugated Matrix/Adjoint Matrix/Hermitian Matrix

1. Let $A \in M_{n\times m}(\mathbb{C})$. We write $\overline{A}$ for the matrix whose entry at i, j is $\overline{(A_{ij})}$, i.e.,

$$\big(\overline{A}\big)_{ij} := \overline{(A_{ij})}.$$

2. Let $A \in M_{n\times n}(\mathbb{C})$. Define

$$A^* := \overline{A}^T.$$

   A^* is called the **adjoint matrix of** A. (Has nothing to do with the classical adjoint $\operatorname{adj} A$).

3. $A \in M_{n\times n}(\mathbb{C})$ is called **Hermitian** if $A^* = A$ ($\iff A^T = \overline{A}$).

Important Example: Let $A \in M_{n\times n}(\mathbb{R})$. Define

$$\langle\cdot,\cdot\rangle_A : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}, \quad \langle v, w\rangle_A = v^T A\, w.$$

We can quickly see that $\langle\cdot,\cdot\rangle_A$ is bilinear. Assume now that A is symmteric. Then $\langle v, w\rangle_A = \langle w, v\rangle_A \quad \forall v, w \in V$. Indeed,

$$\langle w, v\rangle_A = w^T A\, v = w^T A^T v = \big(v^T A\, w\big)^T = \big(\langle v, w\rangle_A\big)^T = \langle v, w\rangle_A,$$

where we've used that a scalar can be viewed as a 1×1 matrix.

Is $\langle\cdot,\cdot\rangle_A$ in general a scalar product?

In general it is not, e.g., for

$$A = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

we have that

$$\langle \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \rangle_A = -1.$$

But for a scalar product we need that $\langle v, v\rangle > 0 \quad \forall\, v \neq 0$.

Definition 7.8: Positive Definite

A symmetric matrix $A \in M_{n\times n}(\mathbb{R})$ is called **positive definite** if

$$v^T A\, v > 0 \qquad \forall\, v \in \mathbb{R}^n \setminus \{0\}.$$

Proposition 7.9: Scalar Product with Matrices

Let $A \in M_{n\times n}(\mathbb{R})$. Then $\langle \cdot, \cdot \rangle_A$ is a scalar product if and only if A is positive definite.

The situation for Hermitian products is similar.

Definition 7.10: Positive Definite for Hermitian Matrices

Let $A \in M_{n\times n}(\mathbb{C})$ be a Hermitian matrix. We say that A is **positive definite** if

$$v^T A \overline{v} > 0 \qquad \forall\, v \in \mathbb{C}^n \setminus \{0\}.$$

Note that $v^T A\overline{v} \in \mathbb{R}$, b.c.

$$\overline{(v^T A \overline{v})} = \overline{v}^T \overline{A}\, v = \left(\overline{v}^T \overline{A}\, v\right)^T = v^T \overline{A}^T\, \overline{v} = v^T A\, \overline{v}.$$

Given a positive definite complex matrix A we can define a Hermitian product on $\mathbb{C}^n$, i.e.,

$$\langle v, w \rangle_A = v^T A\, \overline{w}.$$

Proposition 7.11: Hermitian Product with Matrices

Let $A \in M_{n\times n}(\mathbb{C})$. Then $\langle \cdot, \cdot \rangle_A$ is a Hermitian product if and only if A is positive definite.

7.1 Norms & Angles

Definition 7.12: Norm

Let V be a vector space over K, where $K = \mathbb{R}$ or $\mathbb{C}$. A **norm** on V is a function $\|\cdot\| : V \to \mathbb{R}$ s.t.:

1. Triangle inequality:
$$\|u + v\| \le \|u\| + \|v\| \qquad \forall\, u, v \in V.$$
2. Homogeneity:
$$\|\alpha v\| = |\alpha| \cdot \|v\| \qquad \forall\, \alpha \in K, v \in V.$$
3. Non-degeneracy:
$$\|v\| = 0 \quad \Longrightarrow \quad v = 0.$$

Theorem 7.13: Cauchy-Schwarz Inequality

Let $(V, \langle \cdot, \rangle)$ be a vector space endowed with an inner product $\langle \cdot, \cdot \rangle$. Then

$$|\langle u, v \rangle| \le \|u\| \cdot \|v\| \qquad \forall\, u, v \in V,$$

where equality holds if and only if u and v are linearly dependent.

Corollary 7.14: Induced Norm

Let V be a vector space and $\langle\cdot,\cdot\rangle$ and inner product on V, then

$$V \ni v \mapsto \|v\| := \sqrt{\langle v, v\rangle}$$

is a norm on V.

Definition 7.15: Angle

Let $(V, \langle\cdot,\cdot\rangle)$ be an Euclidean space. Let $u, v \in V$ be two non-zero vectors. By the Cauchy-Schwarz inequality 7.13, we have that

$$-1 \leq \frac{\langle u, v\rangle}{\|u\| \cdot \|v\|} \leq 1.$$

We define the **angle** between u and v to be the unique $\alpha \in [0, \pi]$ s.t.

$$\cos\alpha = \frac{\langle u, v\rangle}{\|u\| \cdot \|v\|} \left(= \langle \frac{u}{\|u\|}, \frac{v}{\|v\|} \rangle \right)$$

7.2 The Gram-Schmidt Orthogonalization Process

Theorem 7.16: Properties of Orthogonal Systems

Let $(V, \langle\cdot,\cdot\rangle)$ be an inner product space.

1. *If $S \subseteq V$ is an orthogonal system then S is linearly independent.*
2. *If $v_1, \ldots, v_n$ is an orthogonal system and $v \in \mathrm{Sp}(v_1, \ldots, v_n)$ and write $v = a_1 v_1 + \ldots + a_n v_n$, for $a_j \in K$. Then the coefficients a_j are given by*

$$a_j = \frac{\langle v, v_j\rangle}{\langle v_j, v_j\rangle} \qquad \forall\, 1 \leq j \leq n.$$

Corollary 7.17: Matrix for an Orthonormal Basis

Let V, W be a vector space over $K = \mathbb{R}$ or $\mathbb{C}$ and let $\langle\cdot,\cdot\rangle$ be an inner product on W. Let $\mathcal{B} = (v_1, \ldots, v_n)$ be a basis for V and $\mathcal{C} = (w_1, \ldots, w_m)$ be an orthonormal basis for W. Let $T : V \to W$ be a linear map. Write $[T]^{\mathcal{B}}_{\mathcal{C}} = (a_{ij})$. Then

$$a_{ij} = \langle Tv_j, w_i\rangle \qquad \forall\, i, j.$$

(If $\mathcal{C}$ is only a orthogonal basis then $a_{ij} = \langle Tv_j, w_i\rangle / \langle w_i, w_i\rangle$).

So having an ortho-gonal/normal basis for out spaces is useful!

Theorem 7.18: Gram-Schmidt Orthogonalization Process

Let $(V, \langle\cdot,\cdot\rangle)$ be an inner product space over $K = \mathbb{R}$ or $\mathbb{C}$ and let $n = \dim V$. Let $(v_1, \dots, v_n)$ be a basis for V. Define a new list of vectors $w_1, \dots, w_n$ by induction, as follows

$$w_1 := v_1,$$
$$w_j := v_j - \sum_{i=1}^{j-1} \frac{\langle v_j, w_i\rangle}{\langle w_i, w_i\rangle} \cdot w_i \quad \textit{for } j = 2, \dots, n.$$

Then:

1. *$(w_1, \dots, w_n)$ is an orthogonal basis for V.*
2. *$(\frac{w_1}{\|w_1\|}, \dots, \frac{w_n}{\|w_n\|})$ is an orthonormal basis for V.*
3. *$\forall 1 \le j \le n$ we have $\mathrm{Sp}(v_1, \dots, v_j) = \mathrm{Sp}(w_1, \dots, w_j) = \mathrm{Sp}(\frac{w_1}{\|w_1\|}, \dots, \frac{w_j}{\|w_j\|})$.*

Corollary 7.19: Existence of an Orthonormal Basis for Finite Dimensional Spaces

Every finite dimensional inner product space has an orthonormal basis.

7.3 The Orthogonal Complement

Let $(V, \langle\cdot,\cdot\rangle)$ be an inner product space over $K = \mathbb{R}$ or $\mathbb{C}$.

Definition 7.20: Orthogonal Complement

Let $\emptyset \neq S \subseteq V$ be a subset. We define the **orthogonal complement of** S as

$$S^\perp := \{v \in V \mid v \perp s \quad \forall s \in S\} = \{v \in V \mid \langle v, s\rangle = 0 \quad \forall s \in S\}.$$

When $S = \{v\}$ we write $v^\perp := \{v\}^\perp$.

Lemma 7.21: The Orthogonal Complement is a Subspace

Let $\emptyset \neq S \subseteq V$ be a subset. Then $S^\perp$ is a linear subspace.

Proof. $0 \in S^\perp$ b.c. $\langle 0, s\rangle = 0$ for every $s \in S$. Furthermore, if $\alpha, \beta \in K$ and $v, w \in S^\perp$ then for every $s \in S$ we have that

$$\langle \alpha v + \beta w, s\rangle = \alpha \underbrace{\langle v, s\rangle}_{=0} + \beta \underbrace{\langle w, s\rangle}_{=0} = 0.$$

Therefore, $\alpha v + \beta w \in S^\perp$. $\square$

Lemma 7.22: Further Properties of the Orthogonal Complement

Let $\emptyset \neq S, T \subseteq V$. Then:

1. $0^{\perp} = V$, $V^{\perp} = \{0\}$.
2. $S \cap S^{\perp}$ *is either* $\emptyset$ *or equal to* $\{0\}$.
3. *If* $S \subseteq T$ *then* $T^{\perp} \subseteq S^{\perp}$.
4. $\big(\mathrm{Sp}(S)\big)^{\perp} = S^{\perp}$.
5. $S \subseteq \big(S^{\perp}\big)^{\perp}$.

Theorem 7.23: The Orthogonal Complement is a Complement

Let V be an inner product space (possibly of infinte dimension), and $U \subseteq V$ a ***finite*** *dimensional subspace. Then $U^{\perp}$ is a complement of U in V (, i.e., $U + U^{\perp} = V$ and $U \cap U^{\perp} = \{0\}$).*

Proof. By Lemma 7.22 2., we have that $U \cap U^{\perp} = \{0\}$, since $U \cap U^{\perp} \neq \emptyset$. So it's enough to show that $U + U^{\perp} = V$.

Let $r := \dim U$ and pick an orthonormal basis $(e_1, \ldots, e_r)$ for U, which we can do by the Gram-Schmidt Orthogonalization Process 7.18 Let $v \in V$ be arbitrary. Define

$$P_U(v) := \langle v, e_1 \rangle e_1 + \ldots + \langle v, e_r \rangle e_r \in U.$$

We have that

$$v = \underbrace{P_U(v)}_{\in U} + (v - P_U(v)). \tag{7.1}$$

We claim that $v - P_U(v) \in U^{\perp}$. Indeed,

$$\langle v - P_U(v), e_i \rangle = \langle v, e_i \rangle - \sum_{k=1}^{r} \langle v, e_k \rangle \cdot \langle e_k, e_i \rangle = \langle v, e_i \rangle - \langle v, e_i \rangle = 0.$$

This holds for every $1 \leq i \leq r$, hence

$$v - P_U(v) \perp \mathrm{Sp}(e_1, \ldots, e_r) = U,$$

i.e., $v - P_U(v) \in U^{\perp}$. Going back to Equation (7.1) we see that

$$v = \underbrace{P_U(v)}_{\in U} + \underbrace{(v - P_U(v))}_{\in U^{\perp}}.$$

Since $v \in V$ was arbitrary, we have that $U + U^{\perp} = V$. □

Remark 7.24. *If $U \subseteq V$ is a finite dimensional subspace, then since $U^{\perp}$ is a complement of U in V, we have a canonical isomorphism $V \cong U \oplus U^{\perp}$.*

Definition 7.25: The Orthogonal Projection onto a Subspace

Let V be an inner prodct space and $U \subseteq V$ a finite dimensional subspace. We have seen that $U^\perp$ is a complement of U in V ($\Longrightarrow \forall\, v \in V,\ \exists!\, u \in U, w \in U^\perp$ s.t. $v = u + w$).
Define a map $P_U : V \to U$ as follows:

$$\forall\, v \in V,\ \exists!\, u \in U, w \in U^\perp \text{ s.t. } v = u + w.$$

We define

$$P_U(v) := u.$$

P_U is called **the orthogonal projection onto** U.

Proposition 7.26: Properties of P_U

P_U has the following properties:

1. *P_U is a linear map.*
2. $\operatorname{Im} P_U = U$, $\ker P_U = U^\perp$.
3. $\forall v \in V:\ v - P_U(v) \in U^\perp$, *i.e.,* $P_U + P_{U^\perp} = id_V$.
4. $\forall u \in U:\ P_U(u) = u$.
5. *If $e_1, \ldots, e_r$ is any orthonormal basis for U, then*

$$P_U(v) = \sum_{i=1}^{r} \langle v, e_i \rangle e_i.$$

Corollary 7.27: The Complement of the Complement of a Subspace

Let V be an inner product space and $U \subseteq V$ a finite dimensional subspace. Then

$$\left(U^\perp\right)^\perp = U.$$

Corollary 7.28: Dimension of a Subspace and its Complement

If V is a finite dimensional inner product space and $U \subseteq V$ is a subspace, then

$$\dim U + \dim U^\perp = \dim V.$$

7.4 Orthogonal & Unitary Matrices

Definition 7.29: Orthogonal & Unitary Matrices

1. A matrix $A \in M_{n\times n}(\mathbb{R})$ is called **orthogonal** if its columns form an orthonormal basis in $\mathbb{R}^n$, w.r.t. the standard scalar product of $\mathbb{R}^n$. We denote the set of orthognal $n \times n$ matrices by $O(n) \subseteq M_{n\times n}(\mathbb{R})$.

2. A matrix $A \in M_{n\times n}(\mathbb{C})$ is called **unitary** if its columns form an orthonormal basis in $\mathbb{C}^n$, w.r.t. the standard hermitian product on $\mathbb{C}^n$. We denote the set of all unitary $n \times n$ matrices by $U(n) \subseteq M_{n\times n}(\mathbb{C})$.

Lemma 7.30: Equivalent Conditions for Orthogonal/Unitary Matrices

- $A \in M_{n\times n}(\mathbb{R})$ *is orthogonal* $\iff A^T A = I_n \iff AA^T = I_n \iff A^{-1} = A^T$.
- $A \in M_{n\times n}(\mathcal{C})$ *is unitary* $\iff A^* A = I_n \iff AA^* = I_n \iff A^{-1} = A^*$.

Corollary 7.31

1. *Let $A \in M_{n\times n}(\mathbb{R})$. Then $A \in O(n) \iff$ the rows of A form an orthonormal basis for $\mathbb{R}^n$.*

2. *Let $A \in M_{n\times n}(\mathbb{C})$. Then $A \in U(n) \iff$ the rows of A form an orthonormal basis for $\mathbb{C}^n$.*

7.5 QR-Decomposition

Theorem 7.32: QR-Decomposition

1. *Let $A \in GL_n(\mathbb{R})$. Then $\exists Q \in O(n)$ and an upper triangular matrix $R \in M_{n\times n}(\mathbb{R})$ s.t.*
$$A = QR.$$

2. *Let $A \in GL_n(\mathbb{C})$. Then $\exists Q \in U(n)$ and an upper triangular matrix $R \in M_{n\times n}(\mathbb{C})$ s.t.*
$$A = QR.$$

Proof. We'll prove 1 & 2 together.

Write
$$A = \begin{pmatrix} | & & | \\ v_1 & \dots & v_n \\ | & & | \end{pmatrix}$$
with $v_k \in K^n_{\text{col}}$, where $K = \mathbb{R}$ or $\mathbb{C}$. Since $A \in GL_n(K)$, $v_1, \dots, v_n$ form a basis for K^n_{col}. Apply the Gram-Schmidt orthogonalization process 7.18 to $v_1, \dots, v_n$ and obtain an orthonormal basis $e_1, \dots, e_n$ for K^n_{col}. Recall that $\forall 1 \le j \le n :\ \mathrm{Sp}(v_1, \dots, v_j) = \mathrm{Sp}(e_1, \dots, e_j)$. By Theorem 7.16 we also have that
$$v_j = \sum_{i=1}^{j} \langle v_j, e_i \rangle e_i \qquad \forall 1 \le j \le n.$$

Therefore, we can write

$$\underbrace{\begin{pmatrix} | & & | & & | \\ v_1 & \dots & v_j & \dots & v_n \\ | & & | & & | \end{pmatrix}}_{=A} = \underbrace{\begin{pmatrix} | & & | & & | \\ e_1 & \dots & e_j & \dots & e_n \\ | & & | & & | \end{pmatrix}}_{=Q} \cdot \underbrace{\begin{pmatrix} \langle v_1, e_1\rangle & \langle v_2, e_1\rangle & \dots & \langle v_j, e_1\rangle & \dots & \langle v_n, e_1\rangle \\ 0 & \langle v_2, e_2\rangle & \dots & \langle v_j, e_2\rangle & \dots & \langle v_n, e_2\rangle \\ \vdots & & \ddots & \ddots & \vdots & & \vdots \\ \vdots & & & \ddots & \langle v_j, e_j\rangle & \dots & \langle v_n, e_j\rangle \\ \vdots & & & & \ddots & \ddots & \vdots \\ 0 & \dots & \dots & \dots & 0 & \langle v_n, e_n\rangle \end{pmatrix}}_{=R},$$

as desired. $\square$

Extension of QR-Decomposition for <u>all</u> matrices $A \in M_{n\times n}(K)$.

Theorem 7.33: Extension of QR-Decomposition

Let $A \in M_{n\times n}(K)$, where $K = \mathbb{R}$ or $\mathbb{C}$ and let $r := \operatorname{rank}(A)$. Then $\exists Q \in O(n)/U(n)$ and

$$R = \begin{pmatrix} C & * \\ 0 & 0 \end{pmatrix} \in M_{n\times n}(K),$$

where $C \in M_{r\times r}(K)$ is upper triangular, s.t.

$$A = QR.$$

8 Dual Spaces & Inner Products

Let V be a vector space over K (any field). Recall that we have the dual space $V^* := \operatorname{Hom}(V, K) =$ the space of linear maps $\varphi : V \to K$ (functionals). V^* is also a vector space over K.

Consider now an inner product space $(V, \langle\cdot,\cdot\rangle)$ over $K = \mathbb{R}$ or $\mathbb{C}$. Let $u \in V$. Define

$$\varphi_u : V \to K, \quad V \ni v \mapsto \langle v, u\rangle \in K.$$

φ_u is linear since, v is the first argument in the scalar/hermitian product.

Theorem 8.1: Canonical 'Almost' Isomorphism between V and V^*

*Let $(V, \langle\cdot,\cdot\rangle)$ be a finite dimensional inner product space over $K = \mathbb{R}$ or $\mathbb{C}$. Let $\varphi \in V^*$. Then there exists a **unique** $u \in V$ s.t. $\varphi = \varphi_u$, i.e., $\varphi(v) = \langle v, u\rangle \quad \forall v \in V$.*
In fact, for $K = \mathbb{R}$ the map $\Phi : V \to V^*$, given by $\Phi(u) := \varphi_u$ is an isomorphism.
For $K = \mathbb{C}$, this map is injective and surjective but only $\mathbb{C}$-anti linear, i.e.,

$$\Phi(u_1 + u_2) = \Phi(u_1) + \Phi(u_2) \qquad \forall u_1, u_2 \in V$$
$$\Phi(cu) = \bar{c} \cdot \Phi(u) \qquad \forall c \in \mathbb{C}, u \in V.$$

Proof. Let $n := \dim V$ and fix an orthonormal basis $e_1, \dots, e_n$ for V, which exists by the Gram-

Schmidt orthogonalization process 7.18. Let $\varphi \in V^*$. Let $v \in V$. By Theorem 7.16 we have that

$$v = \sum_{i=1}^{n} \langle v, e_i \rangle e_i.$$

Therefore,

$$\varphi(v) = \varphi\Big(\sum_{i=1}^{n} \langle v, e_i \rangle e_i\Big) = \sum_{i=1}^{n} \langle v, e_i \rangle\, \varphi(e_i) = \langle v, \sum_{i=1}^{n} \overline{\varphi(e_i)}\, e_i \rangle.$$

Define

$$u := \sum_{i=1}^{n} \overline{\varphi(e_i)}\, e_i.$$

Then we have that $\varphi(v) = \varphi_u(v) \quad \forall v \in V$.

We'll show now the uniqueness of u. Suppose that $\varphi_{u_1} = \varphi_{u_2}$ for some $u_1, u_2 \in V$. Then we have that

$$\langle v, u_1 \rangle = \langle v, u_2 \rangle \qquad \forall v \in V.$$

By a previous result, this implies that $u_1 = u_2$. This shows that Φ is surjective and injective.

For $K = \mathbb{R}$, we have that the scalar product $\langle \cdot, \cdot \rangle$ is bilinear. So it follows that Φ is also linear, hence Φ is an isomorphism.

For $K = \mathbb{C}$, we have that the hermitian product $\langle \cdot, \cdot \rangle$ is $\mathbb{C}$-anti linear in the second variable. So, it follows that Φ is also $\mathbb{C}$-anti linear. $\square$

So, at least for $K = \mathbb{R}$, a scalar product on V gives us a canonical isomorphism $V \cong V^*$. (But this isomorphism depends on the scalar product!)

8.1 The Adjoint Map

(Everything in this Subsection should be learned by heart.) Let V, W be vector spaces over K. Let $T : V \to W$ be a linear map. We have seen in Subsection 4.2 that T induces a linear map (the dual map)

$$T^* : W^* \to V^*, \quad T^*(\varphi) := \varphi \circ T.$$

Assume now that $(V, \langle \cdot, \cdot \rangle_V)$ and $(W, \langle \cdot, \cdot \rangle_W)$ are inner product spaces over $K = \mathbb{R}$ or $\mathbb{C}$. Assume also that $\underline{\dim V < \infty}$. Given T as above, we can define $T' : W \to V$ using our 'canonical' identifications $W \cong W^*$, $V \cong V^*$. (No problem for $K = \mathbb{R}$, but we have to be careful with $K = \mathbb{C}$.)

$$\begin{tikzcd}
W^* \arrow[r, "T^*"] & V^* \arrow[d, "\Phi_V^{-1}"] \\
W \arrow[u, "\Phi_W"] \arrow[r, dashed, "T'"'] & V
\end{tikzcd}$$

i.e. $T' := \Phi_V^{-1} \circ T^* \circ \Phi_W$. We've also used that $\dim V < \infty$, b.c. then Φ_V is bijective.

What properties does T' have? Note that T' is uniquely determined by

$$\begin{aligned}
\Phi_V \circ T' = T^* \circ \Phi_W \quad &\iff \quad \forall w \in W: \ \Phi_V \circ T'(w) = T^* \circ \Phi_W(w) \\
&\iff \quad \varphi_{T'(w)} = T^* \circ \varphi_w \ (\in V^*) \\
&\iff \quad \forall v \in V: \ \varphi_{T'(w)}(v) = \big(T^* \circ \varphi_w\big)(v) \\
&\iff \quad \langle v, T'(w)\rangle_V = \varphi_w\big(T(v)\big) = \langle T(v), w\rangle_W.
\end{aligned}$$

In other words,

$$\langle Tv, w\rangle_W = \langle v, T'w\rangle_V \qquad \forall\, v \in V, w \in W.$$

Claim: T' is a linear map.

Proof. For $K = \mathbb{R}$, Φ_W, Φ_V and T^* are linear. Therefore, T' is linear too.

For $K = \mathbb{C}$, Φ_W, Φ_V are additive but only $\mathbb{C}$-anti linear. Therefore, Φ_V^{-1} is also additive and $\mathbb{C}$-anti linear. Indeed, let $y_1, y_2 \in V^*$. Since Φ_V is bijective, we can find unique $x_1, x_2 \in V$ s.t. $\Phi_V(x_1) = y_1$ and $\Phi_V(x_2) = y_2$. Since Φ_V is additive we have that

$$\Phi_V^{-1}(y_1 + y_2) = \Phi_V^{-1}\big(\Phi_V(x_1) + \Phi_V(x_2)\big) = \Phi_V^{-1}\big(\Phi_V(x_1 + x_2)\big) = x_1 + x_2 = \Phi_V^{-1}(y_1) + \Phi_V^{-1}(y_2).$$

Similarly, let $c \in \mathbb{C}, y \in V^*$. We again can find a unique $x \in V$ s.t. $\Phi_V(x) = y$. Since Φ_V is $\mathbb{C}$-anti linear we have that

$$\Phi_V^{-1}(cy) = \Phi_V^{-1}\big(c \cdot \Phi_V(x)\big) = \Phi_V^{-1}\big(\Phi_V(\bar{c}x)\big) = \bar{c} \cdot x = \bar{c} \cdot \Phi_V^{-1}(y).$$

So, we have shown that Φ_V^{-1} is additive and $\mathbb{C}$-anti linear as well.

Now, we return to the proof of the claim. We recall that T^* is linear. Let $w_1, w_2 \in W$ be arbitrary. Since T^* is in particular additive and $T' = \Phi_V^{-1} \circ T^* \circ \Phi_W$, we can conclude that T' is also additive. Now, let $c \in \mathbb{C}, w \in W$. Since T^* is linear and Φ_V^{-1}, Φ_W are $\mathbb{C}$-anti linear we have that

$$\begin{aligned}
T'(cw) = \Phi_V^{-1} \circ T^* \circ \Phi_W(cw) &= \Phi_V^{-1}\Big(T^*\big(\bar{c} \cdot \Phi_W(w)\big)\Big) = \Phi_V^{-1}\Big(\bar{c} \cdot T^*\big(\Phi_W(w)\big)\Big) \\
&= c \cdot \Phi_V^{-1}\Big(T^*\big(\Phi_W(w)\big)\Big) = c \cdot T'(w).
\end{aligned}$$

This proves the claim. $\square$

Notation: The linear map $T' : W \to V$ is denoted many times by $T^* : W \to V$. This makes sense b.c. of the 'canonical' identifications. But we have to assume additionally that $\dim W < \infty$. The map $T^* : W \to V$ is called the **adjoint** of T.

8.2 Basic Properties of Adjoint Maps

Lemma 8.2: Properties of Adjoint Maps

Let V, W, U be finite dimensional inner product spaces over $K = \mathbb{R}$ or $\mathbb{C}$. Let $S, T : V \to W$ and $R : W \to U$ be linear maps. Then:

1. T^* *is a linear map.*
2. $(S + T)^* = S^* + T^*$.
3. $\forall \lambda \in K : (\lambda T)^* = \overline{\lambda} T^*$.
4. $(T^*)^* = T$.
5. $(id_V)^* = id_V$.
6. $(R \circ T)^* = T^* \circ R^*$.

Proof. Let $y_1, y_2 \in V$. We'll repeatedly use that if $\forall x \in V : \langle x, y_1\rangle = \langle x, y_2\rangle$, then $y_1 = y_2$. This is because

$$\begin{aligned} \forall x \in V : \langle x, y_1\rangle = \langle x, y_2\rangle &\implies \forall x \in V : \langle x, y_1 - y_2\rangle = 0 \\ \implies y_1 - y_2 = 0 &\implies y_1 = y_2, \end{aligned}$$

where we've used 2. of Lemma 7.3

1) Has already been proven in Subsection 8.1.

2) Let $w \in W, v \in V$. By the defining properties of T^*, S^* we have that

$$\langle v, (S + T)^* w\rangle = \langle (S + T)v, w\rangle = \langle Sv, w\rangle + \langle Tv, w\rangle = \langle v, S^* w\rangle + \langle v, T^* w\rangle = \langle v, (S^* + T^*)w\rangle.$$

This holds for every $v \in V$. Therefore, we have that

$$(S + T)^* w = (S^* + T^*)w.$$

Since, this holds for every $w \in W$, we conclude that $(S + T)^* = (S^* + T^*)$.

3) Let $\lambda \in K$, $v \in V$ and $w \in W$. Then we have that

$$\begin{aligned} \langle v, (\lambda T)^* w\rangle &= \langle (\lambda T)v, w\rangle = \lambda\langle Tv, w\rangle = \lambda\langle v, T^* w\rangle \\ &= \langle v, \overline{\lambda} T^* w\rangle. \end{aligned}$$

Since this holds for every $v \in V$, we get that

$$(\lambda T)^* w = \overline{\lambda} T^* w.$$

Since this holds for every $w \in W$, we conclude that $(\lambda T)^* = \overline{\lambda} T^*$.

4) Let $v \in V$, $w \in W$. Then we have that

$$\langle w, (T^*)^* v\rangle = \langle T^* w, v\rangle = \overline{\langle v, T^* w\rangle} = \overline{\langle Tv, w\rangle} = \langle w, Tv\rangle.$$

This holds for every $w \in W$. Therefore, we have that $(T^*)^* v = Tv$. And since this holds for every $v \in V$, we get that $(T^*)^* = T$.

5) Let $v \in V$. Then we have that

$$\langle v, (id_V)^* v\rangle = \langle id_V v, v\rangle = \langle v, v\rangle = \langle v, id_V v\rangle.$$

Since this holds for every $v \in V$ we conclude that

$$(id_V)^* v = id_V v \qquad \Longrightarrow \qquad (id_V)^* = id_V.$$

6) Let $v \in V$, $u \in U$. Then we have that

$$\begin{aligned}\langle v, (R \circ T)^* u\rangle &= \langle (R \circ T)v, u\rangle = \langle R\big(T(v)\big), u\rangle = \langle Tv, R^* u\rangle \\ &= \langle v, T^*\big(R^*(u)\big)\rangle = \langle v, T^* \circ R^* u\rangle.\end{aligned}$$

This holds for every $v \in V$. Therefore,

$$(R \circ T)^* u = T^* \circ R^* u.$$

Since this holds for every $u \in U$, we conclude that $(R \circ T)^* = T^* \circ R^*$. □

Lemma 8.3: Kernel & Image of Adjoint Maps

Let V, W be finite dimensional inner product spaces. Let $T : V \to W$ and $T^* : W \to V$ be the adjoint of T. Then

1. $\ker T^* = (\operatorname{Im} T)^\perp$.
2. $\operatorname{Im} T^* = (\ker T)^\perp$.
3. $\ker T = (\operatorname{Im} T^*)^\perp$.
4. $\operatorname{Im} T = (\ker T^*)^\perp$.

8.3 Matrix Representation of Adjoint Maps

Proposition 8.4: Matrix Representation of Adjoint Maps

Let V, W be finite dimensional inner product spaces and $TV \to W$ a linear map. Let $\mathcal{B} = (e_1, \dots, e_n)$ be an orthonormal basis for V and $\mathcal{C} = (f_1, \dots, f_m)$ an orthonormal basis for W. Then

$$[T^*]^{\mathcal{C}}_{\mathcal{B}} = \left(\overline{[T]^{\mathcal{B}}_{\mathcal{C}}}\right)^T = \left([T]^{\mathcal{B}}_{\mathcal{C}}\right)^*,$$

where the last asterisk denotes the adjoint matrix.

Corollary 8.5

Let $A \in M_{m<timesn}(K)$, where $K = \mathbb{R}$ or $\mathbb{C}$ and let $T_A : K^n \to K^m$ be the linear map $T_A(v) = Av$. Endow K^n and K^m with their standard inner product. Then

$$(T_A)^* = T_{A^*}.$$

Proposition 8.6: Properties of the Adjoint Matrix

Let $K = \mathbb{R}$ or $\mathbb{C}$, $A, B \in M_{n\times m}(K)$, $C \in M_{m\times p}(K)$. Then:

1. $(A+B)^* = A^* + B^*$.
2. $(\lambda A)^* = \overline{\lambda} A^* \qquad \forall\, \lambda \in K$.
3. $(A^*)^* = A$.
4. $I_n^* = I_n$.
5. $(AC)^* = C^* A^*$.

9 Spectral Theorems

Definition 9.1: Orthogonal Diagonalizability

Let V be an inner product space and $T : V \to V$ and endomorphism. We say that T is **orthogonally diagonalizable** if there exists an orthonormal basis for V consisting of eigenvectors of T.

Lemma 9.2: Consequence of Orthogonal Diagonalizability

If T is orthogonally diagonalizable, then

$$T \circ T^* = T^* \circ T.$$

Definition 9.3: Normal Endomorphism/Matrix

Let V be an inner product space. An endomorphism $T : V \to V$ is called **normal** if

$$T \circ T^* = T^* \circ T.$$

A matrix $A \in M_{n\times n}(K)$ $(K = \mathbb{R}, \mathbb{C})$ is called **normal** if

$$AA^* = A^*A.$$

This is equivalent to saying that $T_A : K^n \to K^n$ is normal, if we endow K^n with its standard inner product.

We've seen in Lemma 9.2 that if $T : V \to V$ is orthogonally diagonalizable, then T is normal. Is the converse statement also true?

Over $K = \mathbb{R}$ the answer is **no!**, e.g.,

$$A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

$T_A : \mathbb{R}^2 \to \mathbb{R}^2$ is a clockwise rotation by 90°, so there exist no eigenvalues in $\mathbb{R}$. But $AA^* = A^*A = I$.

Lemma 9.4: Properties of a Normal Endomorphism

Let V be a finite dimensional inner product space over $K = \mathbb{R}$ or $\mathbb{C}$. Let $T : V \to V$ be a normal endomorphism. Then:

1. $\forall\, v \in V : \ \|Tv\| = \|T^*v\|$.
2. $\forall\, \lambda \in K$ *the endomorphism* $T - \lambda \cdot id_V$ *is also normal.*
3. *If v is an eigenvector of T with eigenvalue λ, then v is also an eigenvector of T^* with eigenvalue $\overline{\lambda}$.*
4. *Let v_1, v_2 be eigenvectors of T with eigenvalues λ_1, λ_2 respectively. If $\lambda_1 \neq \lambda_2$, then* $\langle v_1, v_2 \rangle = 0$.

Proof. **1)** Let $v \in V$. Then by the properties of the adjoint map T^* and since T is normal, we have that

$$\begin{aligned} \|Tv\|^2 &= \langle Tv, Tv \rangle = \langle v, T^* \circ Tv \rangle = \langle v, T \circ T^* v \rangle \\ &= \overline{\langle T \circ T^* v, v \rangle} = \overline{\langle T^* v, T^* v \rangle} = \langle T^* v, T^* v \rangle = \|T^* v\|^2. \end{aligned}$$

Since the norm is non-negative, we conclude that

$$\|Tv\| = \|T^*v\|.$$

2) Let $\lambda \in K$. By properties of adjoint maps we have that

$$(T - \lambda \cdot id_V) \circ (T - \lambda \cdot id_V)^* = (T - \lambda \cdot id_V) \circ (T^* - \overline{\lambda} \cdot id_V) = T \circ T^* - \overline{\lambda} \cdot T - \lambda \cdot T^* + \lambda\overline{\lambda} \cdot id_V, \tag{9.1}$$

and

$$(T - \lambda \cdot id_V)^* \circ (T - \lambda \cdot id_V) = (T^* - \overline{\lambda} \cdot id_V) \circ (T - \lambda \cdot id_V) = T^* \circ T - \lambda \cdot T^* - \overline{\lambda} \cdot T + \overline{\lambda}\lambda \cdot id_V. \tag{9.2}$$

By the normality of T we get that Equation (9.1) = (9.2), so $(T - \lambda \cdot id_V)$ is normal.

3) Let v be an eigenvector of T with eigenvalue λ. Then $(T - \lambda \cdot id_V)v = 0$. By **2)**, $(T - \lambda \cdot id_V)$ is normal. Hence by **1)** we have

$$\begin{aligned} & \|(T^* - \overline{\lambda} \cdot id_V)v\| = \|(T - \lambda \cdot id_V)^* v\| = \|(T - \lambda \cdot id_V)v\| = 0 \\ \implies \quad & (T^* - \overline{\lambda} \cdot id_V)v = 0. \end{aligned}$$

This proves that v is an eigenvector of T^* with eigenvalue $\overline{\lambda}$.

4) Let v_1, v_2 be eigenvectors of T with eigenvalues λ_1, λ_2 respectively, where $\lambda_1 \neq \lambda_2$. Then, by properties of adjoint maps and **3)**, we have that

$$\begin{aligned} & \lambda_1 \langle v_1, v_2 \rangle = \langle \lambda_1 v_1, v_2 \rangle = \langle T v_1, v_2 \rangle = \langle v_1, T^* v_2 \rangle = \langle v_1, \overline{\lambda}_2 v_2 \rangle = \lambda_2 \langle v_1, v_2 \rangle \\ \implies \quad & (\lambda_1 - \lambda_2)\langle v_1, v_2 \rangle = 0. \end{aligned}$$

Since $\lambda_1 \neq \lambda_2$, we conclude that $\langle v_1, v_2 \rangle = 0$. ∎

Theorem 9.5: Spectral Theorem over $\mathbb{C}$

Let V be a finite dimensional inner product space over $\mathbb{C}$ and $T : V \to V$ a linear map. Then T is orthogonally diagonalizable if and only if T is normal.

Proof. $\underline{\Longrightarrow}$: Let $\mathcal{B}$ be an orthonormal basis for V s.t. all the elements in $\mathcal{B}$ are eigenvectors of T. Therefore, $[T]^{\mathcal{B}}_{\mathcal{B}}$ is a diagonal matrix. By Proposition 8.4 we have that

$$[T^*]^{\mathcal{B}}_{\mathcal{B}} = \left([T]^{\mathcal{B}}_{\mathcal{B}}\right)^*.$$

So, $[T^*]^{\mathcal{B}}_{\mathcal{B}}$ is also diagonal. Since diagonal matrices commute and by a property of matrix multiplication, we have that

$$\begin{aligned} & [T]^{\mathcal{B}}_{\mathcal{B}} \cdot [T^*]^{\mathcal{B}}_{\mathcal{B}} = [T^*]^{\mathcal{B}}_{\mathcal{B}} \cdot [T]^{\mathcal{B}}_{\mathcal{B}} \\ \Longrightarrow \quad & [T \circ T^*]^{\mathcal{B}}_{\mathcal{B}} = [T^* \circ T]^{\mathcal{B}}_{\mathcal{B}}. \end{aligned}$$

By a result from the lecture, we conclude that $T \circ T^* = T^* \circ T$.

$\underline{\Longleftarrow}$: By the fundamental Theorem of Algebra 6.26, T is trigonalizable, i.e., there exists a basis $\mathcal{C} = (u_1, \dots, u_n)$ for V s.t. $[T]^{\mathcal{C}}_{\mathcal{C}}$ is an upper triangular matrix. By applying the Gram-Schmidt orthogonalization process 7.18 to the basis $\mathcal{C}$ we obtain a new basis $\mathcal{B} = (v_1, \dots, v_n)$ which is orthonormal and s.t. $[T]^{\mathcal{B}}_{\mathcal{B}}$ is still upper triangular, b.c. $\mathrm{Sp}(u_1, \dots, u_j) = \mathrm{Sp}(v_1, \dots, v_j)$ for every $1 \le j \le n$.

We claim that $[T]^{\mathcal{B}}_{\mathcal{B}}$ is actually diagonal. To see this write $[T]^{\mathcal{B}}_{\mathcal{B}} = (a_{ij})$. Since $\mathcal{B}$ is orthonormal, we have $\|Tv_1\|^2 = |a_{11}|^2$. Indeed,

$$\langle Tv_1, Tv_1 \rangle = \langle a_{11}v_1, a_{11}v_1 \rangle = a_{11}\,\overline{a_{11}}\,\underbrace{\langle v_1, v_1 \rangle}_{=1} = |a_{11}|^2.$$

Since $[T^*]^{\mathcal{B}}_{\mathcal{B}} = \left([T]^{\mathcal{B}}_{\mathcal{B}}\right)^*$ we also have

$$\begin{aligned} & [T^*]^{\mathcal{B}}_{\mathcal{B}} = \begin{pmatrix} \bar{a}_{11} & 0 & \dots & 0 \\ \bar{a}_{12} & \bar{a}_{22} & \ddots & \vdots \\ \vdots & \vdots & \ddots & 0 \\ \bar{a}_{1n} & \bar{a}_{2n} & \dots & \bar{a}_{nn} \end{pmatrix} \\ \Longrightarrow \quad & \|T^*v_1\|^2 = \sum_{k=1}^{n} |\bar{a}_{1k}|^2 = |a_{11}|^2 + |a_{12}|^2 + \dots + |a_{1n}|^2. \end{aligned}$$

As T is normal, we have that $\|Tv_1\|^2 = \|T^*v_1\|^2$, hence $a_{12} = \dots = a_{1n} = 0$. This shows that

$$[T]^{\mathcal{B}}_{\mathcal{B}} = \begin{pmatrix} \bar{a}_{11} & 0 & \dots & 0 \\ 0 & * & \dots & * \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \dots & 0 & * \end{pmatrix}.$$

By the same argumentation, we have that $\|Tv_2\|^2 = |a_{22}|^2$ and $\|T^*v_2\|^2 = |a_{22}|^2 + |a_{23}|^2 + \dots |a_{2n}|^2$. As $\|Tv_2\| = \|T^*v_2\|$ we obtain $a_{23} = \dots = a_{2n} = 0$.

We can continue the proof by induction. Suppose the claim holds for k. So, we have that

$\|Tv_{k+1}\|^2 = |a_{k+1,k+1}|^2$, as $a_{l,k+1} = 0 \quad \forall l < k+1$. Since $[T^*]_{\mathcal{B}}^{\mathcal{B}} = \left([T]_{\mathcal{B}}^{\mathcal{B}}\right)^*$, we have that

$$\|T^* v_{k+1}\|^2 = |a_{k+1,k+1}|^2 + \ldots + |a_{k+1,n}|^2.$$

Since T is normal, we get that $\|T^* v_{k+1}\| = \|Tv_{k+1}\|$. Therefore, we conclude that

$$|a_{k+1,k+2}| = \ldots = |a_{k+1,n}| = 0.$$

This proves the claim. □

Remark 9.6. *In the proof above we used $K = \mathbb{C}$ only, when we said that T is trigonalizable. Therefore, we arrive at the following corollary.*

Corollary 9.7

Let V be a finite dimensional inner product space over $\mathbb{R}$ and $T : V \to V$ an endomorphism, which is normal. If T is trigonalizable, then T is orthogonally diagonalizable.

We've seen that over $\mathbb{R}$ $T \in \mathrm{End}(V)$ is normal $\not\Longrightarrow$ T is orthogonally diagonalizable.

Definition 9.8: Self Adjoint

Let V be an inner product space over $K = \mathbb{R}$ or $\mathbb{C}$, and $T \in \mathrm{End}(V)$. T is called **self adjoint** if $T^* = T$.
A matrix $A \in M_{n\times n}(K)$ is called **self adjoint** if $A^* = A$. (Over $\mathbb{R}$ a self adjoint matrix is a symmetric matrix.)

Lemma 9.9: A Condition for Self Adjoint

Let V be a finite dimensional inner product space over $\mathbb{R}$ and $T \in \mathrm{End}(V)$ which is orthogonally diagonalizable. Then T is self adjoint.

Lemma 9.10

Let V be an inner product space over $K = \mathbb{R}$ or $\mathbb{C}$. Then $T \in \mathrm{End}(V)$ is self adjoint if and only if for every orthonormal basis $\mathcal{B}$ of V, the matrix $[T]_{\mathcal{B}}^{\mathcal{B}}$ is self adjoint.

Theorem 9.11: Spectral Theorem over $\mathbb{R}$

Let V be an Euclidean space(, i.e., inner prduct space over $\mathbb{R}$) of finite dimension. Then T is orthogonally diagonalizable if and only if T is self adjoint.

For the proof of the above Theorem the following Lemma is needed.

Lemma 9.12: Properties of Self Adjoint Maps

Let V be an inner productd space over $K = \mathbb{R}$ or $\mathbb{C}$ and $T \in \mathrm{End}(V)$ be self adjoint. Then

1. *All eigenvalues of T are real.*

2. *p_T factors into a product of linear factors over K.*

In the Figure below we illustrated the logical relations regarding the Spectral theorem.

$\mathbb{C}$

$$\begin{tikzcd} & \text{orthogonally diagonalizable} \arrow[dl, Leftrightarrow] & \\ \text{normal} & & \text{self adjoint} \arrow[ll, Rightarrow] \arrow[ul, Rightarrow] \end{tikzcd}$$

$\mathbb{R}$

$$\begin{tikzcd} & \text{orthogonally diagonalizable} \arrow[dl, Rightarrow] \arrow[dr, Leftrightarrow] & \\ \text{normal} & & \text{self adjoint} \arrow[ll, Rightarrow] \end{tikzcd}$$

Theorem 9.13: Spectral Theorem for Matrices

1. *If $A \in M_{n\times n}(\mathbb{C})$ is normal, then $\exists U \in U(n)$ s.t.*

$$\underbrace{U^{-1}}_{=U^*} AU$$

is diagonal.

2. *If $A \in M_{n\times n}(\mathbb{R})$ is a symmetric matrix, then $\exists O \in O(n)$ s.t.*

$$\underbrace{O^{-1}}_{=O^T} AO$$

is diagonal.

10 Orthogonal & Unitary maps / Isometries

Definition 10.1: Linear Isomerty

Let V, W be inner product spaces over $K = \mathbb{R}$ or $\mathbb{C}$. A map $T : V \to W$ is called a **linear isometry** if T is linear and

$$\forall v_1, v_2 \in V : \langle Tv_1, Tv_2\rangle_W = \langle v_1, v_2\rangle_V,$$

i.e., T preserves the inner product.

Remark 10.2. *If $T : V \to W$ is a linear isometry, then:*

1. $\forall v \in V : \|Tv\| = \|v\|$.
2. $\forall p, q \in V : d(Tp, Tq) = d(p, q)$, *where* $d(p, q) := \|p - q\|$.
3. *T is injective. Indeed, if $v \in \ker T$, then*

$$0 = \|Tv\| = \|v\| \implies v = 0.$$

So, $\ker T = \{0\}$.

From now on we'll concentrate on the case $W = V$, i.e., $T \in \mathrm{End}(V)$. If such a map T is an isometry we also call it **orthogonal endomorphism** if $K = \mathbb{R}$, or **unitary endomorphism** if

$K = \mathbb{C}$.

Theorem 10.3: Equivalent Statements to Isometry

Let V be a finite dimensional inner product space over $K = \mathbb{R}$ or $\mathbb{C}$, and $T \in \mathrm{End}(V)$. The following are equivalent:

1. *T is an isometry.*
2. *$\forall\, v \in V :\ \|Tv\| = \|v\|$.*
3. *For every orthonormal system $e_1, \ldots, e_k$ of V, the list of vectors $Te_1, \ldots, Te_k$ is also an orthonormal system.*
4. *There exists an orthonormal basis $e_1, \ldots, e_n$ of V s.t. $Te_1, \ldots, Te_n$ is also an orthonormal basis.*
5. *$T\,T^* = id_V$.*
6. *$T^*T = id_V$.*
7. *$T^* = T^{-1}$.*
8. *T^* is an isometry.*

Back to matrices.

Proposition 10.4: Equivalent Condition for a Orthogonal/Unitary Map

Let V be an n-dimensional inner product space over $K = \mathbb{R}$ (respectively $\mathbb{C}$). Let $\mathcal{B}$ be an orthonormal basis for V. Let $T \in \mathrm{End}(V)$. Then T is orthogonal (respectively unitary) if and only if $[T]^{\mathcal{B}}_{\mathcal{B}} \in O(n)$ (respectively $[T]^{\mathcal{B}}_{\mathcal{B}} \in U(n)$).

Note that $\forall\, A \in O(n)$ we have $\det A = \pm 1$. Indeed,

$$1 = \det I = \det(A\,A^T) = \det A \cdot \det A.$$

Similarly, $\forall\, A \in U(n)$, $|\underbrace{\det A}_{\in\,\mathbb{C}}| = 1$. Indeed,

$$1 = \det(A\,A^*) = \det A \cdot \det \overline{A}^T = \det A \cdot \det \overline{A} = \det A \cdot \overline{\det A} = |\det A|^2.$$

Definition 10.5: Special Orthogonal/Unitary Group

The set

$$SO(n) := \{A \in O(n) \mid \det A = 1\} \subseteq O(n) \subseteq GL_n(\mathbb{R})$$

is called the **special orthogonal group**.
The set

$$SU(n) := \{A \in U(n) \mid \det A = 1\} \subseteq U(n) \subseteq GL_n(\mathbb{C})$$

is called the **special unitary group**.

Proposition 10.6: Group Structure of $O(n)/SO(n)/U(n)/SU(n)$

Let $G = O(n), SO(n), U(n), SU(n)$. Then:

1. $I_n \in G$.
2. $\forall\, A, B \in G:\ A \cdot B \in G$.
3. $\forall\, A \in G:\ A^{-1} \in G$.

10.1 Classification of the Elements of $O(2), O(3)$ etc.

Lemma 10.7: Eigenvalue of an Orthogonal Matrix

Let $A \in O(n)$. If λ is an eigenvalue of A, then $\lambda = \pm 1$.

Proposition 10.8: Classification of the Elements of $O(2)$

Let $A \in O(2)$.

1. *If $\det A = 1$, then $\exists\, \theta \in [0, 2\pi)$ s.t.*

$$A = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} := R_\theta.$$

The map $T_A : \mathbb{R}^2 \to \mathbb{R}^2$ is an anti-clockwise rotation by the angle θ.

2. *If $\det A = -1$, then there exists a basis $\mathcal{B} = (v_1, v_2)$ for $\mathbb{R}^2$ s.t. $T_A : \mathbb{R}^2 \to \mathbb{R}^2$ has the following representation in the basis $\mathcal{B}$*

$$[T_A]^{\mathcal{B}}_{\mathcal{B}} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{10.1}$$

In other words T_A performs a reflection w.r.t. the line $l = \mathrm{Sp}(v_1)$. Moreover, $\exists\, \theta \in [0, 2\pi)$ s.t.

$$A = \begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix} = R_\theta \cdot \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The basis $\mathcal{B}$ for which Equation (10.1) holds, is given by

$$v_1 = \begin{pmatrix} \cos\frac{\theta}{2} \\ \sin\frac{\theta}{2} \end{pmatrix},\ v_2 = \begin{pmatrix} \cos(\frac{\theta}{2} + \frac{\pi}{2}) \\ \sin(\frac{\theta}{2} + \frac{\pi}{2}) \end{pmatrix}.$$

Summary:

$$\begin{aligned} SO(2) &= \text{ rotations} \\ O(2) \setminus SO(2) &= \text{ reflections.} \end{aligned}$$

Corollary 10.9

Let V be a 2-dimensional Euclidean space and $T : V \to V$ a linear isometry. Then $\det T = \pm 1$. Moreover, there exists an orthonormal basis $\mathcal{B}$ for V s.t.

1. *If $\det T = 1$, then $[T]^{\mathcal{B}}_{\mathcal{B}} = R_\theta$.*
2. *If $\det T = -1$, then*
$$[T]^{\mathcal{B}}_{\mathcal{B}} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Proposition 10.10: Classification of the Elements of $SO(3)$

Let $A \in SO(3)$. Then 1 is an eigenvalue of A. Moreover, there exists an orthonormal basis $\mathcal{B} = (v_1, v_2, v_3)$ and $\theta \in [0, 2\pi)$ s.t.

$$[T_A]^{\mathcal{B}}_{\mathcal{B}} = \left(\begin{array}{c|cc} 1 & 0 & 0 \\ \hline 0 & \multicolumn{2}{c}{\multirow{2}{*}{R_θ}} \\ 0 & & \end{array}\right).$$

So, T_A is a rotation with angle θ around the axis $\mathrm{Sp}(v_1)$. Moreover,

(a) If -1 is an eigenvalue of A, then $\theta = \pi$, hence
$$[T_A]^{\mathcal{B}}_{\mathcal{B}} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

*(b) If -1 is **not** an eigenvalue of A ($\Longleftrightarrow$ 1 is the only eigenvalue), then $\theta \neq \pi$.*

Proof. The characteristic polynomial $p_A(x)$ of A is a polynomial of degree 3 and with real coeffiecients. By the intermediate value theorem (Analysis I) there exists a real zero $\lambda \in \mathbb{R}$(,i.e., $p_A(\lambda) = 0$). Therefore, A has at least one real eigenvalue. By Lemma 10.7, $\lambda = \pm 1$. Let $v_1 \in \mathbb{R}^3 \setminus \{0\}$ be an eigenvector of λ. We can normalize v_1 and still have an eigenvector for λ. So, let's assume $\|v\| = 1$. Consider the subspace
$$L := v_1^\perp = \{v \in \mathbb{R}^3 \mid \langle v, v_1 \rangle = 0\}.$$
Since A is orthogonal, we have that
$$\forall v \in L: \ Av \in L,$$
i.e., $T_A(L) \subseteq L$, i.e., L is invariant under T_A. Indeed, if $v \in L$, then since $A \in O(3)$
$$\begin{aligned} 0 &= \langle v, v_1 \rangle = \langle Av, Av_1 \rangle = \langle Av, \pm v_1 \rangle = \pm \langle Av, v_1 \rangle \\ \Longrightarrow \quad & Av \in L. \end{aligned}$$

We also have $\mathbb{R}^3 = \mathrm{Sp}(v_1) \oplus L$, hence $\dim L = 2$.

Now $T_A|_L : L \to L$ is a linear isometry. Let $\mathcal{L} = (w_1, w_2)$ be an orthonormal basis for L. By Proposition 10.4, we have that
$$Q := [T_A|_L]^{\mathcal{L}}_{\mathcal{L}} \in O(2).$$

Case (a): $\lambda = 1$. Take $\mathcal{B} = (v_1, w_1, w_2)$. Then

$$[T_A]_{\mathcal{B}}^{\mathcal{B}} = \left(\begin{array}{c|cc} 1 & 0 & 0 \\ \hline 0 & \multicolumn{2}{c}{\multirow{2}{*}{Q}} \\ 0 & & \end{array}\right).$$

Now, since $A \in SO(3)$, we have

$$\begin{aligned} && 1 = \det A = \det(T_A) = 1 \cdot \det Q \\ \implies && \det Q = 1. \end{aligned}$$

So, $Q \in SO(2)$. By Proposition 10.10, $\exists\, \theta \in [0, 2\pi)$ s.t. $Q = R_\theta$. Therefore,

$$[T_A]_{\mathcal{B}}^{\mathcal{B}} = \left(\begin{array}{c|cc} 1 & 0 & 0 \\ \hline 0 & \multicolumn{2}{c}{\multirow{2}{*}{R_θ}} \\ 0 & & \end{array}\right).$$

Case (b): $\lambda = -1$. By the same argument as for case (a), we get that

$$\begin{aligned} && 1 = \det A = (-1) \cdot \det(T_A|_L) \\ \implies && \det(T_A|_L) = -1. \end{aligned}$$

By Corollary 10.9, there exists an orthonormal basis $\mathcal{L} = (w_1, w_2)$ for L s.t.

$$[T_A|_L]_{\mathcal{L}}^{\mathcal{L}} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Take $\mathcal{B}' = (v_1, w_1, w_2)$. Then

$$[T_A]_{\mathcal{B}'}^{\mathcal{B}'} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

And for $\mathcal{B} = (w_1, v_1, w_2)$ we have

$$[T_A]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix} = \left(\begin{array}{c|cc} 1 & 0 & 0 \\ \hline 0 & \multicolumn{2}{c}{\multirow{2}{*}{R_π}} \\ 0 & & \end{array}\right).$$

Note that in case (b), 1 is an eigenvalue of A as well. $\square$

Proposition 10.11: Classification of $O(3) \setminus SO(3)$

Let $A \in O(3) \setminus SO(3)$. Then there exists an orthonormal basis $\mathcal{B} = (v_1, v_2, v_3)$ for $\mathbb{R}^3$ s.t.

$$[T_A]_{\mathcal{B}}^{\mathcal{B}} = \left(\begin{array}{c|cc} -1 & 0 & 0 \\ \hline 0 & \multicolumn{2}{c}{\multirow{2}{*}{R_θ}} \\ 0 & & \end{array}\right)$$

for some $\theta \in [0, 2\pi)$. (So T_A does a rations by θ around $\mathrm{Sp}(v_1)$, followed by a reflection w.r.t. $\mathrm{Sp}(v_2, v_3)$.)

11 Bilinear & Quadratic Forms

Definition 11.1: Bilinear Form

Let V be a vector space over K. A function $B : V \times V \to K$ is called a **bilinear form** if the following two conditions are satisfied:

1. $\forall\, w \in V$ the map $B(\cdot, w) : V \to K, \quad v \mapsto B(v,w)$ is linear.
2. $\forall\, v \in V$ the map $B(v, \cdot) : V \to K, \quad w \mapsto B(v,w)$ is linear.

Lemma 11.2: Matrix Representation of a Bilinear Form

Let V be a vector space over K and $\mathcal{C} = (e_1, \dots, e_n)$ a basis for V. Let $B : V \times V \to K$ be a bilinear form. Then $\exists\, [B]_{\mathcal{C}} \in M_{n\times n}(K)$ s.t.

$$B\Big(\sum_{i=1}^{n} c_i e_i, \sum_{j=1}^{n} d_j e_j\Big) = \begin{pmatrix} c_1 & \dots & c_n \end{pmatrix} \cdot [B]_{\mathcal{C}} \cdot \begin{pmatrix} d_1 \\ \vdots \\ d_n \end{pmatrix} \qquad \forall\, c_i, d_j \in K.$$

The matrix $[B]_{\mathcal{C}} = B(e_i, e_j)$ is called the matrix representation of B in the basis $\mathcal{C}$.

We saw also that for $T \in \mathrm{End}(V)$ and $\mathcal{C}$ a basis for V we have a matrix representation $[T]_{\mathcal{C}}^{\mathcal{C}} \in M_{n\times n}(K)$, and that if $\mathcal{D}$ is another basis for V, then

$$[T]_{\mathcal{D}}^{\mathcal{D}} = \underbrace{[id_V]_{\mathcal{D}}^{\mathcal{C}}}_{= \left([id_V]_{\mathcal{C}}^{\mathcal{D}}\right)^{-1}} \cdot [T]_{\mathcal{C}}^{\mathcal{C}} \cdot [id_V]_{\mathcal{C}}^{\mathcal{D}}. \tag{11.1}$$

(Corollary 3.17). What happens for bilinear forms?

Lemma 11.3: Change of Bases for a Bilinear Form

Let V be a finite dimensional vector space over K and $\mathcal{C}, \mathcal{D}$ be two bases for V. Let $B : V \times V \to K$ be a bilinear form. Then

$$[B]_{\mathcal{D}} = \left([id_V]_{\mathcal{C}}^{\mathcal{D}}\right)^T \cdot [B]_{\mathcal{C}} \cdot [id_V]_{\mathcal{C}}^{\mathcal{D}}.$$

(This is quite different from Equation (11.1)).

Definition 11.4: Symmetric/Positive Definite Bilinear Form & Quadratic Form

1. A bilinear form $B : V \times V \to K$ is called **symmetric** if

$$\forall\, v, w \in V : \ B(v, w) = B(w, v).$$

2. Assume that $K = \mathbb{R}$. A bilinear form $B : V \times V \to K$ is called **positive definite** if

$$\forall\, v \in V : \ B(v, v) > 0.$$

3. For a bilinear form $B : V \times V \to K$, we define

$$q_B : V \to K, \quad q_B(v) := B(v, v).$$

We call q_B the **quadratic form** associated to B. Sometimes we just denote it by q.

Remark 11.5. *Note that $B \in M_{n\times n}(K)$ and $B^T \in M_{n\times n}(K)$ give the same quadratic form $q : K^n \to K$. Indeed, $\forall\, v \in K^n$*

$$q_B(v) = v^T B\, v = (v^T B\, v)^T = v^T B^T v = q_{B^T}(v),$$

where we used that $q_B(v)$ is a scalar.

Assume now that $2 \neq 0$ in K, then (i.e., $1 + 1 \neq 0$). Then also $\frac{1}{2}(B + B^T)$ gives the same quadratic form as B.

Conclusion: if $2 \neq 0$ in K, then for the discussion on quadratic forms $q : K^n \to K$, we can assume they all come from symmetric matrices / bilinear forms.

Proposition 11.6: Polarization Identity

Assume $2 \neq 0$ in K. (note that then also $4 \neq 0$.) If B is symmetric and defines $q := q_B$, then $\forall\, v, w \in V$ we have

1.

$$B(v, w) = \frac{1}{4} q(v + w) - \frac{1}{4} q(v - w).$$

2.

$$B(v, w) = \frac{1}{2}\big(q(v + w) - q(v) - q(w)\big).$$

Theorem 11.7: Sylvester's Inertia Theorem

Let $B : V \times V \to K$ be a symmetric bilinear form on a vector space V over $\mathbb{R}$, of dimension n. Then there exists a basis $\mathcal{C}$ for V s.t.

$$[B]_{\mathcal{C}} = \begin{pmatrix} I_k & 0 & 0 \\ 0 & -I_l & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

*for some $k, l \in \mathbb{Z}_{\geq 0}$ with $k + l \neq n$. Moreover, k and l do **not** depend on the basis $\mathcal{C}$ (they depend only on B). In fact*

$$\begin{aligned} k &= \max\{\dim W \mid W \subseteq V \text{ is a subspace and } B|_{W \times W} \text{ is positive definite}\}, \\ l &= \max\{\dim U \mid U \subseteq V \text{ is a subspace and } B|_{U \times U} \text{ is positive definite}\}, \\ n - (k + l) &= \max\{\dim Z \mid Z \subseteq V \text{ is a subspace and } B|_{Z \times V} \equiv 0\}. \end{aligned}$$

Remark 11.8. *1. In the basis $\mathcal{C}$, the quadratic form q associated to B looks like*

$$q(x_1, \ldots, x_n) = x_1^2 + \ldots + x_k^2 - x_{k+1}^2 - \ldots - x_{k+l}^2.$$

*2. k is called the positivity index of B, l the negativity index, n the nullity and $\sigma(B) := k - l$ is called the **signature of** B. We say B is a bilinear form of type (k, l).*

Theorem 11.9: Sylvester's Inertia Theorem for Euclidean Spaces

*Let V be a finite dimensional Euclidean space of dimension n. Let $B : V \times V \to K$ be a symmetric bilinear form. Then there exists an **orthonormal** basis $\mathcal{C}$ for V in which*

$$[B]_{\mathcal{C}} = \begin{pmatrix} \eta_1 & 0 & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & 0 \\ 0 & \ddots & \ddots & & & & & & \vdots \\ \vdots & \ddots & \eta_k & \ddots & & & & & \vdots \\ \vdots & & \ddots & \eta_{k+1} & \ddots & & & & \vdots \\ \vdots & & & \ddots & \ddots & \ddots & & & \vdots \\ \vdots & & & & \ddots & \eta_{k+l} & \ddots & & \vdots \\ \vdots & & & & & \ddots & 0 & \ddots & \vdots \\ \vdots & & & & & & \ddots & \ddots & 0 \\ 0 & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & 0 & 0 \end{pmatrix},$$

*where $k.l \in \mathbb{Z}_{\geq 0}$ and $k + l \leq n$, $\eta_1, \ldots, \eta_k > 0$, $\eta_{k+1}, \ldots, \eta_{k+l} < 0$. As before, k, l are the positivity and negativity indices of B. Moreover, the scalars $\eta_1, \ldots, \eta_k, \eta_{k+1}, \ldots, \eta_{k+l}, 0, \ldots, 0$ are independent of $\mathcal{C}$. They depend only on B (and on the scalar product on V). They are the eigenvalues of $[B]_{\mathcal{D}}$ for every **orthonormal** basis $\mathcal{D}$ of V (independent of $\mathcal{D}$ and on whether or not $[B]_{\mathcal{D}}$ is diagonal).*

Remark 11.10. *This theorem has a geometric meaning. If $q : \mathbb{R}^n \to \mathbb{R}$ is a quadratic form*

$$q(x_1, \ldots x_n) = \sum_{i=1}^{n} a_{ij} x_i x_j$$

for some $a_{ij} \in \mathbb{R}$, *then the set*

$$Q = \{(x_1, \dots, x_n) \in \mathbb{R}^n \mid q(x_1, \dots, x_n) = C \in \mathbb{R}\}$$

is called a quadric. The theorem tells us that there exists a linear isometry $T : \mathbb{R}^n \to \mathbb{R}^n$ *s.t.*

$$T(Q) = \{(y_1, \dots, y_n) \in \mathbb{R}^n \mid \underbrace{\eta_1 y_1^2 + \dots + \eta_k y_k^2}_{\textit{pos. coeffs.}} + \underbrace{\eta_{k+1} y_{k+1}^2 + \dots + \eta_{k+l} y_{k+l}^2}_{\textit{neg. coeffs.}} = C \in \mathbb{R}\}.$$

Theorem 11.11: Sylvester's Theorem for 'almost' general Fields

Let V *be a finite dimensional vecotr space over a field* K*, in which* $2 \neq 0$*. Let* $B : V \times V \to K$ *be a symmetric bilinear form. Then there exists a basis* $\mathcal{C}$ *of* V *s.t.* $[B]_{\mathcal{C}}$ *is diagonal.*

What happens over $\mathbb{C}$?

Theorem 11.12: Sylvester's Theorem over $\mathbb{C}$

Let $B : V \times V \to \mathbb{C}$ *be a symmetric bilinear form on a vector space over* $\mathbb{C}$ *of dimension* n*. Then there exists a basis* $\mathcal{C}$ *of* V *s.t.*

$$[B]_{\mathcal{C}} = \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix},$$

for some $0 \leq r \leq n$*. The number* r *is independent of the basis* $\mathcal{C}$*. It depends only on* B*. In fact*

$$n - r = \max\{\dim W \mid W \subseteq V \textit{ is a subspace s.t. } B|_{W \times V} \equiv 0\}.$$

r *is sometimes called the rank of* B*.*

11.1 More on positive definite Bilinear Forms

Theorem 11.13: Equivalent Conditions for a Positive Definite Bilinear Form

Let V *be a finite dimensional vector space over* $\mathbb{R}$ *and* $B : V \times V \to \mathbb{R}$ *a symmetric bilinear form. Then the following conditions are equivalent:*

1. B *is positive definite*
2. *There exists a basis* $\mathcal{C} = (e_1, \dots, e_n)$ *for* V *s.t. the principal minors* $\det\big(B(e_j, e_k)\big)_{j,k=1,\dots,l}$ *for all* $1 \leq l \leq n$ *are positive.*
3. *Same as 2. But instead of* $\exists$ *a basis put for* $\forall$ *bases.*

12 Jordan Normal Form

Definition 12.1: Jordan Matrix

Let $\lambda \in K, n \in \mathbb{Z}_{\geq 1}$. Define

$$J_{\lambda,n} = \begin{pmatrix} \lambda & 1 & 0 & \dots & 0 \\ 0 & \ddots & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 \\ \vdots & & \ddots & \ddots & 1 \\ 0 & \dots & \dots & 0 & \lambda \end{pmatrix} \in M_{n\times n}(K).$$

For $n = 1$, $J_{\lambda,1} = (\lambda)$. For $n = 2$,

$$J_{\lambda,2} = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}.$$

For $n = 3$,

$$J_{\lambda,3} = \begin{pmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{pmatrix}.$$

$J_{\lambda,n}$ is called the n-dimensional **Jordan matrix** with eigenvalue λ.

Lemma 12.2: Properties of the Jordan Matrix

1. *λ is the only eignevalue of $J_{\lambda,n}$*
2. *$p_{J_{\lambda,n}}(x) = (\lambda - x)^n$, and $m_a(J_{\lambda,n}, \lambda) = n$.*
3. *$m_g(J_{\lambda,n}, \lambda) = 1$. In fact $\mathrm{Eig}_\lambda(J_{\lambda,n}) = \mathrm{Sp}(e_1)$.*

Theorem 12.3: Jordan Normal Form (JNF)

Let V be an n-dimensional vector space over K and $T \in \mathrm{End}(V)$ s.t. $p_T(x)$ splits as a product of linear factors in $K[x]$ (e.g. for $K = \mathbb{C}$ this always happens). Then there exists a basis $\mathcal{B}$ for V s.t.

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} J_{\lambda_1,n_1} & 0 & \dots & 0 \\ 0 & J_{\lambda_2,n_2} & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \dots & 0 & J_{\lambda_k,n_k} \end{pmatrix},$$

*where $k \geq 1, \lambda_1, \dots, \lambda_k \in K, n_1, \dots, n_k \geq 1, n_1 + \dots + n_k = n$. $\mathcal{B}$ is called **a** Jordan basis for T. Moreover, up to permutation, the blocks $J_{\lambda_1,n_1}, \dots, J_{\lambda_k,n_k}$ are unique.*

For an example see the lecture notes.

We can write $J_{\lambda,n} = \lambda \cdot I_n + N$, where

$$N = \begin{pmatrix} 0 & 1 & 0 & \dots & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ \vdots & & \ddots & \ddots & 0 \\ \vdots & & & \ddots & 1 \\ 0 & \dots & \dots & \dots & 0 \end{pmatrix} = J_{0,n}$$

Lemma 12.4: Powers of N

1.

$$N^2 = \begin{pmatrix} 0 & 0 & 1 & \dots & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ \vdots & & \ddots & \ddots & 1 \\ \vdots & & & \ddots & 0 \\ 0 & \dots & \dots & \dots & 0 \end{pmatrix}, \; N^3 = \dots, \quad \dots, N^n = 0,$$

$N^k = 0 \quad \forall k \geq n$. *In other words,* N^k *has 1's in the* k*-th diagonal above the central diagonal and 0's everywhere else. Note also that* N *is nilpotent.*

2. $\forall k \geq 1$ *we have*

$$J_{\lambda,n}^k = \sum_{j=0}^{k} \binom{k}{j} \lambda^{k-j} N^j.$$

Corollary 12.5: Geometric Multiplicity and Minimal Polynomial of JNF

Suppose $T \in \operatorname{End}(V)$, *admits a matrix representation*

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} J_{\lambda_1,n_1} & 0 & \dots & 0 \\ 0 & J_{\lambda_2,n_2} & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \dots & 0 & J_{\lambda_k,n_k} \end{pmatrix}$$

in some basis $\mathcal{B}$. *Then for every eigenvalue* λ *of* T *the geometric multiplicity* $m_g(\lambda)$ *equals the number of blocks in* $[T]_{\mathcal{B}}^{\mathcal{B}}$ *labeled with* λ. *In other words,*

$$m_g(\lambda) = |\{j \mid \lambda_j = \lambda, \quad 1 \leq j \leq k\}|.$$

The minimal polynomial $\mu_T(x)$ *is*

$$\mu_T(x) = \prod_{\lambda = eigenvalue} (x - \lambda)^{S(\lambda)},$$

where

$$\begin{aligned} S(\lambda) &= \text{size of the largest block labeled by } \lambda \\ &= \max\{n_j \mid \lambda_j = \lambda, \quad 1 \leq j \leq k\}. \end{aligned}$$